

Chapter II: Electoral Effects of Spatial Proximity to Protests on Voting in Russia

In this chapter, I investigate how localized protests in authoritarian regimes impact electoral outcomes, focusing on Russia between the 2016 and 2021 Parliamentary elections. Using data from anti-corruption protests led by Alexei Navalny in 2017 and pension reform protests in 2018, I analyze the effects of protest proximity on the ruling party's vote share, voter turnout, and opposition support. The findings indicate that protests reduce both the ruling party's vote share and turnout in affected areas but do not significantly increase support for opposition parties. The ruling party's losses are primarily driven by lower turnout among its supporters, with protests in the ruling party strongholds and larger protests showing more pronounced effects. Demonstrations met with crackdown on protesters appear to have somewhat larger negative effect on both turnout and the United Russia vote share compared to protests without repressions, although the difference is not statistically significant. These findings align with prior research while extending its scope to a harsher authoritarian context and different protest agendas. Moreover, the analysis suggests that protests influence attitudes not necessarily by revealing new information but by signaling declining regime support.

Introduction

Ensuring mass compliance has always been a critical task for authoritarian leaders. While rulers of the past relied on the power of tradition and brute military force to keep the population at bay, dictators of the twentieth century weaponized the sprawling machinery of the modern state to build all-penetrating repressive apparatus and establish strict control over mass media. In the age of information, however, dictators that rule through fear and totalitarian propaganda give their way to a different kind of autocrats: those cultivating an image of competence and even democratic legitimacy through populism and manipulation of information (Guriev and Treisman 2020a, 2022). For such new dictators, the main goal becomes mimicking democratic leaders that are able to provide economic prosperity, hence a limited use of overt violence. Among other things, this democratic masquerade necessitates a different approach to managing mass dissent (Robertson 2010). Protests activity is not prohibited officially, and demonstrations of public dissent can happen significantly more often and with significantly fewer risks for the participants than in traditional dictatorships of fear. At the same time, allowing even a very limited degree of freedom of assembly may have significant ramifications for dictators that build their legitimacy through manipulation of information. Even in democratic contexts, costly political actions, such as protests, can act as mechanisms for information aggregation and affect the beliefs of non-participants regarding the state of the world and the policies of the government (Lohmann 1994a); in information environment typical for authoritarian regimes, public displays of dissent may have an amplified effect and even set off a cascade effect undermining regime support, eventually leading to the complete collapse of the regime (Kuran 1991; Lohmann 1994b). But even if futile from the perspective of the regime change – as, for example, recent protests in Hong Kong, Venezuela, Belarus, and Iran – localized dissent can signal an overall growing discontent, policy incompetency, and, depending on the regime’s response, authoritarian nature of the ruler, and therefore affect voters’ behavior.

In this paper, I investigate how local protests in an authoritarian regime, specifically Russia, affect the ruling party’s vote share, voter turnout, and support for opposition parties. Specifically, I use data on protest activity in Russia between the 2016 and the 2021 Parliamentary elections – namely anti-corruption protests organized by Alexei Navalny in 2017 and protests against the increase on the retirement age in 2018 – to investigate whether localities near protest sites register lower shares of votes for the regime party than those farther away, as well as the effect on the

turnout and vote shares of opposition parties. Additionally, I conduct a set of heterogeneity tests to see if the effects in question change conditional on (1) the ruling party pre-treatment support in the region, (2) the magnitude of a protest, and (3) the authorities' use of repression against the protesters. The results suggest that protests negatively affect both the ruling party's total vote share and voter turnout. However, they do not appear to increase support for opposition parties, as no effect is observed on their combined total vote share. The analysis indicates that the ruling party's decreased vote share can be mainly attributed to decreased turnout in affected areas. In other words, local protests may deter regime supporters more than they mobilize potential opposition supporters. The effects are significantly more pronounced in the United Russia strongholds, and for protests with a larger number of participants and for those met with arrests by authorities, though the latter difference is not statistically significant.

Theoretically, the paper contributes to the literature on attitudinal consequences of protests, and political behavior and attitudes in authoritarian regimes in general. The result lend further support to the findings by Tertytchnaya (2020). Similarly to this study, I document how anti-governmental protests demobilize the regime supporters more than they mobilize pro-opposition voters. Importantly, while Tertytchnaya's study focused on the short-term effects of the 2011 Russian protests on reported vote choice in the 2012 Presidential election, the results presented in the current paper suggest that such effects can manifest even several years after the protests, within a significantly harsher authoritarian environment, and for protests with agendas markedly different from those of 2011–2012. Thus, I find that both the anti-corruption protests organized by Navalny and the protests against the increase in the retirement age, primarily led by the Communist Party of the Russian Federation (CPRF), produced essentially the same outcomes. Importantly, the pension protests revealed little about the content of the government's policy, as the issue had been of nationwide salience since the reform was announced by then Prime Minister Medvedev. Therefore, the findings provide tentative evidence that the mechanism through which protests influence bystanders' attitudes is not necessarily linked to their ability to reveal substantively new information about the regime or its leaders. Instead, it may be related to their role in signaling declining regime support and solidifying voters' views (Pop-Eleches, Robertson, and Rosenfeld 2022). In addition to that, I find that protest have a significantly more pronounced effect in regions where United Russia is traditionally strong.

The second contribution is empirical. To my knowledge, it is the first study that analyzes

electoral outcomes in Russia on the lowest possible level of aggregation, that is one the level of electoral precincts and their territories. Using the data from Open Streets Maps, I was able to link more than 3,500,000 buildings to polling stations and reconstruct geographical borders of more than 30,000 electoral precincts in 403 largest settlements in Russia. This allowed me to calculate several key variables, such as median real estate price within the territory of the precinct and the mean night light emission. Coupled with doubly robust difference-in-differences estimator proposed by Sant’Anna and Zhao (2020), this allowed me to plausibly account for potential violation of the parallel trends assumption.

This research builds on these insights by examining the dynamics of protests in authoritarian settings, where repression is more common, and the government propaganda often frames protests as organized from abroad and aimed at the destruction of the country, which could dampen any long-lasting effect on people’s attitudes (Arnon, Edwards, and Li 2023). By focusing on protests that do not result in sweeping change, my study contributes to understanding indirect ways in which protest activity can influence political dynamics in authoritarian regimes, with implications for the long-term stability and adaptability of these systems.

The remainder of the paper is organized as follows. In the next two sections, I briefly engage with the literature on protests and information in authoritarian regimes. The third section introduces to the empirical context, while the fourth discusses the research design and data. The section five present main results, and the sixth concludes.

Political protests, regime stability and information in autocracies

Anti-government demonstrations in authoritarian regimes are often studied for their effects on large-scale political outcomes. Research on the macro-level consequences of protests typically examines why some protests succeed in overthrowing regimes while others fail (Chenoweth and Belgioioso 2019), why some lead to concessions while others are met with repression (Klein and Regan 2018; Mueller 2024; Yuen and Cheng 2017), which types of protests are most effective for democratization (Chenoweth and Schock 2015; Kim and Kroeger 2019), or under what conditions they can trigger coups (Apolte 2022; Gerling 2017). But what happens when protests fail to trigger an immediate change? As recent protests in Hong Kong, Venezuela, Belarus, and Iran remind us, this outcome is significantly more probable than the others, as in most cases political protests result in neither regime change nor even concessions (Carothers and Youngs 2015). In

fact, opposition leaders rarely proclaim regime change as their aim when calling people out on the streets, yet they organize protests on a relatively frequent basis regardless. This persistence suggests that even without achieving immediate regime change or concessions, protests serve crucial functions, notably acting as tools to raise the salience of specific political issues and shape public opinion among the broader population.

The potential to shift public opinion can therefore be a key reason protests occur, and these demonstrations can act as significant informational events. Even if unsuccessful in achieving macro-level outcomes, anti-regime demonstrations can have less direct effects. For instance, Frye and Borisova (2019) show that post-electoral protests in Russia in 2011 paradoxically increased trust in government. Using variation in survey interview timing, they found respondents interviewed after the protests reported higher trust in institutions. They attribute this finding to the regime's unexpected decision to authorize these demonstrations and a generally measured response: protests were peaceful, participants were unharmed, and demands appeared to be acknowledged. However, such favorable surprises are rare in authoritarian regimes, where harassment of dissidents is the norm. All else being equal, it is reasonable to expect protests to exacerbate negative attitudes toward the regime. In African countries, for example, protests generally seem to lead to a decrease in trust in government and leaders, and even more so when they are harshly repressed (Sangnier and Zylberberg 2017). As Tertychnaya and Lankina (2020) show, the protests of 2011-2012 in Russia increased support for the protesters' demands among wider segments of society: in the first weeks of the anti-regime demonstrations, respondents expressed more sympathy for the protesters' demands. In a separate paper, Tertychnaya (2020) examines how anti-regime protests influence voter behavior. Focusing on Russia's 2011-2012 protest wave, she finds that while opposition protests reduce support for the ruling regime and decrease political engagement among its supporters, they do not increase support for opposition parties.

Beyond the authoritarian context central to this study, a body of literature also examines the effects of protests in established democracies. Research on movements like the Civil Rights marches, immigrant rights rallies, or Black Lives Matter protests in the United States, for instance, explores their influence on voters, elites, and policy (Branton et al. 2015; Wasow 2020). However, the evidence on their attitudinal effects is mixed. Thus, recent studies focusing on Western democracies suggest that while protests can effectively increase issue salience or affect

short-term behavioral responses, their capacity for prompting lasting shifts in political attitudes among the broader public may be limited (Gethin and Pons 2024; Haas et al. 2024). At the same time, the dynamics and potential impact of protests in authoritarian regimes may diverge significantly from those in democratic settings. Specifically, the very act of protesting under repressive conditions, where participation carries high personal risk, can lend the protest signal greater weight and credibility (Kricheli, Livne, and Magaloni 2011). Furthermore, in environments with tightly controlled media landscapes, protests often represent rare public displays of dissent, potentially carrying greater informational value for bystanders. Unlike in democracies, where protest is a more routine and institutionalized form of political expression, demonstrations in autocracies can constitute more potent, non-routine informational shocks. Consequently, it is plausible that protests under authoritarianism possess a greater potential to alter political attitudes compared to their counterparts in more open regime.

However, despite the few notable exceptions mentioned above, there is comparatively little research on the local effects of protests on the political attitudes and behavior of ordinary citizens in authoritarian regimes. This gap is particularly surprising given the influential literature suggesting that protests can lead to the regime collapse by changing the attitudes of bystanders. Thus, the literature on the diffusion of protests emphasizes the importance of anti-government demonstrations for encouraging larger numbers of citizens to openly oppose the regime. According to Kuran (1991), in autocracies, people often falsify their preferences to align with socially acceptable norms due to the costs of expressing dissent. However, individuals vary in their tolerance for the psychological costs of suppression. When these costs surpass a certain threshold, they reveal their true preferences. Signs of discontent lead people to reassess the regime's support and the acceptability of dissent, altering the payoff of preference falsification. This triggers a chain reaction where others follow suit, initiating a "revolutionary bandwagon" that can culminate in a regime's collapse. As Lohmann (1994a) argues, costly political actions - signing petitions, demonstrations, riots, etc. - can serve as a vehicle for disseminating information, whether about the regime's popularity or the broader conditions affecting individuals' preferences and policy decisions. Such methods can prove to be especially effective in a society where information flows are strictly controlled by the government, as happened in the GDR in 1989 (Lohmann 1994b). Thus, mass exodus from GDR to FRG and anti-regime demonstrations in Leipzig revealed to broader segments of the population the extent of the general discontent with the communist government, which led to the cascade of other demonstrations and eventually

translated into mass protest and regime change.

Both Kuran's and Lohmann's models offer valuable insights into the dynamics of mass dissent in authoritarian regimes and serve as key conceptual foundations for this paper. All the more important are the differences between the two accounts. In Kuran's model, signs of dissent prompt people to reveal their true preferences by altering their beliefs about others' opinions and their perceptions of repression risks; the preferences themselves are largely not affected, only the decision to reveal (or conceal) them. In contrast, Lohmann's model emphasizes how protests can reveal new information that leads bystanders to update their beliefs about various aspects of the world, such as overall regime support and the state of the economy; while repression reduces the amount of information revealed by suppressing the number of participants and occurrences, in itself it plays a secondary role in Lohmann's model. In short, Kuran focuses on fear and social conformity, while Lohmann puts more emphasis on cues and learning.

These theoretical frameworks highlight why protests can shift attitudes, but understanding the impact also requires considering how the relevant signals and information diffuse through the population. Generally, individuals can learn about protests, their demands, the participants, and the regime's response through several key channels. First, media coverage can provide broad visibility, though often with specific framing. On the other hand, direct experience, such as witnessing events firsthand or participating, offers unmediated information but reaches a more limited audience. In between lies the diffusion of information through interpersonal networks: conversations with friends, family, colleagues, and neighbours. At the same time, mechanisms driving attitudinal shifts can be particularly relevant for those in the immediate vicinity of the events. Specifically, spatial proximity can intensify the exposure: it increases the likelihood of direct witnessing of the protest itself and the regime's response (seeing it near one's home or workplace), facilitates the spread of information and interpretations through local networks (hearing from neighbours, local friends, and colleagues), and enhances the personal salience of the event compared to receiving information remotely. Therefore, the theoretical effects proposed by Kuran and Lohmann are likely amplified or shaped by these real-world transmission processes, particularly for those geographically closer to the events. This study focuses specifically on these spatially mediated effects, and aims to investigate under what circumstances protests' attitudinal effects are likely to be most pronounced.

In this respect, classic theories by Lohmann and Kuran can again provide a useful starting point.

Their main distinction with regards to empirical expectations pertains to the effect of repression: if we live in Kuran’s world, protests met with arrests and detentions signal higher costs of aligning with opposition, potentially leading to more preference falsification; In Lohmann’s, such response could expose the malign nature of the regime, potentially making people more likely to defect from it. In addition to that, the very occurrence of protests in a more repressive environment sends a stronger signal about the level of discontent, which aligns with a more rigorous model proposed by Kricheli, Livne, and Magaloni (2011). According to its main propositions, although “more repressive autocratic regimes are in principle more stable, in part because they are better able to deter civil opposition,” ... “protest that takes place in a more repressive autocratic regime reaches its maximum information-revealing potential and hence is more likely to cascade into a successful uprising” (Kricheli, Livne, and Magaloni 2011). Similarly, in a working paper dedicated to electoral consequences of protests in Chile and Bolivia, Castro and Retamal (2022) find that in districts where the protesters were repressed, the incumbent parties received lower shares of votes.

As for the other key factors that can moderate the effect of protests on political behavior and attitudes, the models developed by Kuran and Lohmann generate largely the same expectations. Thus, both models predict that:

- (1) More frequent and more attended protests should have a stronger effect.
- (2) Protests can have a larger impact when they are least expected, particularly when the perceived support for the regime is initially high.

It is worth noting that the mechanisms outline in both models are not mutually exclusive. In reality, different individuals may respond to protests in ways that align with either model. For example, some individuals might reveal their true preferences only when they perceive a shift in the risks of dissent or the social acceptability of opposing the regime. On the other hand, others might be influenced less by the perceived risks of repression and more by the new knowledge they gain from observing the protests. In practice, both mechanisms may operate simultaneously within the same society, with some people acting based on fear and conformity, while others respond to cues and learning. Together, these dynamics can interact to amplify the effects of protests, making them both a signal of discontent and a catalyst for belief updating. Thus, according to Tucker (2007), post-election protests in Serbia, Ukraine, and Georgia in the 2000s disseminated information about electoral fraud, as well as lowered the perceived costs of

participating in anti-regime activities.

Drawing on Kuran's and Lohmann's theoretical models, and supported by Tertychnaya's work on Russia, the literature suggests that protests in authoritarian regimes can influence public attitudes and behavior even in the absence of immediate regime change. Kuran's approach emphasizes how visible dissent reduces the perceived costs of expressing opposition, while Lohmann's model views protests as informational events that prompt individuals to update their beliefs about regime strength. Tertychnaya's findings demonstrate that such shifts can alter support for the regime and the opposition, ultimately affecting electoral outcomes.

These insights imply that protest frequency, size, timing, and the intensity of repression can shape the extent to which protest signals translate into changes in citizens' political views and actions. This study will test these expectations by examining data on protests, elections, vote shares, and turnout in Russia, providing a clearer understanding of how protests influence political landscapes under authoritarian conditions.

In the next sections, I will discuss the context and outline the empirical design used to test these propositions.

Background

Protests in Russia

Before discussing the data, a few words about the protests in Russia are in order. Since Vladimir Putin was elected in 2000, there were four waves of protests. The first wave occurred in 2004 and was triggered by the monetization of social benefits. The government replaced fare-free public transit, medicines, and other services for disabled individuals, military personnel, and pensioners with disproportionately lower monetary compensations. This policy sparked widespread discontent, resulting in anti-government demonstrations, primarily in Moscow and Saint Petersburg. The second wave began in December 2011 following massive electoral fraud during the State Duma elections. These protests marked the first large-scale, explicitly political demonstrations since the 1990s, with up to 120,000 participants rallying in Moscow. Protesters demanded re-election, the release of political prisoners, and the resignation of the head of the Central Election Committee. Although the demonstrations continued into the spring of 2012, they were eventually suppressed by the regime.

In the current paper, I focus of the third wave of the protest that started on March 26, 2017, with anti-corruption protests organized by Alexei Navalny and his team across Russia. Following the 2018 presidential election, protests against the announced increase in the retirement age added to this wave. While Navalny’s team played a significant role, the Communist Party of the Russian Federation (CPRF) organized the majority of these anti-reform demonstrations. In total, I have the data on 1108 protest events, collected by Rumiantseva (2021), which include their precise location within a city, the number of participants and whether or not any of the participants was detained.

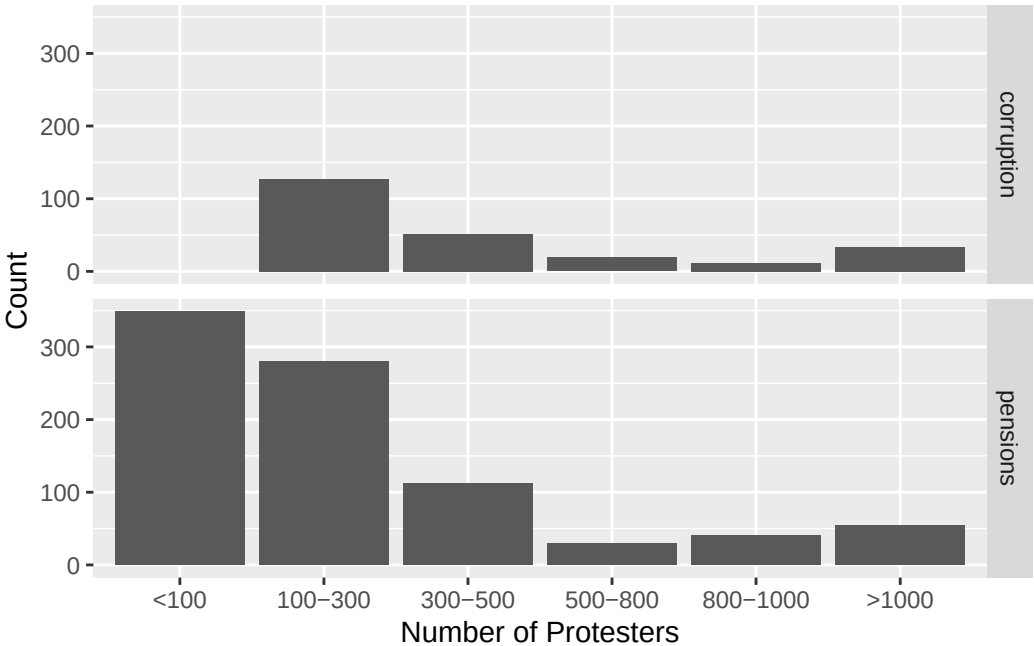
The only nationwide protests not included to the sample are the January 2021 protests that took place after Alexei Navalny’s detention upon his return to Russia, and the April 2021 protests organized by Navalny’s team in response to his imprisonment conditions. While these protests attracted thousands of participants, their exclusion is unlikely to significantly affect the results. First, their spatial distribution likely mirrors the 2017 protests, which were also organized by the Anti-Corruption Foundation (AFC) and were similarly unsanctioned by authorities. Second, if protests occurred in previously unaccounted locations, this would contaminate the treatment group and bias the results toward zero. Consequently, the analysis may underestimate the true effect, providing a conservative lower-bound estimate.

In addition, it is important to acknowledge potential limitations inherent in protest event datasets like this one, which rely on media reports or monitoring efforts. A common concern is selection bias: the likelihood that a protest is recorded may correlate with characteristics such as its size or the authorities’ response, potentially leading to underreporting of smaller, less visible protests. As a result, some small protests may go unrecorded and thus appear in the control group in our analysis. Again, if such treated observations are present in the control group, this would likely bias the estimated effects toward zero. However, this bias is expected to be minimal, as smaller protests are unlikely to produce strong effects.

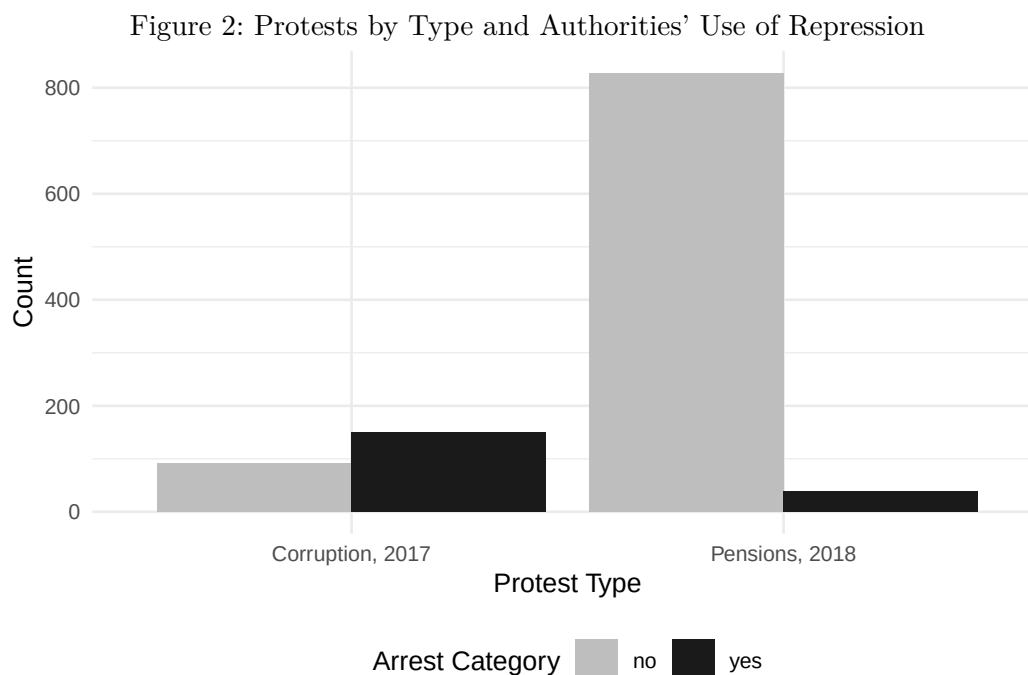
Several things are worth noting regarding the protests of 2017-2018. First, the majority of the 1108 protest events in the sample were organized in response to the announcement of the Pension Reform. Thus, out of 1108 protest event, 242 focused on corruption and took place in 2017, while 866 were held to oppose the pension reform in 2018. In both cases, there were substantial variations in the number of protesters. As shown in Figure 1, the Pension Reform protests attracted a significantly larger number of participants overall. Notably, protests with

fewer than 100 participants accounted for over one-third of all Pension Reform protests.

Figure 1: The number of protesters across Anti-corruption and Pension reform protests



In addition, as reported in Figure 2 the authorities adopted a mixed approach: some protests were officially permitted or occurred without arrests, while others faced crackdowns. For example, out of 242 anti-corruption protests in the sample, 150 involved arrests of participants, while 92 did not. In contrast, protests against the increase in the retirement age saw significantly fewer crackdowns, with detentions reported in only 38 out of 866 cases. Importantly, the number of protesters is not correlated with whether authorities resorted to detentions or not.



Second, when the authorities granted official permission for protests, the designated locations were often in hard-to-reach areas far from city centers. This extract from the regional newspaper’s website illustrates the overall strategy of the authorities well:

The protest rally in Tver on July 28 was organized by the local Communist Party. They planned to hold a public event on Saturday at Slava Square [the central square - I. F.]. For the rally to be sanctioned, it had to be approved by the city administration, which the organizers duly requested. In response, they received a “proposal letter,” which did not explicitly ban the rally but relocated it from Slava Square to... the Cheminstitute [the district 9 km away from the city center - I. F.], citing that all significant public spaces in the city center and nearby areas would be occupied that day for the “Youth Matters” event. <...> Communist and deputy of the Tver Regional Legislative Assembly, Artyom Goncharov, posted on Facebook, stating that the rally in Tver would take place regardless:

“The administration of Tver decided to suppress the protest against raising the retirement age by scheduling children’s events at all approved venues for public gatherings, except for the Cheminstitute. Well, we’ve decided to hold the event anyway, even if it has to be there.”

Because of this approach, the protests were not concentrated exclusively in central, potentially more affluent districts. Instead, they were distributed across localities with varying levels of wealth and centrality, creating a more balanced geographic spread than might be expected in a democratic context. While this likely reduced the number of participants and the potential city-wide impact of these protests, it also helps mitigate concerns about selection into treatment. In this case, selection into treatment could threaten the parallel trends assumption if protests were disproportionately held in areas with specific socioeconomic characteristics, such as higher wealth or greater political engagement, as these factors might lead to divergent pre-treatment trends in the outcome variable. By distributing protests more evenly across districts, the likelihood of such pre-existing differences driving the results is reduced, increasing confidence that any observed effects can be attributed to the protests rather than violations of the parallel trends assumption.

Third, the protests differed in their main organizers, agendas and the kind of information they could reveal. The anti-corruption protests were organized by the Anti-Corruption Foundation (ACF), led by Alexei Navalny. They began shortly after the release of a journalistic investigation by the ACF, in which Navalny and his team accused then-Prime Minister Dmitry Medvedev of controlling a network of ostensibly charitable foundations used to conceal ownership of luxurious villas, yachts, and other assets. In this regard, these protests could be seen as potential transmitters of information previously unknown to the general population. The 2018 pension reform protests, in their turn, began after Dmitry Medvedev, who was still a Prime Minister, announced the reform. Organized primarily by the Communist Party of the Russian Federation (CPRF) and other left-wing groups, these protests revealed little new information beyond the general public discontent with the reform itself.

Geographic boundaries of electoral precincts

A particular challenge in analyzing the local effects of protests is the absence of readily available data on the boundaries of electoral precinct territories. In Russia, the only readily available information is the location of polling stations. Individuals can vote only at their designated polling station, which is assigned based on their residential address⁴. While this data makes it possible to identify the buildings where votes are cast, it does not indicate which residential

⁴It is possible to obtain a voting absentee certificate and register to vote at a different polling station, though this option is not commonly used and does not allow one to vote for single-member district candidates.

buildings are associated with each polling station. Since electoral territories vary in size and shape, understanding which protests might influence voting outcomes at a specific polling station requires knowing whether voters assigned to that station lived near the protest location. Without this information, it becomes impossible to determine the proximity of voters to protest events and, consequently, to analyze spatial effects of protests.

To address this issue, I reconstructed the boundaries of electoral precinct territories using data from OpenStreetMap and polling station-building associations for 2022, which were scraped from the Central Election Committee website and provided by the Anti-Corruption Foundation. Specifically, I linked over 3.5 million buildings to their respective polling stations and reconstructed the geographical boundaries of 30,000 electoral precincts. These boundaries reflect those used during the 2021 State Duma election across 403 of Russia’s largest settlements.

The second empirical problem for this research is the difficulty in constructing a stable unit of observation, which is essential for methods like Difference-in-Differences. DiD relies on tracking the same observational units over time to isolate the causal effects of an intervention or event. Without stable units, changes in the structure or boundaries of the units themselves can confound the results, making it impossible to attribute observed changes to the intervention rather than to shifts in the units of analysis. Unfortunately, polling station IDs are not stable and can change from one election to another. For example, in cases of partial redistricting, a polling station №100 can be split into №100 and №101, causing all subsequent polling station IDs to shift. Thus, a polling station №101 in 2016 might become №102 in 2021. Polling station locations are also not fixed, making it difficult to identify units based on their coordinates or addresses, as polling stations can relocate between elections. Finally, the boundaries of electoral precincts are also subject to change and can be modified between elections. This instability, combined with changes in precinct boundaries, necessitates a method that accounts for such variations.

To mitigate these issues, I cluster electoral precincts by grouping polling stations that are closely connected — either by being in the same building or through geographical proximity. As long as changes in polling station locations or precinct boundaries occur within the same cluster, the units remain stable for analysis. Thus, electoral precincts were clustered using the following criteria: (1) polling stations located in the same building were assigned to the same cluster, and (2) whenever possible, each cluster included at least two polling stations. This approach resulted in an average cluster size of 2.85 polling stations and provided a robust framework for

mitigating the effects of changes in polling station IDs, locations, and boundaries.

The primary reason for clustering is to ensure that I can accurately determine voting patterns in specific territories, as my analysis relies on the geographic location of electoral precincts. Since the location of precincts can shift slightly between elections, even if the underlying territory remains the same, clustering precincts helps protect against these fluctuations. By grouping precincts into clusters, I also minimize the risk of distortions in my results caused by changes in precinct boundaries, as long as these changes occur within the same cluster. This method allows me to plausibly associate polling stations from 2016 with their counterparts from 2021 and ensures that the geographic integrity of the data is preserved. As a result, I am left with 10,333 polling station clusters in each election period, which serve as the core units of my analysis.

In all specifications, I compare the change in the outcome — total opposition vote share, United Russia vote share, or turnout — in the units within a specified threshold distance from the protest location against units farther away. This difference-in-differences approach relies on several assumptions for causal identification. A key assumption is the Stable Unit Treatment Value Assumption (SUTVA), which posits “no interference” between units. In other words, the treatment status of one unit (being near a protest) does not affect the potential outcomes of other units, particularly those assigned to the control group. Violations could arise if protests generated spatial spillover effects — for example, if information diffused from treated areas into neighbouring control areas. Importantly, to mitigate such potential violations of the “no interference” assumption, I explicitly exclude polling station clusters that border the treated units from the primary control group. However, as reported in Appendix Section , the analysis remains robust when buffer clusters are included in the control group. As a robustness check, I also implemented an alternative specification in which all clusters within 1 km of protest sites, along with their immediate neighbors, were excluded from the control group. This more conservative buffer aims to further minimize potential spillover effects. The results of this analysis, presented in the Appendix Section , are substantively similar to those reported in the main analysis, lending additional confidence to the findings. The Figure 3 illustrates the resulting setup.

The map displays the Danube delta region, showing the distribution of three cluster types: buffer (light gray), control (dark gray), and treatment (500m) (black). The map includes geographical features like the Danube river, settlements (e.g., Petrodubrovka, Petrodubrovskaya, Petrodubrovskiy), and administrative boundaries. A legend at the bottom identifies the cluster types.

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level variations. As reported in Appendix Section , stable and unstable clusters are generally balanced with respect to the covariates used in the analysis.

Empirical strategy

Estimation

Using these data, along with official election results, I employ doubly robust difference-in-differences estimator proposed by (Sant’Anna and Zhao 2020) to compare the change in turnout, total opposition vote share, and the ruling party vote share between localities within a specified threshold and outside of it. I use the DR estimator because no data on the location of polling stations are available before 2014. This makes it impossible to interrogate the parallel trends assumption in a standard way, either by running placebo tests or by visualizing the trends. Instead, the DR estimator relies on the conditional parallel trends assumption, which requires that trends are parallel only after conditioning on observed covariates. This provides a more flexible and robust framework for causal inference.

The key advantage of this estimator lies in being “doubly robust”, meaning it provides unbiased estimates of treatment effects if either the outcome model or the propensity score model is correctly specified (but not necessarily both). The general formula is:

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n \left[\frac{Z_i Y_i}{e(X_i)} - \frac{(1 - Z_i) Y_i}{1 - e(X_i)} + (1 - Z_i) \hat{\mu}_1(X_i) + Z_i \hat{\mu}_0(X_i) \right],$$

where Z_i is the treatment indicator, Y_i is the outcome, $e(X_i)$ is the propensity score, and $\hat{\mu}_1(X_i)$ and $\hat{\mu}_0(X_i)$ are the predicted outcomes for the treated and control groups.

Following Abadie et al. (2023), I compute bootstrapped standard errors clustered on the level of the treatment assignment. In other words, all adjacent electoral precincts within the specified threshold (discussed below) are assigned the same cluster ID, which is then used to calculate the standard errors.

Outcome variables

The analysis focuses on three key outcome variables:

1. *Turnout:*

- Measured as the percentage of registered voters who cast their ballots in a given cluster.

2. *Combined Total Vote Share of Opposition Parties:*

- Calculated as the total vote share of all opposition parties combined, expressed as a percentage of the total number of registered voters in the cluster.

3. *Total Vote Share of United Russia:*

- The vote share of United Russia, relative to the total number of registered voters in the cluster.

Measuring vote share relative to the total number of registered voters in the district, rather than relative to valid votes cast, provides a clearer and more comprehensive measure of a party's overall support. This approach is particularly advantageous in contexts where turnout may vary significantly across clusters or be influenced by the treatment. If vote shares were calculated only relative to valid votes, changes in turnout could confound the results by altering the denominator, making it difficult to disentangle whether observed effects are due to shifts in voter preferences, turnout dynamics, or both.

By using the total number of registered voters as the denominator, this measure captures changes in both the absolute level of support for a party and any turnout-related dynamics. For instance, a decline in United Russia's vote share relative to total registered voters could reflect either a reduction in their core support or an increase in abstention among their voters, both of which are relevant outcomes. Similarly, it ensures that the combined vote share of opposition parties is not artificially inflated or deflated by variations in turnout, providing a more stable and interpretable metric. This method thus allows for isolating the treatment effect on electoral outcomes without conflating it with turnout-related factors or shifts in the behavior of other parties' supporters.

These outcome variables provide insights into the effects on voter participation, support for opposition parties, and support for the ruling party.

Treatment variables

The treatment variables are defined based on proximity to protest sites, using three distinct formulations:

- 500-Meter Radius: Clusters located within 500 meters of a protest site are considered treated.
- 1000-Meter Radius: Clusters within 1000 meters of a protest site are classified as treated.
- Magnitude-Based Threshold: The treatment threshold is dynamically determined by the magnitude of the protest. For protests with up to 400 participants, the threshold increases linearly with the number of protesters. Beyond 400 participants, the threshold distance decays logarithmically. This approach results in a treatment threshold ranging between 1 and 600 meters.

For each distance threshold, clusters within the specified distance are assigned to the treatment group, neighboring clusters are placed in a buffer zone excluded from the analysis, and all remaining clusters are designated as the control group. This approach enables the analysis of the effects of proximity to protests on the defined outcome variables at varying spatial scales, while minimizing the risk of underestimating the effect due to spillovers between treated and neighboring districts.

Covariates

A key methodological concern when comparing places in the vicinity of protests and those farther away is the potential systematic differences between treated and untreated units, which may exhibit divergent trends in their support for the regime and the opposition. Protests, for instance, could be more likely to occur in central, wealthier areas, where patterns of support differ, and disillusionment with the regime tends to rise more rapidly compared to peripheral regions, where support for the regime remains more resilient. To address potential violations of the assumption of unconditional parallel trends, I include a comprehensive set of covariates that capture the economic, demographic, and geographic characteristics of the studied clusters, such as real estate prices, voter density, nightlight intensity, and geographic coordinates

The primary data source consists of over 1.8 million geo-referenced real estate advertisements obtained from Avito, a leading online classified advertisements platform. From this dataset, I extract two key metrics: the median price per square meter at the cluster level and the ratio of this median price to the city-wide average. The price ratio is particularly valuable as it highlights whether a cluster's real estate values exceed or fall below the city average, serving as a proxy for local economic conditions and desirability. In addition to real estate data, I incorporate nightlight intensity metrics derived from BlackMarble satellite imagery, which act as a proxy for economic activity and urban development, capturing fine-grained variations in urbanization and vibrancy across clusters.

To further account for demographic and spatial factors, I include voter density, calculated as the number of registered voters per unit area, to measure population concentration. Geographic coordinates, specifically latitude and longitude, are also incorporated to control for spatial patterns and regional variations that might influence outcomes. Together, these covariates allow for a detailed analysis of cluster-level differences, helping to mitigate potential violations of the parallel trends assumption and ensuring the robustness of the results.

Results

Effect of protests on turnout and support for political parties

Table 1 reports the main results for the effect on the turnout. Here, as well as in the following tables, coefficient estimate represent the difference in the change in the outcome of interest for treated and untreated units. In all specifications, the combined effects of the protests of both types are statistically significant, indicating a turnout decrease of 1.3 to 1.65 percentage points in the affected localities. When estimating the effects of the two types of protests separately, a substantial difference emerges: across various specifications of the treatment, Anti-Corruption protests lead to a turnout decrease that is, on average, twice as large as that associated with Pension Reform protests.

Table 1: Estimated difference between treated and untreated units in the turnout change (2021 vs. 2016 elections)

	Treatment					
	500m		1000m		Relative	
All Protests						
Estimate	-1.289**	-1.122*	-1.655***	-1.645***	-1.505***	-1.44**
SE	(0.526)	(0.592)	(0.409)	(0.47)	(0.548)	(0.635)
Obs.	8358	8264	8564	8472	8382	8280
Pension Reform						
Estimate	-0.84*	-0.575	-1.417***	-1.342***	-0.917	-0.724
SE	(0.506)	(0.52)	(0.42)	(0.488)	(0.593)	(0.613)
Obs.	8176	8082	8382	8290	8200	8098
Anti-Corruption						
Estimate	-1.675**	-1.76**	-1.885***	-1.989***	-2.569***	-3.01***
SE	(0.667)	(0.697)	(0.471)	(0.523)	(0.943)	(0.991)
Obs.	7796	7702	8002	7910	7820	7718
Covariates	No	Yes	No	Yes	No	Yes

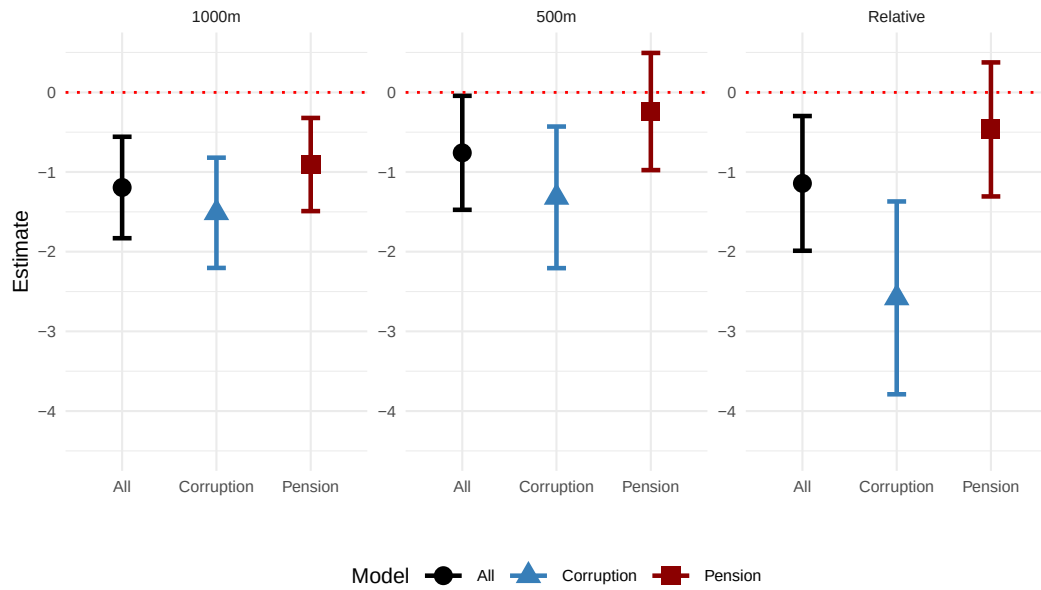
Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

However, as Figure 4 show, the 83% confidence intervals for the effects intersect in all but one case, providing weak evidence in favor of statistically significant differences.



83% confidence intervals are displayed; non-overlapping intervals indicate a statistically significant difference at the 5% level.

Figure 4: Estimated difference between treated and untreated units in the turnout change (2021 vs. 2016 elections)

The next question is whether the turnout decline impacted pro-regime voters, pro-opposition voters, or both. As Table 2 demonstrates, there is a consistent negative effect on United Russia’s total vote share across all specifications when considering both types of protests combined. Notably, in all specifications the effect on United Russia’s vote share closely mirrors the turnout decline. As before, Anti-Corruption protests have a more pronounced impact on voter behavior compared to Pension Reform protests.

Table 2: Estimated difference between treated and untreated units in the change of United Russia vote share (2021 vs. 2016 elections)

	Treatment					
	500m		1000m		Relative	
All Protests						
	-1.304***	-1.133**	-1.41***	-1.342***	-1.402***	-1.347**
Estimate						
SE	(0.45)	(0.454)	(0.367)	(0.395)	(0.526)	(0.525)
Obs.	8358	8264	8564	8472	8382	8280
Pension Reform						
	-0.899*	-0.663	-1.18***	-1.067***	-0.839	-0.698
Estimate						
SE	(0.483)	(0.492)	(0.355)	(0.407)	(0.537)	(0.528)
Obs.	8176	8082	8382	8290	8200	8098
Anti-Corruption						
	-1.79***	-1.748***	-1.635***	-1.632***	-2.466***	-2.679***
Estimate						
SE	(0.546)	(0.593)	(0.397)	(0.47)	(0.766)	(0.851)
Obs.	7796	7702	8002	7910	7820	7718
Covariates	No	Yes	No	Yes	No	Yes

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

As for the effect on the total opposition vote share, as Table 3 shows, we essentially observe no statistically significant differences between treated and untreated units. This goes in line with previous research, where opposition protests were shown to disengage pro-regime voters while neither making them more likely to switch to opposition, nor leading to increased mobilization on the sided of pro-opposition voters.

Table 3: Estimated difference between treated and untreated units in the change of total opposition vote share (2021 vs. 2016 elections)

		Treatment					
		500m		1000m		Relative	
All Protests							
		0.016	0.012	-0.244	-0.302	-0.103	-0.093
Estimate							
SE	(0.195)	(0.222)	(0.169)	(0.187)	(0.23)	(0.259)	
Obs.	8358	8264	8564	8472	8382	8280	
Pension Reform							
		0.059	0.088	-0.237	-0.275	-0.078	-0.026
Estimate							
SE	(0.199)	(0.234)	(0.178)	(0.198)	(0.244)	(0.266)	
Obs.	8176	8082	8382	8290	8200	8098	
Anti-Corruption							
		0.115	-0.012	-0.25	-0.357	-0.103	-0.331
Estimate							
SE	(0.257)	(0.275)	(0.177)	(0.222)	(0.305)	(0.348)	
Obs.	7796	7702	8002	7910	7820	7718	
Covariates	No	Yes	No	Yes	No	Yes	

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Heterogeneity

Electoral vulnerability

While exposure to protests appears to influence voting behavior in Russia, particularly by disengaging regime supporters, the questions remains: why? One explanation could be that protests fail to mobilize people because election outcomes are largely predetermined, and disillusioned voters may see little point in shifting their support to the opposition. In that case, we may expect observing a more pronounced effect on the opposition vote share in places where people expect the ruling party to perform poorly. At the same time, the opposite may be true for the effect on turnout: in places where the ruling party is traditionally perceived to be dominant, protests may have a larger effect on turnout, but still fail to increase opposition vote share. In the latter case, the effect on turnout can simply be explained mathematically: if protests are informative primary for regime supporters, the larger the pool of such supporters is, the more

of them will disengage. In addition to that, protests can have a larger information revelation potential in places where most people believe the party to be extremely popular.

To investigate the role of the regime party's electoral position, I split the sample based on the regional UR share in the pre-treated period (2016). Specifically, I treat regions in which the UR vote share in the 2016 election was higher than the county average of 48% as UR strongholds (approximately one third of the sample), and compare the effect in such regions to the effect in regions where UR performed poorly. The complete list of regions in both groups is reported in the Appendix Section . My first expectation is that the effect on turnout will be larger in the UR strongholds than everywhere else.

The results reported in Table 4 indeed support this intuition. In fact, the overall negative effect on turnout appears to be driven almost exclusively by the observations from regions with above-average United Russia vote share: while the effect in the regions with low UR vote share hovers around 0.5 and does not reach statistical significance in most specification, in the UR strongholds, localities in the vicinity of protests register 4-7% lower turnout compared to the control group. As shown in the Figure 5, this difference is statistically significant.

Table 4: Estimated difference between treated and untreated units in the turnout change in regions with above- and below-average national UR vote share

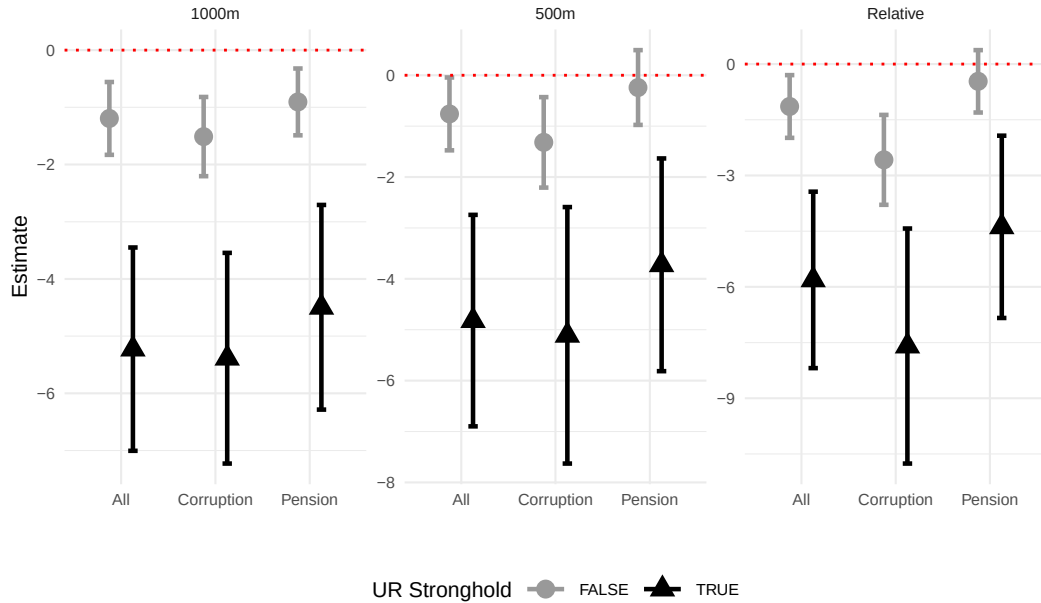
UR Stronghold	Treatment					
	500m		1000m		Relative	
	Yes	No	Yes	No	Yes	No
All Protests	-4.821***	-0.509	-5.228***	-1.039***	-5.813***	-0.288
Estimate						
SE	(1.516)	(0.393)	(1.297)	(0.326)	(1.734)	(0.404)
Obs.	2446	5818	2470	6002	2456	5824
Pension Reform	-3.724**	-0.532	-4.496***	-1.076***	-4.384**	-0.3
Estimate						
SE	(1.525)	(0.392)	(1.306)	(0.322)	(1.791)	(0.424)
Obs.	2394	5688	2418	5872	2404	5694
Anti-Corruption	-5.109***	-0.689	-5.387***	-1.216***	-7.594***	-0.516
Estimate						
SE	(1.84)	(0.431)	(1.344)	(0.34)	(2.31)	(0.538)
Obs.	2278	5424	2302	5608	2288	5430
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.



.4% confidence intervals are displayed; non-overlapping intervals indicate a statistically significant difference between the estimates at the 5% level.

Figure 5: Estimated difference between treated and untreated units in the turnout change in regions with above- and below-average national UR vote share

And yet, the second part of the proposed mechanism does not work: regardless of the baseline support for the United Russia, protests do not seem to increase opposition vote share, as reported in Table 5. Just as before, regardless of the formulation of the treatment variable or the type of the protest, we observe no effect on the total opposition vote share.

Overall, these results suggest two main implications. First, protests can have an amplified effect in situations when their occurrence is least expected. In our case, even accounting for the fact that the United Russia's electoral strongholds had a larger pool of voters that could potentially be affected by the protests, the effect on turnout and, therefore, United Russia's vote share, is around 10 times larger than in regions where the ruling party's dominance was contested more successfully. Second, the failure of protests to mobilize disengaged United Russia supporters or attract additional pro-opposition voters, even in areas where United Russia appears to be more vulnerable, suggests that the conditions required for such mobilization are likely much more demanding than those needed for individuals to simply defect from the ruling party. It could be, for example, that exposure to protests, while making people more skeptical of the ruling party, do not affect their beliefs about the opposition. Ultimately, the protests in question were organized by various groups that rarely collaborated with one another. For instance, the Pension protests did not result in any kind of political alliance between A Just Russia and the Communist Party, the two largest left-wing parties, nor did it elevate any new force to a prominent position on the political stage. All in all, it can be that exposing the current government's lacks of support or incompetence is not enough. Equally important is presenting a credible and viable alternative.

Table 5: Estimated opposition vote share change between treated and untreated units in the turnout change in regions with above- and below-average national UR vote share

UR Stronghold	Treatment					
	500m		1000m		Relative	
	Yes	No	Yes	No	Yes	No
All Protests						
	-0.406	-0.016	-0.687	-0.247	-0.513	-0.023
Estimate						
SE	(0.492)	(0.223)	(0.433)	(0.211)	(0.576)	(0.269)
Obs.	2446	5818	2470	6002	2456	5824
Pension Reform						
	-0.297	-0.051	-0.681	-0.276	-0.448	-0.089
Estimate						
SE	(0.502)	(0.234)	(0.471)	(0.201)	(0.63)	(0.254)
Obs.	2394	5688	2418	5872	2404	5694
Anti-Corruption						
	0.064	-0.1	-0.485	-0.315	0.199	-0.273
Estimate						
SE	(0.587)	(0.286)	(0.499)	(0.217)	(0.749)	(0.353)
Obs.	2278	5424	2302	5608	2288	5430
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Protest magnitude

Additionally, the relatively moderate size of most protest events under examination could partly account for the observed effects. While there is little reason to believe that the size of the protests alone would alter the key pattern – namely, their limited mobilization potential – it may still play a role in understanding the mechanisms through which protests influence public behavior. In general, the literature suggests that large-scale demonstrations should have a larger information revelation potential. At the same time, poorly attended protests may provide little information about “the state of the world”, or might even signal the weakness of dissenters, thereby reinforcing the stability of the regime. Whether this is the case is ultimately an empirical question, which this subsection aims to address.

Specifically, I estimate the effect separately for protests with participant numbers above and below the median threshold of 500. In both cases, I compare the change in the outcome of

interest between elections for the restricted treatment group and the full control group. The results for turnout are presented in Table 6. As expected, protests with a larger number of participants demonstrate a significantly greater effect compared to those below the median.

Table 6: Estimated change in turnout between treated and untreated units in regions with above- and below-median protest magnitude

Above Median N.	Treatment					
	500m		1000m		Relative	
	Yes	No	Yes	No	Yes	No
All Protests						
Estimate	-1.74**	-0.119	-1.987***	-1.152**	-3.023***	-0.004
SE	(0.748)	(0.583)	(0.544)	(0.495)	(1.019)	(0.687)
Obs.	7710	7696	7918	7902	7726	7710
Pension Reform						
Estimate	-0.921	-0.073	-1.605***	-1.166**	-1.687*	0.099
SE	(0.675)	(0.655)	(0.535)	(0.506)	(0.939)	(0.826)
Obs.	7626	7612	7834	7818	7642	7626
Anti-Corruption						
Estimate	-1.951**	-0.4	-2.06***	-1.42***	-4.059***	-0.989
SE	(0.851)	(0.775)	(0.555)	(0.524)	(1.415)	(1.17)
Obs.	7550	7344	7758	7550	7566	7358
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

As for the effect for the total opposition vote share, once again, there is no evidence that larger protests lead to increased mobilization. If anything, the effect is negative in some cases, although it is not consistent across specification.

Table 7: Estimated change in opposition vote share between treated and untreated units in regions with above- and below-median protest magnitude

Above Median N.	Treatment					
	500m		1000m		Relative	
	Yes	No	Yes	No	Yes	No
All Protests						
	-0.036	0.351	-0.375*	-0.178	-0.394	0.331
Estimate						
SE	(0.263)	(0.24)	(0.223)	(0.204)	(0.362)	(0.312)
Obs.	7710	7696	7918	7902	7726	7710
Pension Reform						
	0.12	0.323	-0.317	-0.212	-0.192	0.252
Estimate						
SE	(0.255)	(0.259)	(0.219)	(0.199)	(0.378)	(0.367)
Obs.	7626	7612	7834	7818	7642	7626
Anti-Corruption						
	-0.035	0.382	-0.386*	-0.261	-0.577	0.255
Estimate						
SE	(0.28)	(0.322)	(0.228)	(0.234)	(0.463)	(0.554)
Obs.	7550	7344	7758	7550	7566	7358
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

However, it is important to acknowledge potential confounding factors when interpreting the relationship between protest size and public attitudes. For example, Whether a protest is officially permitted or banned can significantly influences both its potential turnout and the likelihood of state repression. Protests denied authorization may simultaneously be smaller and face harsher crackdowns. This complicates the empirical analysis, as simple correlations between size or repression and attitudes might be partially driven by this underlying authorization dynamic. Therefore, while this and the next sections explore the patterns related to protest size and repression based on available data, a definitive causal assessment of size and repression's independent effects would require a more advanced approach.

Repression

Finally, I compare the effect separately for protests that were met with arrests and for those that concluded peacefully. As discussed before, repression could expose the regime's malign

nature, potentially increasing defections from it, or signal the potential repercussions of dissent for the opposition. Therefore, the aim of this comparison is to determine whether repression deters opposition by raising the costs of dissent or, alternatively, amplifies dissatisfaction with the regime by highlighting its oppressive nature.

Due to the limited variation in this variable, the results should be interpreted as suggestive and treated with caution when drawing broader conclusions. Since only a small fraction of pension protests resulted in arrests (38 out of 866), my analysis focuses on the 2017 anti-corruption protests, where repression was both more prevalent and balanced. In this case, 150 out of 242 protests involved the detention of protesters, while 92 did not. Table 8 report the results. Although, as previously noted, the results are only indicative, there appears to be little difference between protests that faced a crackdown and those that concluded peacefully.

Table 8: Estimated change in opposition vote share between treated and untreated units in regions with above- and below-median protest magnitude

Above Median N.	Treatment					
	500m		1000m		Relative	
	Yes	No	Yes	No	Yes	No
Turnout						
Estimate	-1.174	-0.909	-1.7***	-1.605***	-2.925**	-2.373**
SE	(0.768)	(0.697)	(0.547)	(0.556)	(1.382)	(1.041)
Obs.	7422	7412	7628	7618	7436	7426
ER share						
Estimate	-1.379**	-1.087*	-1.393***	-1.282***	-1.403	-0.728
SE	(0.634)	(0.622)	(0.45)	(0.438)	(0.985)	(1.011)
Obs.	7422	7412	7628	7618	7336	7348
Opposition share						
Estimate	0.205	0.178	-0.307	-0.323	-0.18	-0.337
SE	(0.31)	(0.273)	(0.22)	(0.231)	(0.511)	(0.474)
Obs.	7422	7412	7628	7618	7436	7426
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Discussion

This study examines the impact of localized protest activity on electoral outcomes in Russia between the 2016 and 2021 parliamentary elections. The analysis reveals that exposure to protests leads to a significant decrease in voter turnout and the ruling party's vote share, particularly in areas near protest sites. Importantly, while support for United Russia declines, there is no corresponding increase in the opposition parties' vote share. This suggests that protests in authoritarian regimes primarily demobilize regime supporters without mobilizing opposition voters.

These findings align with theoretical frameworks that emphasize the role of protests as information-revealing events in authoritarian contexts. According to Kuran's theory of preference falsification, individuals may conceal their true preferences out of fear but can be influenced to reveal them upon perceiving growing dissent (Kuran 1991). Similarly, protests act as mechanisms for information aggregation, affecting the beliefs and behaviors of both participants and observers (Lohmann 1994b). In controlled information environments, even local signs of dissent can signal growing dissatisfaction with the regime, prompting supporters to abstain from voting.

The results also build on the work of Tertytchnaya, who found that anti-government protests demobilize regime supporters more than they mobilize opposition voters (Tertytchnaya 2020). While her study focused on immediate effects, this research demonstrates that such effects can persist over time and across different protest agendas, including both Anti-Corruption protests organized by Alexei Navalny and Pension Reform protests led by the Communist Party. This suggests that the mechanism influencing bystanders' attitudes may not be tied to revealing new information but rather to signaling declining regime support (Pop-Eleches, Robertson, and Rosenfeld 2022). However, the evidence presented here is only suggestive, and further research is required to uncover the precise mechanisms through which protests influence bystanders' attitudes and behavior.

Furthermore, the heterogeneity analysis indicates that the negative effect on turnout is significantly larger in regions where United Russia has traditionally been strong. This supports the notion that protests have an amplified impact where they are least expected, consistent with Kricheli et al.'s argument regarding the information-revealing potential of protests in more re-

pressive autocracies (Kricheli, Livne, and Magaloni 2011). Larger protests also have a more substantial negative impact on turnout, aligning with Lohmann's idea that the magnitude of protests enhances their informational value (Lohmann 1994b). However, even significant protests fail to boost the opposition's vote share, indicating that exposure to dissent is insufficient to mobilize voters without a credible alternative. Regarding repression, the analysis finds little difference between protests that faced crackdowns and those that concluded peacefully.

Overall, this study contributes to understanding political behavior in authoritarian regimes by showing that protests can erode support for the ruling party by demobilizing its base, especially in areas where the regime appears strong. However, without a compelling opposition, this disengagement does not necessarily translate into increased support for opposition parties.

More broadly, the study sheds light on the logic behind authoritarian regimes' intolerance to public displays of dissent. Even relatively small protests can undermine the regime's perceived dominance by signaling cracks in its support base, discouraging participation among its nominal supporters. This sensitivity to public dissent explains why authoritarian governments often respond to protests with repression and censorship, seeking to prevent the spread of demobilization effects. By tightly controlling the public sphere, they aim to sustain the illusion of widespread approval and suppress signals that might encourage further disengagement. Ultimately, the findings underscore the precarious nature of manufactured legitimacy in authoritarian systems, where the appearance of stability can be eroded once dissent becomes visible.

Chapter III: War, Propaganda, and Russian Fatalities in Ukraine (with Vladimir Zabolotskiy)

How do military losses impact political behavior and attitudes in autocracies? This paper examines the effects of reports about local soldiers killed in action (KIA) during the ongoing war in Ukraine on online political behavior reflecting patriotic sentiment or regime support. Using more than 20 million posts from 36,000 geo-referenced social media groups, we employ topic-specific engagement metrics as proxies for approval of authorities and patriotism. Leveraging variation in the timing of KIA reports across Russian municipalities and a staggered treatment rollout design, we show that war fatalities lead to a sustained decline in social media engagement with content referencing authorities, with the magnitude of this effect increasing over the course of the war. At the same time, interaction with patriotic and military topics temporarily increases, driven largely by posts referencing the war in Ukraine and fallen soldiers. These effects are stronger in municipalities where local online group administrators actively publish soldiers' obituaries. Analysis of emotional tone further reveals that engagement with propagandistic content — positive in tone and politically relevant — declines after KIA reports, regardless of topical focus. This suggests that direct exposure to war fatalities erodes trust in state-promoted narratives. These findings highlight the critical role of personal exposure to war casualties in shaping public discourse, with implications for understanding political attitudes in conflict settings.

Introduction

Authoritarian leaders tend to be more militant in their international relations than their democratic counterparts (Benoit 1996; Rousseau et al. 1996). With fewer constraints from public opinion and domestic political pressures, they can engage in longer wars (Bennett and Stam III 1996) and show less regard for their costs (Siverson 1995), which gives them a strategic advantage over democracies in prolonged conflicts (Bennett and Stam III 1998). Moreover, because the ‘rally-round-the-flag’ effect tends to be stronger and more pronounced in autocracies at war (Kizilova and Norris 2024), and international pressure often reinforces domestic support for authoritarian leaders (Hellmeier 2021), they may be tempted to initiate military interventions not for the sake of resolving international disputes, but to consolidate societal support and strengthen their position within the regime.

Wars, however, are costly — and their greatest cost is human lives. Autocrats may initiate wars, but ordinary people pay the ultimate price. At the same time, we know very little about the impact of war fatalities on political opinions and the attitudes of ordinary citizens in autocracies. Studies focusing on democracies suggest a significant relationship between war fatalities and public sentiments, but the size and the direction of the effect largely depends on the duration of the conflict (Kuijpers 2019), media coverage (Baker and Oneal 2001; Gartner 2008b), and framing (Berinsky 2007; Boettcher and Cobb 2009). To what extent these results can be generalized to drastically different political and informational environment typical for authoritarian regimes is unclear.

In autocracies, repression against anti-war dissenters and state control over information shape how citizens perceive military losses. When propaganda stirs up nationalist sentiments and frames interventions as defensive actions against external threats, war fatalities can potentially fuel support for continued aggression rather than calls for peace. In such cases, citizens may adopt a ‘don’t let them die in vain’ mentality, reinforcing militaristic attitudes. Consistent with this, Juan et al. (2024) finds that the revanchist Nazi Party received more votes in German localities that experienced higher soldier mortality rates in the First World War. On the other hand, prolonged wars with high human cost may eventually erode the support for the regime, particularly when state narratives lose credibility. Thus, while the rally-round-the-flag effect can bolster authoritarian resilience in the short term, over time, the realities of war may undermine domestic stability and weaken the very foundations the regime seeks to strengthen. It can also

be that war fatalities affect citizens attitudes in both ways, eroding the support for the regime and reinforcing nationalist sentiment at the same time. In this article, we investigate this very question.

Specifically, we explore the dynamic effects of Russian war fatalities in Ukraine on political attitudes in Russia in 2022-2024. In particular, we examine how reports of soldiers killed in action (KIA) influence user engagement with war, patriotic, and regime-related content on social media in their hometowns, using these metrics as proxies for war support, patriotism, and attitudes toward the authorities. To capture social media engagement, we collect over 28 million posts from 36714 geo-referenced online school groups, with users accounting for 12% of Russian adult population, and classify these postings using keyword analysis and a zero-shot large language model (LLM), as well as sentiment analysis. For military losses, we use municipality-level geocoded data from Russian sources (Mediazona and BBC 2023). To measure a more direct exposure to information on fatalities, we also use obituaries posted by the administrators of the school groups in our sample as treatment.

We find a robust and negative effect on engagement with pro-regime content after the first KIA report in a municipality, whereas engagement with patriotic content and content referring to the war in Ukraine increases. Theoretically, the results indicate that war fatalities can erode the legitimacy of the current regime and amplify nationalistic sentiment at the same time.

Additional heterogeneity analysis provides further insights into the mechanism behind these effects. First, we find that the effect is larger in municipalities where obituaries were shared on local school group pages, highlighting the importance of direct exposure to information in shaping attitudes toward the war and the political actors responsible. This finding also underscores the limited spillover of information in autocracies, where state-controlled media and constrained social networks restrict broader dissemination through word-of-mouth or conventional media. Second, by examining the topical content of posts *and* their emotional tone, we show that, following KIA reports, users disproportionately *disengage* with propaganda, i.e., content that is both politically relevant and positive in tone. This is true for both patriotic and pro-government content, suggesting that exposure to information on military losses can undermine the effectiveness of state propaganda.

Importantly, our results are robust to several potential confounding factors: (i) a general decline in social media activity, as the analysis adjusts for engagement with neutral content, such as

posts focused solely on education; (ii) the nationwide shifts in attitudes over the course of the invasion, as month fixed effects control for temporal differences; (iii) cross-municipality variation, as municipality-level fixed effects account for time-invariant differences across municipalities. Finally, we show that (iv) the results are not driven by the users shifting their social media consumption towards more positive entertaining content.

Our study makes three main contributions. First, it advances understanding of the separate effects of war casualties on public attitudes toward the conflict, the governing authorities, and nation. We find that war fatalities degrade support for the autocrat — not unlike in settings where leaders are elected democratically (e.g., Althaus, Bramlett, and Gimpel 2012; Getmansky and Weiss 2023) — but also promote an increase in patriotism (Acemoglu et al. 2022; Carozzi, Pinchbeck, and Repetto 2023; Juan, Haass, Koos, Riaz, and Tichelbaecker 2024). These findings suggest that while war fatalities may decrease support for the ruling elite or the political actors leading the war, they do not necessarily foster anti-war sentiment. Instead, they can amplify patriotic sentiment and, potentially, drive militarism.

Second, our paper contributes to the literature on the effectiveness and limits of propaganda in authoritarian regimes (Alyukov 2023; Carter and Carter 2023; Geddes and Zaller 1989; Gehlbach and Sonin 2014; Yang and Zhu 2024). By looking at how the KIA reports affect users’ reaction to a more explicitly propagandistic content, our study contributes to the existing research on public responses to propaganda (Horz 2021; Little and Nasser 2018), particularly during periods of crisis (Rozenas and Stukal 2019a; Sirotkina and Zavadskaya 2020) and international instability (Weiss and Dafoe 2019). We show that distressing information, even when framed within a state-promoted narrative, can weaken public support for the regime and its leaders while potentially reducing the effectiveness of subsequent propaganda. Specifically, exposure to reports of soldiers killed in the war in Ukraine prompts more negative reactions to subsequent posts featuring political content with positive tone. In other words, we show that sugarcoating political developments becomes more difficult after people encounter the evidence of the detrimental cost of government actions.

Finally, we contribute to the literature on measuring political attitudes by introducing an innovative approach to track public sentiment with high frequency and granularity through online political behavior. We demonstrate that user engagement metrics, such as likes, reposts, and comments, provide advantages that complement traditional measures of political attitudes, espe-

cially in autocratic settings. In such contexts, survey data are often difficult to collect (Libman 2023; Rosenfeld 2023) and may suffer from fear-induced bias (Reisinger, Zaliznaya, and Woo 2023; Yudin 2022) or inflation bias (Frye, Gehlbach, Marquardt, and Reuter 2023). Other activities indicative of anti-war and anti-regime sentiment, such as protests (as in Duvanova, Nikolsko-Rzhevskyy, and Zadorozhna (2023)) or information dissemination, are shaped by repression and persecution, which discourage broader participation and restrict involvement to a narrow subset of the politically active population. This is particularly relevant in the case of Russia, where a law criminalizing “public actions aimed at discrediting” the Russian Armed Forces was enacted shortly after the start of the invasion of Ukraine, resulting in the arrest of more than 7,000 individuals for actions allegedly “discrediting the military” (UN 2023). Finally, electoral outcomes are both rare and not consistently reliable due to fraud and voter demobilization (Enikolopov et al. 2013; McAllister and White 2017; The Moscow Times 2022).

The rest of the paper is organized as follows. Sections 2 and 3 provide a brief overview of the literature and the empirical background. Section 4 describes the data employed in the study. Section 5 outlines the empirical strategy, and the results are presented in Section 6. Section 7 explores the potential mechanisms behind the effect, and Section 8 concludes.

War and political attitudes

War can have profound effects on society and its political landscape. It can help political actors gain support from the public (Hetherington and Nelson 2003; Mesquita and Siverson 1995), foster social cohesion (Bauer et al. 2016), and promote national identity (Acemoglu, De Feo, De Luca, and Russo 2022; Carozzi, Pinchbeck, and Repetto 2023). In this paper, we focus on how one particular aspect of war — its human cost — impacts attitudes towards the authorities, patriotism, and effectiveness of propaganda.

Military losses have been shown to impact public support for the political figures leading the war effort. In the months following the start of a military conflict, war-time leaders frequently experience increased support from the public (Hetherington and Nelson 2003; e.g., Mueller 1985), especially when the public is invested in ‘winning the war’ (Koch 2011). However, as the conflict unfolds and its costs start to accumulate, the initial ‘rally around the flag’ dies out, oftentimes leaving these leaders less popular than before the intervention (Karol and Miguel 2007; Mueller

1985; Voeten and Brewer 2006) and giving way to opposing political forces (Getmansky and Weiss 2023; Kuijpers 2019).

In addition, war fatalities can shape public opinion on whether the intervention itself is worth continuing. On one hand, as the casualties pile up, the public might decrease its support for the intervention, considering it too costly to continue (Gartner and Segura 1998; Mueller 1985; Sullivan 2008). Increasing recent casualties, compared to the cumulative total, and rising casualty trends has been shown to have a more pronounced negative effect on the support for war (Gartner 2008b; Gartner and Segura 1998). Additionally, previous studies have found the effect to be larger for local casualties than for the national body count (Althaus, Bramlett, and Gimpel 2012; Gartner 2008a; Gartner, Segura, and Wilkening 1997; Hayes and Myers 2009). On the other hand, if the goals of the intervention have not been achieved, the military losses already incurred may be perceived as in vain, or sunk cost, lending more support to continuing the intervention (Boettcher and Cobb 2009; Veilleux-Lepage 2013).

These countervailing effects are further complicated by the narratives surrounding the intervention. The support for the intervention is conditional on whether there is a consensus regarding the war, both in the elites and among the public (Baker and Oneal 2001; Berinsky 2007). If there is no consensus in the elites, the public might become more polarized, aligning the attitudes towards the intervention along the partisan lines. Essentially, the degree to which the public reacts to the new information on war casualties depends on individual casualty tolerance and the framing of the reports (e.g., reporting only casualties among your own forces or comparing them to enemy losses) (Boettcher and Cobb 2006), the perceived likelihood of the success of the intervention (Gelpi 2010; Gelpi, Feaver, and Reifler 2006) and whether the war is considered just. Ultimately, this can contribute to an increase in polarization, as those who consider the war just are more likely to increase their support for the war as the new information on casualties is revealed (Boettcher and Cobb 2009).

An important caveat regarding the research mentioned above is that it focuses on how the public responds to military interventions and casualties in democratic contexts, whereas authoritarian regimes operate in a substantively different informational and institutional environment. With no free press and silenced opposition, contesting framings and narratives of the war are not allowed to propagate. Alternative information regarding the course of the war — including the number of casualties, the success of military operations, and humanitarian costs costs of the

war — is significantly more difficult to access than in democracies. It is therefore not clear how, under these circumstances, people would react to reports of military casualties.

On the one hand, this lack of research can be attributed to the limited role public support has played in the survival of the autocrats of the past. However, it became more relevant in the recent years with the rise of the *informational autocrats*, who rely on the support of the public, mimicking democracy and employing a variety of informational techniques to ensure their stay in the office, without resorting to mass-scale repression (Guriev and Treisman 2019b).

On the other hand, studying public opinion in autocracies can be a challenging enterprise even in periods of apparent stability, and even more so in times of crises and international conflicts. The problem is that in authoritarian regimes, the dynamics of political support are often shaped by perceptions of widespread endorsement (Buckley, Marquardt, Reuter, and Tertychnaya 2024). When individuals believe that a leader or regime enjoys significant public backing, this perception can reinforce support, creating a feedback loop that amplifies the appearance of mass approval. Crises can further intensify this dynamic, as fear and uncertainty may lead individuals to publicly profess greater support for the regime, even as their private trust erodes (Jiang and Yang 2016). This renders traditional surveys significantly less capable of registering shifts in public opinion masked by preference falsification, not to mention practical obstacles to carrying research on such a sensitive topic in autocracy during wartime.

At the same time, understanding it becomes crucial as the world faces increasingly belligerent autocrats. The research has shown that they are more resilient in the aftermath of war, even after defeat (Mesquita and Siverson 1995), and are prone to waging longer (Bennett and Stam III 1996) and more costly (Siverson 1995) wars. To a large extent, these empirical regularities can be explained by the fact that authoritarian leaders face fewer constraints from public opinion and domestic political pressures, which gives them a strategic advantage over democracies in prolonged conflicts (Bennett and Stam III 1998). In addition to that, as the ‘rally-round-the-flag’ effect may be more pronounced and last longer in autocracies during wartime (Kizilova and Norris 2024), and international pressure often reinforces domestic support for authoritarian leaders (Hellmeier 2021), they may be tempted to initiate military interventions not only for the sake of resolving international disputes, but to consolidate societal support and strengthen their position within the regime. Against this background, it is important to understand how ordinary people in autocracies react to the inevitable human cost of war: whether they become

disillusioned with the leaders, or, perhaps paradoxically, rally around them even more, and how it affects their views on the conflict and attitudes more generally.

The literature on war casualties and support for incumbents in democracies demonstrates that these effects can be highly context-specific. In the context of the current study, we could reasonably expect any negative effect of the reports of soldiers killed in action on the support for the military actions and the political leadership to be mitigated by state propaganda. While war fatalities can signal incompetence of the leaders and the high costs of war, in the extreme case, government propaganda can be effective enough to weaponize the casualties to promote social cohesion and out-group intolerance, offsetting potential negative effects on the support for the regime. In Russia, where anti-war dissenters are harshly persecuted, and the state narrative of the war dominates everywhere — not only on television that has been under the regime control since 2000s, but also on the streets through billboards advertising military recruitment, and to some extent even in the Russian segment of the internet, where access to opposition websites is blocked, while state-affiliated platforms flood news feeds with propaganda — such an outcome would not be impossible. On the other hand, if propaganda effort is not enough, war fatalities can lead to the erosion of support both for the war and for the regime.

At the same time, inter-state conflicts forge national identities (Hutchinson, Leoussi, and Grosby 2007; see also Kulyk 2024 on the effect of the war on Ukrainian national identity), and, as argued by Juan, Haass, Koos, Riaz, and Tichelbaecker (2024), war fatalities can foster ingroup favoritism and outgroup distrust, leading to stronger preferences for nationalist parties. Similarly, we also expect information on soldiers killed in action to amplify nationalistic sentiment.

Finally, information on casualties might gain an additional informational dimension in autocracy at war. More specifically, casualty reports can carry information not only about the cost of war and competence of the government, but also about the credibility of propaganda that attempts to portray the invasion as a competent and masterfully executed operation. In such a context, revealing the information on war fatalities might have an extra effect on *how* effective the subsequent propaganda is, in addition to those it has on war and regime support. From this perspective, the research on the effectiveness and framing of propaganda in authoritarian regimes becomes particularly relevant to our research (Horz 2021; Little and Nasser 2018). For example, Horz (2021) present a behavioral model in which the extremeness of propaganda is inversely correlated with the likelihood that the citizen believes it (see also Little and Nasser

2018). Similar to how the Russian regime manipulates economic news to divert blame for policy failures and claim credit for positive developments (Rozenas and Stukal 2019a), the regime in Horz’s model decides whether to frame a certain event — which might convey information unfavorable to the regime — in a way that casts it in a more or less positive light. The more extreme the propaganda spin is, the more likely citizens are to become skeptical of the message. The analysis presented in the current paper supports this formal intuition.

In this paper, we explore the link between war fatalities and war and regime support in autocracy. Up to this point, the literature has focused primarily on this relation in the context of military interventions performed by democracies. This relation, however, is likely to differ in war-waging autocracies, as the autocrats’ control over the media and their repressive potential allow them to shape the narrative to a greater extent along with concealing true information on the cost of the intervention. Given the growing number of military interventions and interstate conflicts involving autocracies in recent years, it is all the more important to understand what shapes public support for such interventions in autocracies.

Arguably, capturing public sentiment in autocracies is challenging, even more so for sensitive topics like war. Survey measures of regime support are subject to biases and limitations, limiting the predictive power their findings can provide. Other indicators of regime and war support are either rare and falsifiable, like elections, or not representative, like protests. In this study, we introduce a novel way to capture public sentiment towards war and regime using social media data that we believe is likely to overcome at least some limitations of the other measures.

Background

Since the commencement of the full-scale invasion on February 24, 2022, the Russian government has orchestrated an extensive campaign with the dual purpose of disseminating war propaganda and quashing anti-war dissent. Employing an array of propaganda tools, ranging from prime-time political talk shows to state-controlled telegram channels and troll farms, the government’s intention has been to vilify the perceived enemy and legitimize the invasion.

This multifaceted approach combines disinformation tactics with emotional manipulation to sway public opinion, both within the nation and on the international stage. False narratives have been strategically disseminated, portraying the enemy as a direct threat to national security

while emphasizing alleged atrocities. This deliberate dehumanization of the enemy serves a dual function: it galvanizes support for the invasion while suppressing any critical voices that challenge the government's actions or motivations. The extensive reliance on war propaganda reflects the government's determination to maintain control over the narrative and present a unified front, even as international criticism and condemnation mount.

The Russian authorities have dedicated substantial efforts to suppress and persecute anti-war protesters. As of October 19, 2023, the impact of these actions has been alarming, with over 19,000 individuals incarcerated for participating in anti-war protests. An additional 748 individuals have been designated as suspects in anti-war criminal proceedings, and 8,122 administrative cases have been initiated against those accused of 'discrediting the Russian military forces' (OVD-Info 2023). Most notably, the majority of administrative actions have targeted individuals who have documented and disseminated information about Russian war crimes in Ukraine.

The broader media landscape in Russia concerning the war now paints a bleak picture, where easily accessible non-governmental sources of information are virtually non-existent, allowing the Russian government to tightly control the narrative surrounding the conflict. Many independent media outlets ceased to exist, while independent journalists and activists face severe repercussions and persecution. This stifling of dissent not only obstructs transparency but also perpetuates a one-sided perspective of the conflict, leaving the Russian public uninformed and oblivious to the atrocities committed in their name. The absence of accessible non-governmental information further reinforces the state's control over public opinion, making it increasingly challenging for the truth to reach the Russian populace.

Local reports of soldiers killed in action (KIA) are a notable exception, as they do not undergo the same level of censorship. Such reports are, in most cases, published by local authorities on social media — e.g., on the official pages of governors, city managers, municipal administrations, etc. — and then disseminated by local newspapers and online news outlets. Usually, each report focuses on the death of one or several soldiers and contains a name, often accompanied by a short biographical note. In principle, KIA reports published by local authorities aim to honor deceased soldiers. Although the federal government could have prohibited the publication of such information, it has generally refrained from doing so, and as of January 2025, KIA reports continue to appear routinely on the websites and social media pages of local authorities and news outlets. One possible explanation for why this practice has not been suppressed is that,

Table 1: Keyword topic frequencies

Topic	Mean
Neutral	0.4556261
War	0.3360837
President	0.0548923
Government	0.0565106
Patriotism	0.1213250
WW2	0.1901032
War in Ukraine (SMO)	0.0135293
Ukraine	0.0145750

given the scale of the war, as well as the existence of the internet and social media, the local dissemination of such information would be impossible to prevent regardless of the authorities’ actions. Moreover, an attempt to do so could further undermine the credibility of the regime’s narrative.

Importantly, the information on military losses is in stark contrast with the dominant narrative promoted by the federal government since the onset of the invasion (The Current Time 2024). The Ministry of Defense, in particular, has either claimed minimal losses, concealed them, or attributed military casualties to non-military activities, as in the case of the Moskva battleship (TASS n.d.). This discrepancy between the information from the local and the federal sources, therefore, raises substantial concerns regarding the transparency and credibility of the federal government’s narrative regarding the invasion, both among those who support the invasion and those against it. Moreover, comments under the postings about the KIAs on social media appear to be one of the few spaces where the federal narrative is contested (Urman 2019).

Data

Local social media groups

To proxy political attitudes, we use users’ reactions — “likes” and derivative engagement metrics discussed in more detail in the next section — to social media posts on specific topics. Previous research has theorized engagement through likes as a multifaceted act, serving different purposes based on the user’s attitude toward the content and/or the poster. Most obviously, however, liking reflects an automatic and straightforward response to content the user agrees with or

enjoys (Alhabash et al. 2018; Sude, Pearson, and Knobloch-Westerwick 2021), and this is how we interpret it in the current study.

The data on posts and users’ reactions to them come from *VK.com*, Russia’s most popular social networking site. We gather a universe of posts from official social media accounts of schools, as well as the conventional user interaction metrics — likes, views, shares, and comments — for each post. Our main outcomes of interest are based on the engagement with these posts by the topic of their content in a given municipality.

Using the official social media accounts of schools was a deliberate choice. Fundamentally, we were interested in using social media data to investigate the digital fingerprints of support for the regime and the war in Ukraine. Although there is an abundance of political content on VK.com, analyzing individual users’ behavior is problematic for both ethical and technical reasons. An alternative approach is to focus on local, geographically specific online groups that post both political and neutral content and to analyze aggregate engagement metrics for different types of posts. These groups needed to be comparable in size, audience type, and content. Moreover, they had to be numerous — preferably in the thousands or tens of thousands — and geographically dispersed across the country to ensure broad coverage and granularity. School groups satisfied all the aforementioned criteria. Furthermore, schools in contemporary Russia play a significant role as instruments of state propaganda. Their official online communications are thus important channels through which state propaganda is disseminated and reinforced at a grassroots level. Studying engagement with these accounts therefore provides insight into how local populations react to this propaganda.

Thus, a significant share of the content published in these communities is, on par with postings about school events, either related to matters of politics or patriotism or mentions the authorities (see examples of posts in Appendix Section). Patriotic posts, e.g., those about military holidays and the Great Patriotic war in particular, comprise a large part of our sample. In addition, some schools have engaged in commemoration of the KIA troops via installing commemorative plaques and hero’s desks in the classrooms. Table 1 reports the probabilities of a post falling in a keyword-defined topic. As for the authors of these posts, online content for these school groups is typically produced directly by school administration staff. Usually, there is no dedicated “Social Media Manager” position within a school’s formal structure. Instead, the responsibility for managing the school’s online presence and posting content is assigned to existing staff members.

This could be a teacher, or an administrative employee, in addition to their primary duties.

Second, the school communities are highly similar in terms of content and its timing. As of 2021 Russian government requires all schools to be present online in its attempt to scale up the indoctrination and propaganda among the youth, and a large part of it originates from the Ministry of Education and is sent to the schools simultaneously. Each community in our sample has a Russian government verification mark, and we discard the rest to avoid potential bias in type of content and reduce the noise. We therefore expect the content in our sample to be rather uniform across communities.

Third, we expect school communities to have less partisanship bias than explicitly political online groups, and in general representing a theoretically more relevant population (Appendix Section provides available socio-demographic characteristics of the groups' subscribers). While online news outlets on the VK social network enjoy relatively high levels of engagement, these communities are largely partisan along the opposition-regime line and serve to a subset of population. Most people do not follow communities with political content (Urman 2019). School communities, not being explicitly political, should overcome the problems associated with selective exposure and polarization in news consumption.

In addition, we expect school communities to be less exposed to paid human commentators and automated bots in comparison to larger communities and social media news outlets that are specifically targeted by bot farms (Meduza 2023; Shirikov 2022). For instance, out of more than 36,000 schools only 84 had more than 1 comment made by a bot (see Supplementary Appendix).

Finally, as of now, almost every school in Russia has a presence on social media. Therefore, online school groups exist in most Russian localities, and most community public pages include address information, allowing us to geolocate them precisely. The only exception is the city of Moscow, where official school groups — i. e. schools groups run by school administrations — do not have government verification marks and rarely post anything political. This is no coincidence, but the reflection of a deliberate approach the regime took to pacify the capital.

School-community search criteria

We establish specific search criteria for VK school communities. Initially, we focus on communities containing the term *school* or related synonyms in either the community name or description. Next, we refine the list by filtering out communities associated with extracurricular activities, thereby excluding music, art, and sport schools. Subsequently, we eliminate communities referencing specific grades or graduation years, as these typically cater to smaller groups and do not post much content. Finally, we verify the presence of a government verification mark from the *gosuslugi.ru* website on VK school pages and exclude communities that are not verified.

As of September 2022, Russia had 38,549 public schools (Bondarenko et al. 2024). Of these, we collected data from 36,782 official school pages on social media, each representing a specific school. Using data from the official register of legal entities, we linked 36,420 of these social media pages to specific legal entities, along with associated information such as tax numbers, registration numbers, and, most importantly, official addresses, which we used to geo-locate the pages. As shown in Figure C30, by January 2022, the majority of municipalities had at least one school group on social media. The distribution of these groups closely aligned with population density (see Figure C29). By March 2024, however, school groups had spread to cover all of Russia (Figure C31). Given that school groups' coverage was increasing throughout this period, we focus our analysis on 2,040 of Russia's 2,587 municipalities that had at least one school group by January 2022.

Next, we download all the postings made by the school communities, more than 28 million in total. We exclusively consider posts created directly by school group administrators. We exclude reposts, i.e., posts originating from other communities shared on the school community's page, as well as posts with no text. Our final sample amounts to roughly 4 million posts made from February 2022 to February 2023. All posts contain information on how many likes, shares, comments, and views they received from the users, as well as the text of the post.

Topics of the social media posts

We use multiple techniques to categorize the content of posts, the simplest of which is keyword analysis. In this approach, we identify sets of keywords related to topics such as war and the military, patriotism, the president, and the government. A post is coded as belonging to a

certain topic if it contains a keyword associated with that topic. For example, we classify a post as ‘patriotic’ if it mentions the national anthem, the flag, the Great Patriotic War, patriotic holidays such as Victory Day (9 May) or Defender of the Fatherland Day, and similar themes. We set up all topics except *War in Ukraine* to be not mutually exclusive. This means that if a post is associated with both *Patriotism and Military* and *War in Ukraine*, it is categorized *only* under *War in Ukraine*. However, if a post does not mention the war in Ukraine, it can be categorized under multiple topics, such as both patriotic and related to the authorities. We consider a post *Neutral* if it does not mention any of the keywords from our topics of interest. Topics and their corresponding keywords are detailed in Table C2. A post is categorized into a specific topic if it contains at least one keyword from the predefined set, allowing for a broader selection of relevant posts.

Exploratory keyword analysis confirms that political content makes up a significant share of school postings: More than 30% of the posts in our sample use military vocabulary, with 1% directly related to the ongoing War in Ukraine, or Special military operation (SMO) as the Russian authorities call it, while at least 5% mention either government or president. Patriotic symbols such as the flag, the anthem, or the great patriotic war are mentioned in around 20% of the posts. Only 45% of the posts don’t mention these topics.

In addition to keyword analysis, we use zero-shot classification. The difference between zero-shot classification and typical supervised learning is that a model may classify data into numerous classes without requiring particular training samples. In classical supervised learning, a model is trained on a labelled dataset with examples for each category it must classify. In the context of our study, a pre-trained zero-shot categorization provides a scalable pipeline for estimating the likelihood of a text falling into a topic.

To analyze the thematic content of the posts, we employ RuBERT. RuBERT (Russian BERT) is a deep learning model based on the Transformer architecture, pre-trained on a corpus of Russian language text (e.g., news articles, Wikipedia, and other web data). Its pre-training utilized a Masked Language Model (MLM) objective: the model learned to predict randomly hidden words within sentences, thereby developing a nuanced understanding of Russian grammar, semantics, and contextual word relationships. This gives RuBERT a strong ability to comprehend the meaning of input texts, making it effective for tasks such as topic classification, as it can understand the underlying themes even without explicit task-specific training (zero-shot). We

Table 2: Zero-shot topic probabilities summary statistics

Topic	Mean	SD	Min	Max
Education	0.7445025	0.2771659	0.0010981	0.9979771
War	0.1166711	0.2507715	0.0004749	0.9976617
President	0.0323865	0.1215595	0.0005417	0.9978237
Government	0.1389104	0.2410433	0.0008137	0.9971288
Patriotism	0.2270499	0.2518490	0.0005497	0.9951110
Ukraine	0.0890145	0.1517093	0.0004476	0.9972161

focus on longer posts of more than 100 characters as they are likely to be most informative. The model then assigns probabilities that a given post falls into one of pre-defined topics, or labels. For comparability with the keyword analysis, we set the labels to *President*, *Government*, *Patriotism* and *War*.[^]Due to the specific word markers the Russian authorities use to refer to the Russian-Ukrainian war (e.g., special military operation or SMO), in regard to this topic, we reserve to keyword analysis as zero-shot analysis provides little advantages over it. The zero-shot label War therefore refers to a broader set of content about the military or the Great Patriotic War.] In addition, we select *Education* as a benchmark label, which we expect to be the most prominent topic in local online school groups. We use President and Government labels for pro-authorities content (Authorities), while Patriotism and War labels refer to patriotic content (Patriotism).

Table 2 summarises the estimated probabilities of a post falling into our categories. In line with our keyword analysis, matters of politics, war, and patriotism appear to be frequent topics of the postings on school pages. Wordcloud figures in Appendix show the most frequent words by topic from a sub-sample of posts.

Russian war fatalities

To capture the exposure to war fatalities, we employ data from the Mediazona and BBC database of Russian military casualties (Mediazona and BBC 2023). A group of independent journalists and volunteers manages and regularly updates the database. They manually confirm each fatality using open-source data. To confirm the death of a soldier, the volunteers refer to obituaries, news articles and social media posts published by local news outlets, government officials and relatives of the Russian troops killed in action (KIA), identified by name.

Each entry in the casualties database gives details about the KIA military personnel. Most importantly, it contains an obituary publication date, i.e., when the death of a KIA was confirmed, what source was published by, and where the deceased resided. In addition, most entries include information on the deceased's military rank, military unit and its type, as well as the date and place of birth, death, and burial. Finally, the database includes information on whether the individual was affiliated with the Russian Ministry of Defense, pro-government militias such as the Wagner group, and whether he was a military professional, volunteer, or a conscript. The database does not contain information on the KIA troops of the so-called Luhansk and Donetsk People's Republics (LNR and DNR).

Importantly, the information in this database is limited to the KIA reports published by sources within Russia accessible to the Russian public. While the reports on Russian casualties provided by Ukrainian officials and independent media outside Russia have proven to be credible, people in Russia might not have access to this information or disregard it. Obituaries provided by local authorities, regional media, and relatives, on the other hand, should have more credibility even among the most ardent supporters of the regime and the *party of war*. These reports are in stark contrast to statements made by the federal government, which confirmed only 5,937 casualties by September 2022 and has not provided any updates since. In comparison, BBC and Mediazona identified by name more than 55 thousand Russian troops killed in Ukraine as of May 2024.

In our dataset, we have information on Russian fatalities until March 2024, and we geolocate 45,696 reports about Russian KIAs with municipality-level precision. In addition, we are able to identify 7,762 posts in our sample of 13 thousand school groups that directly mention the KIAs, e.g., obituaries and posts about commemoration plaques. Figure Figure C22 shows the number of KIA reports for the period from February 2022 to March 2024 by month, and Figure Figure C23 shows the distribution of the first KIA reports by municipality over time. Figure Figure C28 shows the number of total KIA reports in municipalities by March 2024.

Empirical Strategy

In this section we describe how we construct the social media engagement metrics and then elaborate on our identification strategy and underlying assumptions.

Relative Engagement Indicator

Our primary outcome of interest is social media engagement with different topics. Specifically, we examine how engagement with topics related to the war, patriotism, or the authorities changes after an obituary of a soldier from a given municipality is published by local authorities, his or her relatives, or media.⁵

The social media data provides various engagement metrics, including likes, views, shares, and comments. However, exploratory analysis reveals that in our sample, shares and comments are extremely rare. Consequently, we use likes as the main metric of engagement, as they are significantly more frequent and easier to interpret. Although VK allows for different types of reactions, the default is the *heart* emoji, and as of the time of data collection, VK’s API did not provide information on reaction types. For simplicity, we refer to reactions of all types as likes throughout the analysis.

To account for variations in posting intensity, we adjust the number of likes by the number of posts. This adjustment ensures that fluctuations in the volume of posts—whether increasing or decreasing after the treatment—do not bias our engagement measures. As a robustness check, we also compute the number of likes per view. However, this metric is more problematic due to endogeneity between likes and views: posts with higher views tend to receive more likes, and vice versa. This issue is less relevant when normalizing likes by posts.

Another challenge is that overall engagement might decrease after the treatment, regardless of the content’s topic. To address this, we calculate the difference in likes per post between politically relevant topics and neutral topics, such as those referring to school events and general education.

For keyword-based topics, the relative engagement for a topic τ in municipality i at month t is calculated as:

$$\begin{aligned}\Delta Engagement_{it,\tau}^{keywords} &= \Delta \log \text{Likes per post}_{it,\tau} \\ &= \log \frac{\sum Likes_{it,\tau}}{\sum Posts_{it,\tau}} - \log \frac{\sum Likes_{it,-\tau}}{\sum Posts_{it,-\tau}},\end{aligned}\tag{4}$$

where $-\tau$ stands for *Neutral* content.

⁵All the obituaries are available online. We cannot control directly who within the municipality saw the obituary.

Zero-shot topic analysis allows for higher flexibility. Specifically, for each post classified with the zero-shot model, we know the estimated quasi-probability of a post falling into a topic, allowing for multiple true classes. This approach captures the varying intensity of each topic within a single post. Given this flexibility, we adjust our relative engagement metric by weighting the sums of likes and posts by the raw zero-shot probabilities, as follows:

$$\begin{aligned}\Delta Engagement_{it,\tau}^{zero-shot} &= \Delta \log \text{ Likes per post}_{it,\tau} \\ &= \log \frac{\sum Likes_{it,\tau} \times \mathbb{P}_\tau}{\sum Posts_{it,\tau} \times \mathbb{P}_\tau} - \log \frac{\sum Likes_{it,-\tau} \times \mathbb{P}_{-\tau}}{\sum Posts_{it,-\tau} \times \mathbb{P}_{-\tau}},\end{aligned}\tag{5}$$

where $\mathbb{P}_\tau$ is the estimated probability of a post falling into topic τ . The inclusion of raw zero-shot probabilities in the engagement metric introduces an important feature: engagement with a single post can simultaneously contribute to both politically relevant and neutral content.

To address concerns that small variations in probabilities could disproportionately influence the engagement metrics, we also construct a complementary metric using binary zero-shot probabilities. In this alternative approach, a post is considered to belong to a topic only if its quasi-probability exceeds a predetermined threshold. This binary classification provides a robustness check against potential over-sensitivity in the weighted metric and allows for more straightforward comparison between results based on zero-shot and keywords categories.

Identification Strategy

The analysis is based on the *Callaway and Sant’Anna (2021) doubly robust (DR) Difference-in-Differences (DiD)* framework (Callaway and Sant’Anna 2021; Sant’Anna and Zhao 2020), which is well-suited for staggered treatment adoption designs. We exploit the idiosyncrasies in the precise timing of KIA reports across towns, which has a substantial random component due to variation in the time required to confirm a soldier’s death. This framework accounts for treatment effect heterogeneity and avoids the limitations of traditional two-way fixed effects (TWFE) models.

The DR-DiD estimator explicitly incorporates unit and time fixed effects through its structure, while remaining more flexible than the TWFE framework. Instead of relying solely on strong linearity assumptions, the DR estimator leverages both *Inverse Probability Weighting (IPW)*

and *Outcome Regression (OR)* components. The parameter of interest is the average treatment effect on the treated (ATT) for each group and time, defined as:

$$\text{ATT}_{g,t} = \mathbb{E}[Y_t(1) - Y_t(0) \mid G = g],$$

where $(Y_t(1))$ and $(Y_t(0))$ denote potential outcomes with and without treatment, respectively, for units treated in group (g) at time (t) .

Given that no covariates beyond unit IDs and time periods are used in our analysis, the DR formula simplifies to effectively adjust for time and unit-specific variation without explicitly modeling covariates. The DR estimator is robust to model misspecification as long as one of the two components (IPW or OR) is correctly specified. The ATT is estimated using the following formula:

$$\widehat{\text{ATT}}_{g,t} = \mathbb{E} \left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\mathbb{E} \left[\frac{p_g(X)C}{1-p_g(X)} \right]}{\mathbb{E}[G_g]} \cdot \frac{p_g(X)C}{1-p_g(X)} \right) \cdot (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{\text{nev}}(X)) \right].$$

This approach ensures that treatment effects are estimated more flexibly and avoids the potential biases and inefficiencies associated with TWFE models, such as inappropriate weighting or homogeneous treatment effect assumptions.

To analyze the dynamic effects of treatment, we aggregate the group-time average treatment effects (GATTs) into event-study-like estimates, which allow us to examine the treatment effects relative to the timing of the first KIA report in each municipality. Specifically, we organize the GATTs by the number of periods since the initial treatment event (event time) and compute average treatment effects for each event-time period. This approach enables a detailed visualization of the temporal evolution of the treatment effects. We use simultaneous confidence intervals for these dynamic effects, following the procedure outlined in (Callaway and Sant’Anna 2021), which accounts for multiple testing across event-time periods and ensures robust inference.

Our identification strategy relies on the assumption that in the absence of news about local military losses, engagement patterns with politically charged content on social media would have followed parallel trends in treated and not treated units. The assumption is solidified given that the precise timing of KIA reports is random and not influenced by pre-war characteristics of the municipalities, which mitigates concerns about self-selection into treatment. Thus, Figure C10

shows substantial variation in the time elapsed between soldiers’ deaths and the confirmation of those deaths among the 18950 soldiers with known dates of death (Median = 14, Mean = 26.761, SD = 41.528). Furthermore, Figure C24 illustrates that, after accounting for region fixed effects, the timing of the first KIA report in a municipality is primarily influenced by its population size, age structure, and average temperature. The latter is likely correlated with population density due to the geographical layout of many Russian regions along meridians. Finally, when Y_{it} is defined as the difference in engagement metrics between neutral and politically relevant content — our preferred specification — we assume that trends in engagement with different types of content would have evolved similarly within units. Figure C25 reports the dynamics of engagement by topic around the date of the first KIA report.

Incorporating time and unit fixed effects using the C&SA DR DiD estimator effectively addresses key concerns about estimating the causal effects of local military losses on municipal-level outcomes. This estimator accounts for differences in demographics, economic conditions, access to alternative information sources, and proximity to the war zone by leveraging covariates to satisfy conditional parallel trends. The DR DiD framework explicitly models potential heterogeneity in untreated potential outcomes over time and across municipalities, ensuring robustness to model misspecification. National trends — such as shifts in political attitudes or economic policies — are implicitly controlled through the doubly robust approach, which allows valid identification even if only the outcome or propensity score model is correctly specified. Unlike traditional fixed effects models, this approach does not assume homogeneous treatment effects or restrict outcome trends to be group-invariant, making it more flexible and suitable for capturing the complex dynamics in this context.

Results

Effect of KIA reports

For the main part of our analysis, the unit of analysis is municipality i observed in month t . The dependent variable is the difference in user engagement between politically relevant content and its neutral counterpart, captured as the difference in number of likes per post (for more detail, refer to the previous section). Treatment is defined with variable *Post KIA*, equal to one for the months after the first obituary was posted on a school page and zero otherwise. We replicate

the analysis separately for a set of pre-defined topics, including *War in Ukraine* and *Patriotism and Military*, — to capture changes in engagement with patriotic, nationalistic, or militarized content, — and *Authorities* — to capture engagement with content directly mentioning the president and local and federal authorities. Note that posts categorized under the *War in Ukraine* topic are excluded from all other topics to avoid contamination. To ensure that the results are not an artifact of a statistical anomaly in the data generating process behind the community administrators’ posting behavior, we exclude municipality-months when no posts of a relevant topic were made, as well as schools with less than 40 members (bottom 5% of all communities). We also exclude municipalities where no school community was created before 2022, i.e. municipalities with no data available for the pre-invasion period.

For the main part of the analysis, the period in consideration is from March 2022 to September 2022, covering the first 7 months of the invasion. This allows us to focus on the effect casualty reports had prior to the “partial mobilization” of the military reservists on September 21, 2022, which was a uniform shock that might have impacted attitudes towards both the invasion and the government. Results for the full period are in Appendix.

Baseline estimates: To evaluate the average effect of KIA reports on engagement with politically relevant content, we first regress the change in engagement on *Post KIA*, effectively accounting for month and municipality as well as for the cohort fixed effects, following Callaway and Sant’Anna (2021).

Figure 1 reports our main results - the average treatment effects on the treated (ATTs) with not yet treated as the control group. The results indicate that engagement with militaristic and patriotic content, relative to neutral, increases after KIA reports across all classifiers and relevant topics. This increase reaches up to 14.2% ($p < 0.1$) for content explicitly referencing the *War in Ukraine*. Content falling into *Patriotism & Military* topic also sees a rise in engagement of 2% for zero-shot classifier and 3% for keywords classifier ($p < 0.05$).

By contrast, engagement with *Authorities* topic decreases by 5% relative to neutral content for a zero-shot based classifier and by 8.4% for a keyword-based classification ($p < 0.05$). Analysis for more detailed topics provides similar results, with a decrease in engagement being the highest for the *Government* and *Local authorities* (9-9.4% decrease, $p < 0.01$).

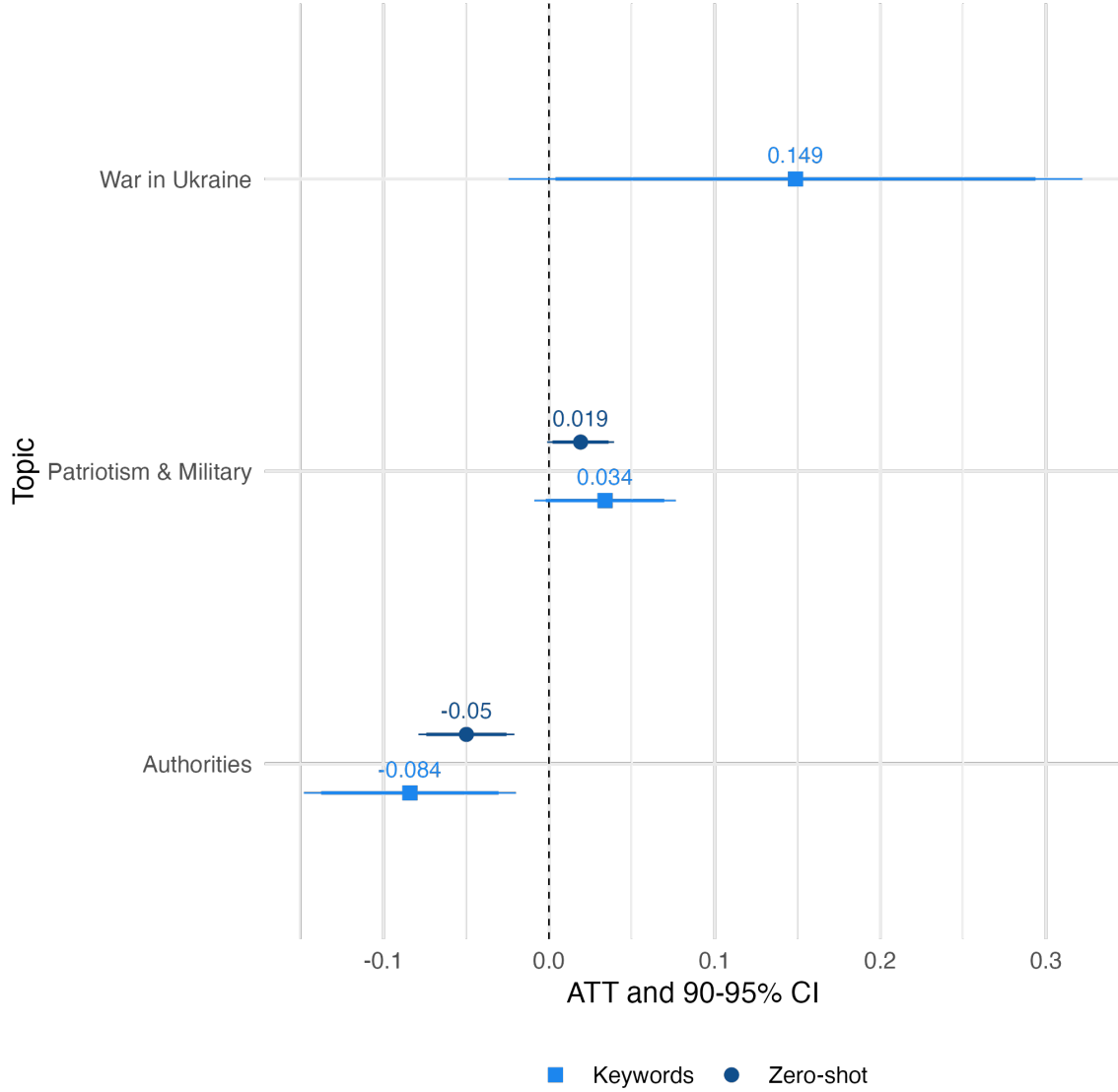


Figure 1: Change in engagement by topic relative to *Neutral* content after the first KIA report. *Note:* Dependent variable: $\Delta \log \text{Likes}$ per post. Treatment: First KIA report month in the municipality. Dots show ATT coefficients from separate Callaway and Sant’Anna (2021) estimations by topic. Topic is defined along the vertical axis. Horizontal bars represent 90% and 95% confidence intervals. Standard errors clustered at the municipality level. Squares and circles indicate classification strategy.

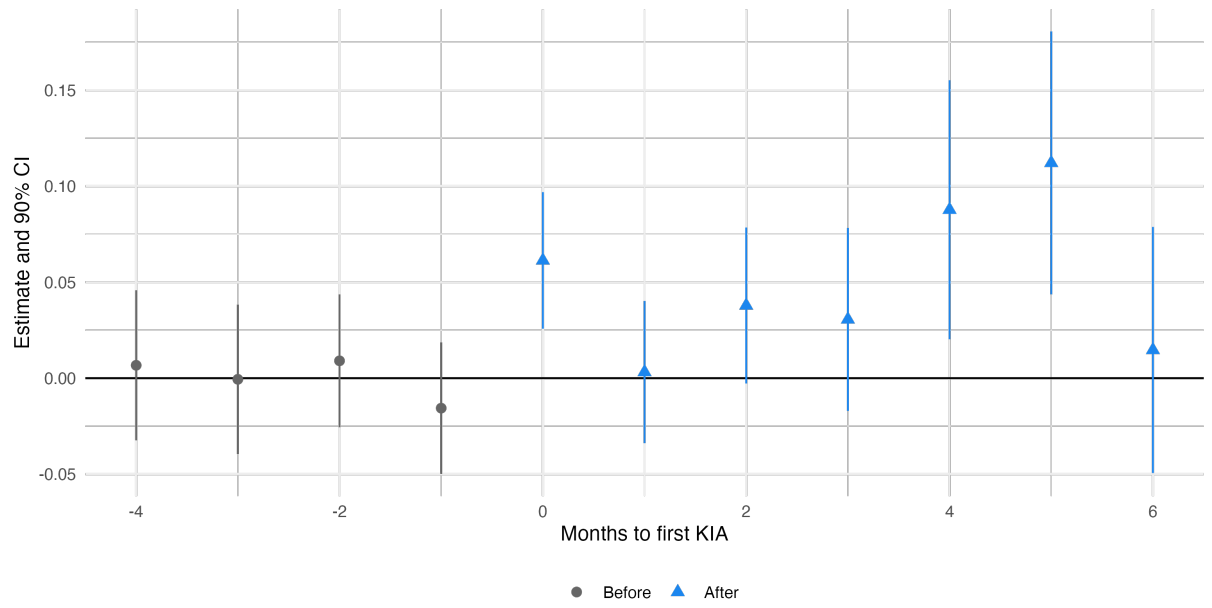
Event study estimates: The literature on the effects of war and war fatalities suggests that the attitudes towards war evolve over time: While in the first months of the intervention the government responsible for the war and the war itself often faces high public support, it starts to shrink later on when the costs of the intervention start to accumulate (Kuijpers 2019). To probe this logic further as well as to get a better understanding of whether the parallel trends assumption holds, we explore the dynamics of the effect of the KIA reports on engagement. We employ the Callaway and Sant’Anna (2021) estimator with not yet treated as the control group.

Figure 2 show the dynamics of the effect by time since the first KIA report for *Patriotism & Military* and *Authorities* topics defined with keywords.⁶

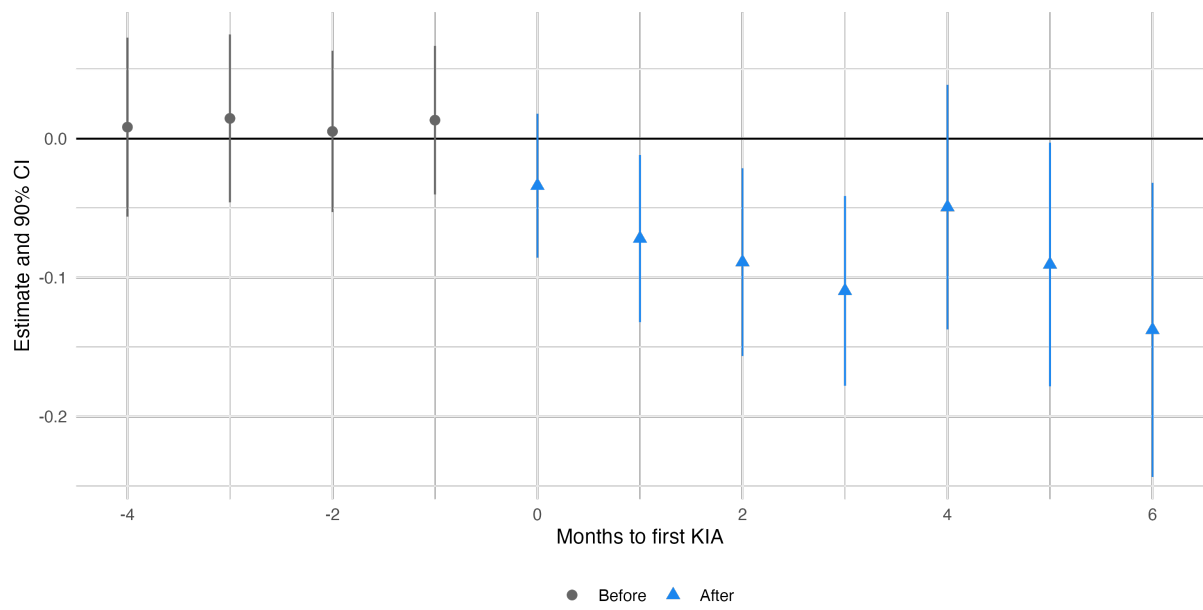
We find no evidence for a significant difference in trends in the months leading up to the publication of an obituary. The pre-treatment period coefficients are not statistically significant, both individually and jointly (Wald test of joint significance: $W = 0.7772$, $p = 0.9415$ for *Patriotism & Military*; $W = 1.1663$, $p = 0.8836$ for *Authorities*), and close to zero in magnitude for both topics. After the publication of an obituary, by contrast, we observe a sharp 6 percentage points increase in engagement with *Patriotism & Military*. The effect, however is not limited to the immediate increase in the first month that can be attributed to the obituaries falling into the topic *Patriotism & Military*, and accumulates over several months, reaching its peak of more than 12 percentage points 5 months after the first KIA report in a municipality. Engagement with *Authorities* steadily decreases after the first KIA report, reaching magnitude of around 14% six months after the initial report.

Finally, our estimates for the keyword-based topics, applied to an extended sample covering the period until March 2024 (two years into the war), reveal a more pronounced decline in engagement with Authorities-related content ($\beta = -0.14$, $p\text{-value} < 0.01$). In contrast, the effect on engagement with Patriotism and Military-related content is indistinguishable from zero (see Figure C13). This suggests that the initial “rally around the flag” effect is relatively short-lived, failing to persist through the prolonged stages of the war, and reflects an increase in patriotic sentiment rather than sustained support for the political figures responsible for initiating and continuing the conflict. Results aggregated by calendar months lend further support to this reasoning: As reported in Figure C14, the increase in patriotic engagement is observed primarily in the months before the “partial mobilization”, while the decline in engagement with *Authorities* continually increases in magnitude over the course of the war. Engagement with War in Ukraine remains elevated throughout the period ($\beta = 0.366$), likely driven by the continued publication of obituaries. We explore the matter of obituary publications directly on the school pages below.

⁶Resolution of the Government of the Russian Federation of 01.09.2023 N 1421 (as amended on 05.04.2024) “On approval of the Rules ... for issuing a certificate of death of a citizen, the form of a certificate on the circumstances of the disappearance of a citizen, the form of a certificate on the circumstances of the disappearance or possible death of a citizen, the form of a certificate of death of a citizen”



(a) Effect on engagement with *Patriotism & Military* topic



(b) Effect on engagement with *Authorities* topic

Figure 2: Event study graphs for the effect of the first KIA report (keywords). *Note:* Dependent variable: Δ log Likes per post. Independent variable: *Post KIA*. Estimator: Callaway and Sant'Anna (2021). Control group: Not yet treated. Dots report the point estimates. Vertical bars report the 90% confidence intervals. Standard errors clustered at municipality level.

Robustness

Detailed topics: To rule out the possibility that the effect we observe is driven by the idiosyncrasies in zero-shot classification or our keyword selection, we perform the above analysis on the posts classified with more detailed sets of keywords and finer zero-shot topics.

Figure C16 reports the results. We find effects similar to baseline across more detailed topics. Specifically, we observe an increase in engagement with *War*, *Patriotism*, and *World War 2* topics, and increase in *War* is the largest in magnitude and statistical significance.

Results concerning the authorities are also in line with our baseline. We document a decrease in engagement with postings mentioning the *President*, the *Government*, and the *Local authorities*. The decrease in log likes per posts relative to neutral content however is not statistically significant for president which might be due to the relative rareness of this topic as only 2% of the posts mention president.

Alternative estimator: As an alternative to the Callaway and Sant’Anna (2021) estimator, we also implement a Sun and Abraham (2021) estimation strategy. Both estimators are designed for the staggered treatment rollout scenarios, with the main difference being the control group: not yet treated in Callaway and Sant’Anna (2021) vs. last treated in Sun and Abraham (2021). The event study estimates obtained with the Sun and Abraham (2021) estimator are comparable to our baseline. Figure Figure C17 reports the results.

Repeated treatment: Another potential concern is that our findings may reflect the cumulative impact of the rising death toll in each municipality rather than the effect of individual KIA reports. To address this, we employ the estimator proposed by Chaisemartin and D’Haultfoeuille (n.d.) that is robust to heterogeneous effects and limits the threat posed by contamination from other treatments. Unlike the approach in Callaway and Sant’Anna (2021), this estimator accommodates multiple treatment levels by comparing changes in outcomes across units with similar treatment histories prior to a given treatment switch. In our context, a treatment switch corresponds to an increase in the cumulative number of KIA reports within a municipality.

The results, presented in Figure C18, indicate that repeated KIA reports do not drive our findings. As in the baseline, we observe an increase in engagement with patriotic and militaristic content following KIA reports, accompanied by a decline in engagement with content that mentions the authorities.

Placebo study: As an alternative way of inference, we randomly reshuffle the month of the first confirmed KIA across the municipalities. The density plot of the baseline ATT placebo coefficients based on 500 permutations shows that the baseline estimated KIA report effect on engagement is larger than the 99th percentile of the distribution of the 500 placebo KIA report effects for *Patriotism and Military*. Similarly, the baseline ATT for *Authorities* is smaller than the bottom 1% of the placebo KIA effects. Figure C19 and Figure C20 report the results for keyword and zero-shot based topics, respectively.

Mechanism

Channels

Information channel To better understand the channels mediating the effect of local war fatalities, we restrict the sample to municipalities where a local soldier died but no obituaries were posted on school group pages. In these cases, the death was confirmed through either a local news outlet, local authorities, or another source. This restriction helps limit the direct exposure of group users to information about military losses. Treatment is defined as before, with the variable *Post KIA* equal to one for the months following the first soldier death in a municipality and zero otherwise. All other elements of the analysis remain consistent with the baseline setup.

Figure 3 presents the results. While the baseline estimates show significant changes in social media engagement with politically relevant content following KIA reports, the effects in this restricted sample, where exposure to information about local military losses is less direct, are smaller and less statistically significant. Specifically, compared to the full sample, the restricted sample estimates indicate an 11.3 percentage point increase in engagement with *War in Ukraine* over a six-month horizon. For *War in Ukraine*, the effect size is about 3 percentage points lower than the baseline estimates (Figure 1) and is not statistically significant at conventional levels ($p\text{-value} > 0.1$). The effect for *Patriotism and Military* and the *Authorities* is similarly reduced. These results are statistically significant only for the zero-shot topic classifier ($p\text{-value} < 0.05$).

These findings suggest that direct exposure to information on war fatalities significantly amplifies the impact of military losses on engagement. The difference between the baseline and restricted sample results is unlikely to be driven by lower elasticity of engagement in school groups that do

not post obituaries. On the contrary, schools posting about military losses after the start of the invasion exhibited lower engagement levels before February 2022 (Mean difference in likes per post = -2.906, t-statistic = -5.764, p-value = 0). Finally, the fact that posting decisions on school pages are made by group administrators suggests that these decisions are likely exogenous to broader municipality-level characteristics or trends. Although indirect exposure, such as through local news outlets or word-of-mouth, cannot be completely ruled out due to the consistent direction of the effects, these results point out the importance of direct access to information for the effect to fully develop.

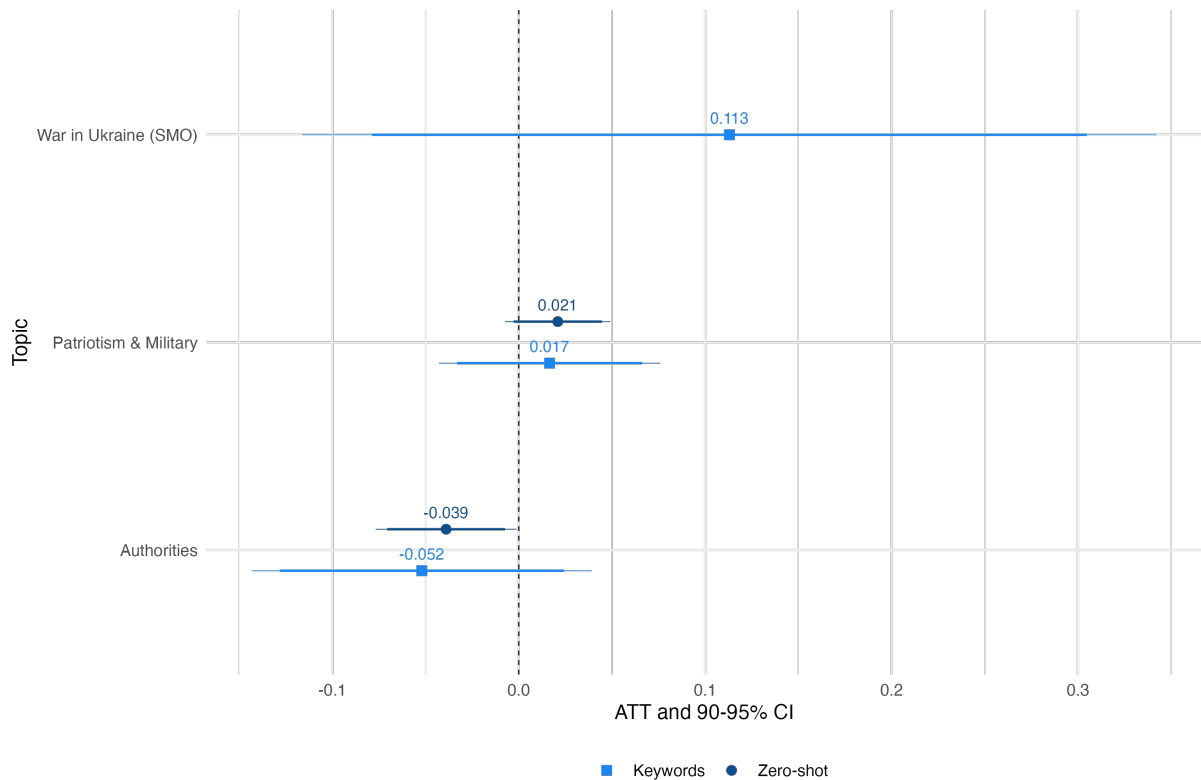


Figure 3: Change in engagement by topic relative to *Neutral* content after the first KIA report, posts with obituaries excluded. *Note:* Dependent variable: $\Delta \log \text{Likes}$ per post. Treatment: First KIA report month in the municipality. Dots show ATT coefficients from separate Callaway and Sant’Anna (2021) estimations by topic. Topic is defined along the vertical axis. Horizontal bars represent 90% and 95% confidence intervals. Standard errors clustered at the municipality level. Squares and circles indicate classification strategy.

Cost-benefit channel Another mechanism frequently discussed in the literature is the cost-benefit channel. As argued by Gartner (2008b), public support for a war and the regime overseeing it is shaped by recent casualties, casualty trends, and the broader context in which these losses occur. Specifically, trends in war fatalities can influence individuals’ perceptions of both

the trajectory of military losses and their expected future magnitude. For instance, when fatalities increase at an accelerating rate, individuals are more likely to revise their estimates upward regarding the future costs of continuing the war. This mechanism is particularly relevant in positional warfare contexts, such as the ongoing conflict in Ukraine, where the battlefield situation remains relatively static over time.⁷

Table C1 presents the results of regressing social media engagement on the monthly changes in the cumulative number of KIA reports, controlling for municipality and month fixed effects. Compared to the baseline results, we observe no significant correlation between military loss dynamics and engagement, except for a decrease in engagement with *Authorities*.

This finding suggests that battlefield loss dynamics *do not* amplify patriotic sentiment beyond the effect of initial fatality reports and provide further evidence that the patriotic rally is limited in duration. However, regime support appears to deteriorate, as ongoing battlefield dynamics likely prompt individuals to reassess the future costs of the war, leading to further erosion of support for the authorities.

Heterogeneity

Content political relevance The content of school postings often spans multiple topics simultaneously. For instance, while some posts exclusively address the authorities or the war, the majority are intertwined with themes related to school life and education. As a result, it is challenging to fully rule out the possibility that the observed effects are driven by changes in the composition of topics rather than genuine shifts in engagement. To validate our baseline results, we examine heterogeneous changes in engagement across content with varying levels of political relevance. Specifically, we compare engagement with content that has a higher or lower concentration of political topics relative to education topic.

To this end, we further investigate how the intensity of political topics in the content influences user engagement following KIA reports. *Topic relevance* is defined as a binary variable equal to 1 if the ratio of the zero-shot probability of a topic to the probability of it being neutral exceeds 0.2, and 0 otherwise. Using this definition, we estimate a TWFE model on the sample

⁷Figure C27 illustrates the changes in area controlled by Russian and Ukrainian forces. Notably, the overall balance of territories controlled by both militaries has shifted minimally throughout 2023, reflecting the highly positional war of attrition.

of individual posts. The outcome of interest is the number of likes, while the treatment variable is specified as *Post KIA* interacted with the *Topic relevance* of a post. The model includes month and school group fixed effects. Excluding schools created after January 2022, our sample contains 13346 schools and roughly 2 million posts.

It is important to note that this analysis relies on the strong form of the parallel trends assumption, which requires that trends are parallel across content with different levels of political relevance. While the inclusion of school fixed effects, rather than municipality fixed effects, alleviates this concern to some extent, as one might expect political relevance to depend on the writing style of the school administrators, our interpretation for this exercise is correlational.

Table 3 presents results that support our baseline interpretation. The changes in engagement following KIA reports are more pronounced for posts with higher political relevance. Specifically, we observe a decrease of 0.709 likes per post for content with high intensity on the *Authorities* topic (p-value < 0.01) after the first KIA report (the interaction term in the regression table). Similarly, the increase in engagement is more pronounced for content with high intensity on the *Patriotism and War* topic ($\beta = 0.578$, p-value < 0.01).

These results also provide a glimpse into how *Topic relevance* and military losses change engagement regardless of each other. We find that content mentioning the authorities gets on average 4.8 likes less — a 23 % decrease relative to the mean, — while patriotic posts get 1.78 likes more (8.5% higher than the mean). News about military losses appear to decrease likes by 0.255-0.623 (0.1-0.3%). This further highlights the ambivalent impact of war fatalities on public attitudes: while exposure to information on military casualties may erode the support for the regime, it does not appear to foster anti-war sentiment.

Content emotional intensity To further explore the mechanisms behind the observed changes in engagement with different topics, we focus on the emotional tone of content. One possible explanation for our baseline findings is that the observed changes in engagement are partially driven by users avoiding uncomfortable content and shifting their consumption towards entertainment as a coping strategy. Psychological studies have demonstrated that higher levels of stress are associated with increased social media consumption (Wolfers, Festl, and Utz 2020; Wolfers and Utz 2022), with stress often serving as a trigger for such behavior (Ingen, Utz, and Toepoel 2016; Veer, Ozanne, and Hall 2015). In addition, social media can function as a

Table 3: Heterogeneity of the KIA reports effects with respect to political relevance of the content

Topic	likes	
	Authorities (1)	Patriotism and Military (2)
Topic relevance	-4.81*** (0.106)	1.78*** (0.077)
Post KIA	-0.255 (0.173)	-0.623*** (0.178)
Topic relevance $\times$ Post KIA	-0.709*** (0.125)	0.578*** (0.103)
Mean Likes	20.9	20.9
Mean Topic Probability	0.086	0.172
N School	13,346	13,346
N Month	13	13
Observations	2,080,875	2,080,875
R ²	0.276	0.272
School FE	✓	✓
Month FE	✓	✓

coping mechanism during stressful events such as war or pandemic. For instance, Wit, Kraan, and Theeuwes (2020) found that Twitch users utilized the platform to manage stress induced by experiencing hardship. Similarly, during the COVID-19 pandemic, individuals with higher stress levels increased their consumption of entertainment, including short videos, TV shows, and video games during lockdowns (e.g., Aghababian et al. 2021; Xu, Wang, and Ma 2023).

Building on this evidence, we hypothesize that in response to arguably traumatic news about military losses, users might shift their social media consumption toward more positive and reassuring content as a way to mitigate stress. To test this, we perform sentiment analysis on our sample of raw posts and calculate *Negativity*, *Neutrality*, and *Positivity* sentiment scores using Rogers et al. (2018) sentiment analysis model for Russian language. Each score ranges from 0 to 1. In addition, we calculate *Negativity-Positivity* score that is equal to the difference between *Positivity* and *Negativity* scores and therefore ranges from -1 to 1. We also calculate overall *Emotionality* as the maximum value of *Positivity* and *Negativity* scores for a posts, as well as *Emotional Complexity* score that is the sum of squared *Positivity* and *Negativity* scores and captures the degree of emotion heterogeneity within a post. Finally, as another measure of text complexity, we use the text length as the number of characters in a post.

Table 4 presents the results of the analysis. On average, a 10% increase in *Negativity* score

Table 4: Heterogeneity of the KIA reports effects with respect to sentiment scores

Score	Likes per post		
	Neg (1)	Pos (2)	NegPos (3)
Post KIA	-0.344* (0.196)	-1.21*** (0.184)	-0.370** (0.171)
Score	-19.9*** (0.733)	27.1*** (0.551)	21.2*** (0.422)
Post KIA $\times$ Score	0.486 (0.916)	8.23*** (0.731)	5.21*** (0.544)
Mean Likes	18.9	18.9	18.9
Mean Score	0.097	0.110	0.012
N School	12,971	12,971	12,971
N Month	13	13	13
Observations	2,014,787	2,014,787	2,014,787
R ²	0.273	0.284	0.284
School FE	✓	✓	✓
Month FE	✓	✓	✓

decreases the number of likes per post by 1.31, respectively. Conversely, a 10% increase in *Positivity* increases engagement by 2.32 likes, equivalent to a 12% increase relative to the mean. Additionally, as shown in Table C3, a 10% increase in *Emotionality* results in 1.44 additional likes per post. Longer posts, however, tend to receive fewer likes overall.

Following the KIA reports, we observe a significant decrease in engagement with posts characterized by higher *Negativity* and *Neutrality* scores. This type of content is more likely associated with government-related topics, as shown in the correlations reported in Figure C26. The effect is most pronounced for purely neutral content, with up to a 6.67 decrease in likes per post (p-value < 0.01). While engagement with negative posts also decreases, this effect is smaller and not statistically significant.

In contrast, engagement with posts scoring higher in *Positivity* and *Emotionality* significantly increases after the KIA reports. For instance, purely positive posts receive up to 11.3 additional likes compared to what they would have received in the absence of military losses (p-value < 0.01). These findings suggest that the change in engagement, at least partly, reflects a coping mechanism, where users turn to uplifting and emotionally engaging posts following the stress-inducing news about military losses in their hometowns.

To further investigate the mechanisms at play, we interact the Negativity-Positivity sentiment

score with the Topic relevance indicator, as shown in Table 3. This approach allows us to explore the heterogeneity in responses to KIA reports, accounting for both the emotional tone of posts and their topical content. In addition, we theorize that users are more likely to perceive content scoring high in both Topic relevance and Negativity-Positivity as more propagandistic due to its simplified framing and the reinforcement of polarized narratives.

As reported in Table 5, the effects of KIA reports on engagement with different topics cannot be fully explained by changes in user interactions with content of varying tones. For posts related to the authorities, engagement declines by 1.12 likes (approximately 5%), significant at the 10% level. For patriotic content, engagement increases by 0.55 likes, though this effect is not statistically significant.

Crucially, tone mediates these effects to a large extent. Consistent with Table 4, we find that engagement with more positive content increases after KIA reports, suggesting a substitution effect between negative and positive content. Strictly positive posts receive 9.64–11.4 more likes ($p\text{-value} < 0.01$) compared to neutral or ambiguous posts, representing a substantial increase of approximately 50%.

Interestingly, the results reveal that highly political content with a less negative tone receives fewer likes following KIA reports. Posts about the authorities that are positive in tone receive about 4 fewer likes compared to similar posts with a neutral tone. Similarly, for posts related to patriotism and military topics, those with higher Negativity-Positivity and Topic relevance also experience reduced engagement after KIA reports. These findings suggest a dampening effect on engagement with overly positive and topically intense content in the aftermath of KIA reports.

One possible explanation is that users perceive such highly positive and politically relevant posts as propaganda.⁸ As pointed out in Miller (1939), Weston (2018), and Da San Martino et al. (2019), one of the key features of propaganda is the extensive use of emotional language. If high emotional appeal combined with topic relevance signals propagandistic content, the decline in engagement could reflect a broader decrease in regime support following the reports of military losses and disillusionment with the narratives perpetrated by the regime.

⁸Since the posts in our sample are from government-curated online groups, we deem it unlikely that they will contain negative sentiment towards the government or the military. In addition, correlations show that overall emotionality is more closely aligned with higher positivity than negativity scores.

Table 5: Heterogeneity of the KIA reports effects with respect to sentiment scores and topic relevance

Topic	Likes per post	
	Authorities (1)	Patriotism and Military (2)
Post KIA	-0.273 (0.171)	-0.469*** (0.176)
Negativity-Positivity	19.5*** (0.429)	23.2*** (0.483)
Topic relevance	-4.04*** (0.103)	1.55*** (0.074)
Post KIA $\times$ Negativity-Positivity	6.27*** (0.576)	6.76*** (0.666)
Post KIA $\times$ Topic relevance	-0.708*** (0.125)	0.327*** (0.101)
Negativity-Positivity $\times$ Topic relevance	3.28*** (0.637)	-5.55*** (0.502)
Post KIA $\times$ Negativity-Positivity $\times$ Topic relevance	-7.44*** (0.868)	-3.85*** (0.774)
Mean Likes	18.9	18.9
Mean Topic relevance	0.213	0.370
Mean Sentiment Score	0.012	0.012
N School	12,971	12,971
N Month	13	13
Observations	2,014,787	2,014,787
R ²	0.288	0.285
School FE	✓	✓
Month FE	✓	✓

Discussion

This paper provides evidence that exposure to information on war fatalities impacts political attitudes in autocracy. To demonstrate this, the paper exploits idiosyncrasies in the timing of reports about Russian military personnel killed in combat during the ongoing invasion of Ukraine across Russian municipalities in 2022. Using a Doubly Robust Difference-in-Difference estimator, we show that social media engagement with authorities-related content drops significantly after local sources publicly announce the deaths of military personnel, whereas engagement with patriotic content increases. In other words, exposure to information on war fatalities may alienate citizens from the regime, but it does not necessarily foster anti-war sentiment. However, given the severe restrictions on expressing anti-war views in Russia since March 2022, the latter finding requires careful consideration. Our interpretation depends on observed engagement patterns with war-related content: a significant rise in anti-war sentiment might be expected to dampen engagement with pro-war or patriotic themes. Yet, we observe a temporary increase in engagement with patriotic and military topics following KIA reports. This suggests that immediate public reactions, as captured online, do not involve an overt rejection of war-related narratives. This inference is therefore necessarily indirect due to the repressive context, where expressing anti-war sentiment openly is harshly prosecuted. Still, this provides evidence that as the war continues, fueling nationalism can help the regime to sustain power even if the support for individual politicians or government institutions falls.

These results bear implications for both the survival of authoritarian leaders and the potential post-regime change political dynamics in a country. When other sources of support diminish, many authoritarian leaders rely on nationalism and traditionalism to shore up their legitimacy, with Vladimir Putin in Russia serving as a notable example. In this context, military ventures and “short victorious wars” often extend this strategy, aiming, among other things, to bolster public support for the regime through the mobilization of nationalistic sentiment. However, if a military campaign goes awry, the human cost of war can offset the initial “rally-round-the-flag” effect. This creates a dual threat for the regime: as casualties mount, public disillusionment with the regime may grow, while simultaneously laying the societal foundations for a more nationalistic challenger to emerge. At the same time, it remains to be seen whether the effect documented in this article will persist and in what form. It is likely that its long-term manifestation will be contingent on the subsequent framing, commercialization, and interpretation of the war in

Russian society.

Next, the sentiment analysis reveals that politically relevant posts — i.e., posts featuring either authorities or patriotic content — that are more positive in tone receive less engagement after the KIA reports. Since these posts can be viewed as more propagandistic, we interpret this effect as evidence of declining effectiveness of state propaganda. This poses a challenge for the regime. As the public becomes increasingly aware of military failures and their human cost, the regime may need to moderate its propaganda strategy to maintain credibility. However, it is during periods of conflict that the regime needs citizens' loyalty the most; therefore, it may find itself constrained in how much it can vary the spin of propaganda. Further research is needed to explore how authoritarian regimes navigate this dilemma — balancing the need to secure loyalty with the difficulties of adapting propaganda strategies in the face of rising public awareness and dissatisfaction.

Another aspect to consider when interpreting these findings is whether they primarily reflect Russia's specific authoritarian context or represent more general responses to war fatalities. The observed decline in engagement with authority-related content could signal a broadly universal reaction to war losses, similar to patterns seen in democracies where independent media and alternative narratives are available. On the other hand, the temporary rise in patriotic and military-focused engagement — though clearly shaped by Russia's tightly controlled media environment — might reflect a more complex dynamic. War losses leading to nationalistic responses are not necessarily unique to autocratic regimes; recent research suggests this can happen in significantly more democratic environments as well (Juan, Haass, Koos, Riaz, and Tichelbaecker 2024). This indicates that while the mechanisms might differ, the effects of casualties on nationalism likely depend on a broader set of factors beyond just regime type, such as historical legacies or prevailing socio-economic conditions. Clarifying exactly how regime type, media environment, and public reactions to war casualties interact, especially regarding nationalist or patriotic sentiment, remains an important area for future research.

Finally, this paper also contributes to the literature on the methodology behind capturing political dynamics. It offers a novel method for capturing political attitudes and public sentiment toward political events, applicable across different contexts. Alongside the recent trend of employing unstructured textual data to capture political and economic phenomena, we propose that how people engage with this unstructured data can serve as a valuable metric. Specifically,

we suggest capturing public sentiment dynamics using anonymized aggregate data from social media. More precisely, we use changes in engagement with political content compared to neutral content and overall levels of engagement to proxy for changes in war and regime support. As the costs associated with traditional survey and election-based metrics for regime support escalate and social media data become more accessible and abundant, such approaches are particularly pertinent in contexts with low accessibility to conventional data, providing an alternative and insightful means of studying political dynamics.

Conclusions

In the three studies presented in this dissertation, I ask questions that informed citizens in autocracies, and in Russia in particular, can ponder on themselves. Why wouldn't people who are against the regime vote collectively against it? Why are authorities so intolerant of even small and apparently harmless acts of dissent? Don't people see the cost of the war, and if they do, why don't they raise their voice against it? By answering these questions, the studies contribute to a deeper understanding of political behavior under authoritarianism, highlighting the conditions under which citizens respond to information about regime weakness, public dissent, and the human costs of war.

Thus, the first study demonstrates that merely signaling the ruling party's electoral vulnerability is insufficient to encourage tactical voting in favor of opposition candidates. Instead, strategic coordination depends on the partisan landscape, particularly the availability of a viable and acceptable alternative. This suggests that authoritarian regimes can manipulate opposition fragmentation to their advantage, shaping party systems in ways that inhibit effective electoral coordination against them. This raises the question: under what conditions can opposition supporters overcome partisan barriers, and could information about the incumbent's electoral vulnerability be more effective in areas where signs of regime weakness are less apparent?

The second study further underscores the limits of opposition mobilization in authoritarian settings by showing that protests, while capable of demobilizing regime supporters, do not necessarily translate into increased opposition votes. The findings suggest that protests function primarily as signals of declining regime support rather than as catalysts for opposition mobilization. Importantly, their effects are most pronounced in areas where the ruling party previously enjoyed strong backing, illustrating how unexpected instances of dissent can disrupt perceptions of regime stability. Future research could explore under what conditions protests lead to sustained political engagement rather than short-term disengagement. What distinguishes protests that catalyze opposition movements from those that primarily signal discontent? Are there specific types of protest messaging, leadership structures, or external factors (e.g., elite defections, international attention) that enhance their mobilizing potential?

The third study, co-authored with Vladimir Zabolotskiy, reveals how exposure to information about war fatalities can alienate citizens from the regime without necessarily fostering anti-war

sentiment. Instead, heightened patriotic engagement suggests that public discontent does not necessarily translate into direct opposition to the regime but can instead reinforce alternative narratives that sustain authoritarian legitimacy. This dynamic has important implications for the long-term trajectory of autocratic stability, as the erosion of trust in individual leaders or institutions does not necessarily signal broader regime collapse but may open the door to nationalist alternatives. Here, the key question that remains unanswered is what long-term political effects does wartime disillusionment produce. Future research could examine whether disillusioned citizens become politically apathetic, align with opposition movements, or shift toward more radical nationalist alternatives. Additionally, studies could investigate how state propaganda adapts to sustain legitimacy in the face of mounting war casualties and whether these strategies successfully retain public loyalty.

Taken together, the findings from the three studies presented in this dissertation advance our understanding of political behavior under authoritarianism by highlighting how citizens respond to information about regime weakness, dissent, and war casualties. They demonstrate that while signals of regime fragility can influence public attitudes, their effects are highly contingent on context. Crucially, the studies reveal that disengagement from the regime does not necessarily translate into active opposition, and that the persistence of authoritarian rule is not merely a function of repression but also of how information is perceived and internalized by different segments of the population. Thus, the studies illuminate which groups within society are most susceptible to information about the regime's vulnerabilities and under what conditions this information is most impactful. While opposition-leaning voters remain difficult to mobilize through electoral or protest-related signals, regime supporters appear more sensitive to contextual cues, particularly when the information challenges their expectations. Ultimately, these findings underscore the contingent nature of political support under autocracy.

Overall, the findings highlight both the potential sources of fragility in authoritarian rule and the resilience of these regimes. While autocrats invest heavily in constructing an image of invincibility, their legitimacy often rests on unstable equilibria that can be disrupted by information shocks — whether electoral, protest-related, or wartime. However, disillusionment with the ruling party does not automatically translate into effective opposition mobilization. This underscores the need for further research on the mechanisms that transform disengagement into active resistance, as well as the role of institutional and social constraints in shaping political

responses to authoritarian weakness.

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Appendix A: Conjoint Survey Experiment

Logit regression results

The tables below report coefficient estimates from logistic regression models predicting candidate selection as a function of candidate characteristics. Coefficients represent log-odds of selection relative to the reference category for each categorical variable. Standard errors are clustered at the respondent level to account for within-respondent correlation.

Table A1: Regression Estimates of Candidate Selection Probability as a Function of SV endorsement

Variable	Model Specification	
	Logit	OLS (LPM)
Smart Voting Endorsement: Yes	1.905*** (0.111)	0.364*** (0.022)
Constant	-1.951*** (0.040)	0.124*** (0.004)
Pseudo R ² /R ²	0.115	0.134

*** p<0.01, ** p<0.05, * p<0.1. Clustered standard errors (by id) in parentheses.

Table A2: Logistic Regression Estimates of Candidate Selection Probability: Basic Model.

Variable	Coefficient
Constant	-2.196*** (0.154)
Smart Voting Endorsement: Yes	1.754*** (0.115)
Party: JUST RUSSIA	-0.111 (0.131)
Party: LDPR	-0.119 (0.132)
Party: UNITED RUSSIA	-0.952*** (0.178)
Party: Yabloko	0.392*** (0.136)
Occupation: Chief Medical Officer	0.393** (0.156)
Occupation: Civil and Environmental Activist	0.572*** (0.144)
Occupation: Member of Legislative Assembly	0.114 (0.146)
Occupation: Municipal Deputy	0.133 (0.145)
Occupation: School Principal	0.3** (0.147)
Occupation: Self-employed	-0.032 (0.147)
Occupation: University Lecturer	0.753*** (0.144)
Nationality: Russian	0.15* (0.088)
Gender: Male	-0.062 (0.074)
Observations	6960
Pseudo R	0.143
AIC	5953.6
BIC	6056.3

*** p<0.01, ** p<0.05, * p<0.1.

Table A3: Logistic Regression Estimates of Candidate Selection Probability: Party Ratings and Partisan Distance.

Variable	Coefficient
Constant	-1.908*** (0.064)
Smart Voting Endorsement: Yes	1.992*** (0.12)
Party Attitude: 2	0.073 (0.105)
Party Attitude: 3	0.309*** (0.087)
Party Attitude: 4	1.092*** (0.117)
Party Attitude: 5	2.162*** (0.258)
Party Attitude: NA	-0.141 (0.123)
Distance: 1	-0.578*** (0.101)
Distance: 2	-0.996*** (0.13)
Distance: 3	-1.23*** (0.176)
Distance: 4	-1.56*** (0.293)
Observations	6960
Pseudo R ²	0.199
AIC	5558.4
BIC	5633.7

Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A4: Logistic Regression Estimates of Candidate Selection Probability: Information Treatment.

Variable	Coefficient
Constant	-2.165*** (0.159)
Smart Voting Endorsement: Yes	1.664*** (0.16)
Image Vulnerability: 1	-0.059 (0.082)
Party: JUST RUSSIA	-0.112 (0.131)
Party: LDPR	-0.12 (0.132)
Party: UNITED RUSSIA	-0.953*** (0.178)
Party: Yabloko	0.393*** (0.136)
Occupation: Chief Medical Officer	0.395** (0.156)
Occupation: Civil and Environmental Activist	0.572*** (0.144)
Occupation: Member of Legislative Assembly	0.113 (0.146)
Occupation: Municipal Deputy	0.133 (0.145)
Occupation: School Principal	0.301** (0.146)
Occupation: Self-employed	-0.034 (0.147)
Occupation: University Lecturer	0.753*** (0.144)
Nationality: Russian	0.15* (0.088)
Gender: Male	-0.063 (0.075)
Interaction: Endorsement * Vulnerability	0.174 (0.227)

Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Politicians ratings

Here I present some descriptive statistics: how respondents rated politicians and, accordingly, how many of them were considered as pro-opposition and pro-regime, the distribution of age, education and income across these two groups, as well as how pro-opposition respondents answered direct questions about the Smart Voting project.

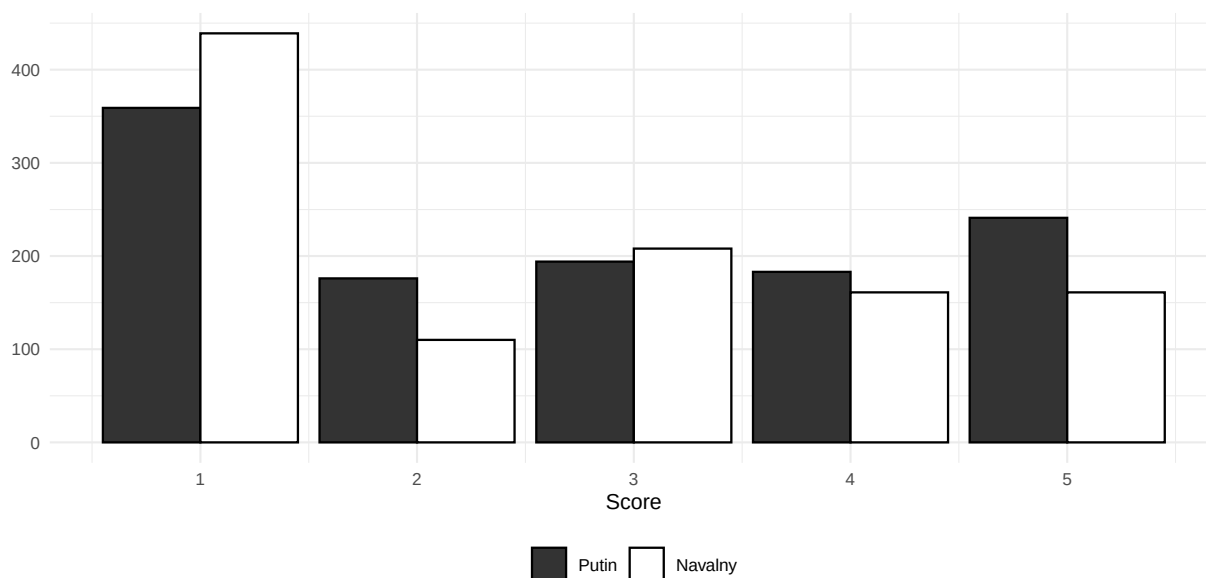


Figure A1

As we can see, the general sample is rather balanced in respect to the number of pro-opposition and pro-regime respondents: of 1205 respondents surveyed 549 were identified as pro-opposition and 656 as pro-regime.

As Figure A2 shows, the age of respondents from these two groups, we can see that the proportion of the supporters of the regime increases with the age group: among the pro-opposition respondents about 40% are between the age of 18 and 29, while in the other group only 28% fall in that age category. However, among those of 30-39 years old the proportions of the opposition minded respondents is only slightly larger than that of the pro-regime ones, and in the higher the age group gets, the larger the share of the sympathizers of the regime is.

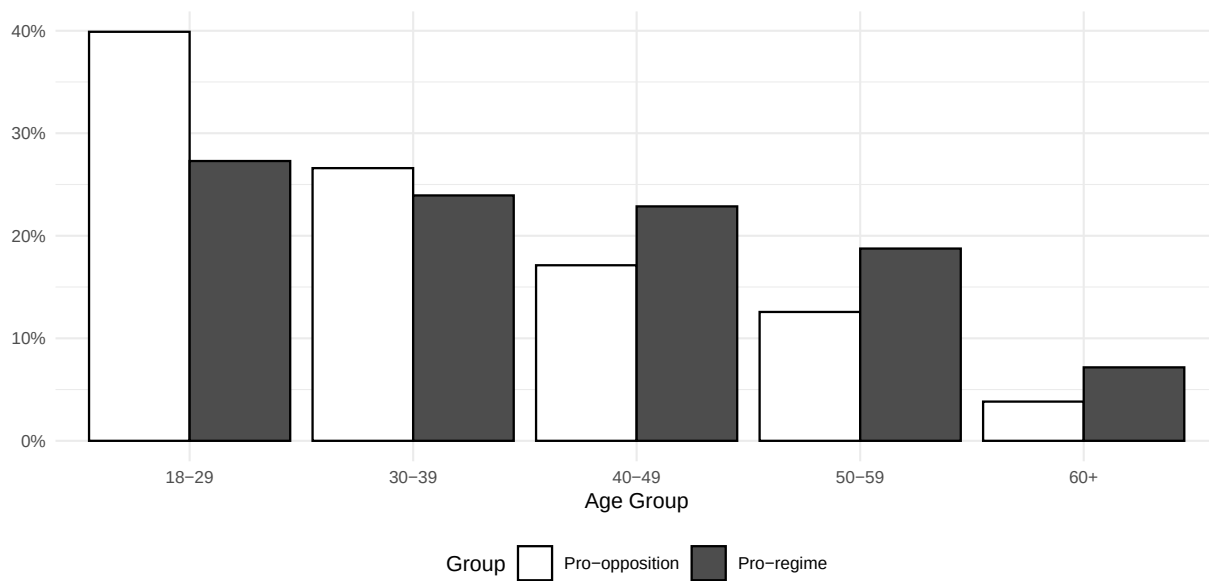


Figure A2

Education and income

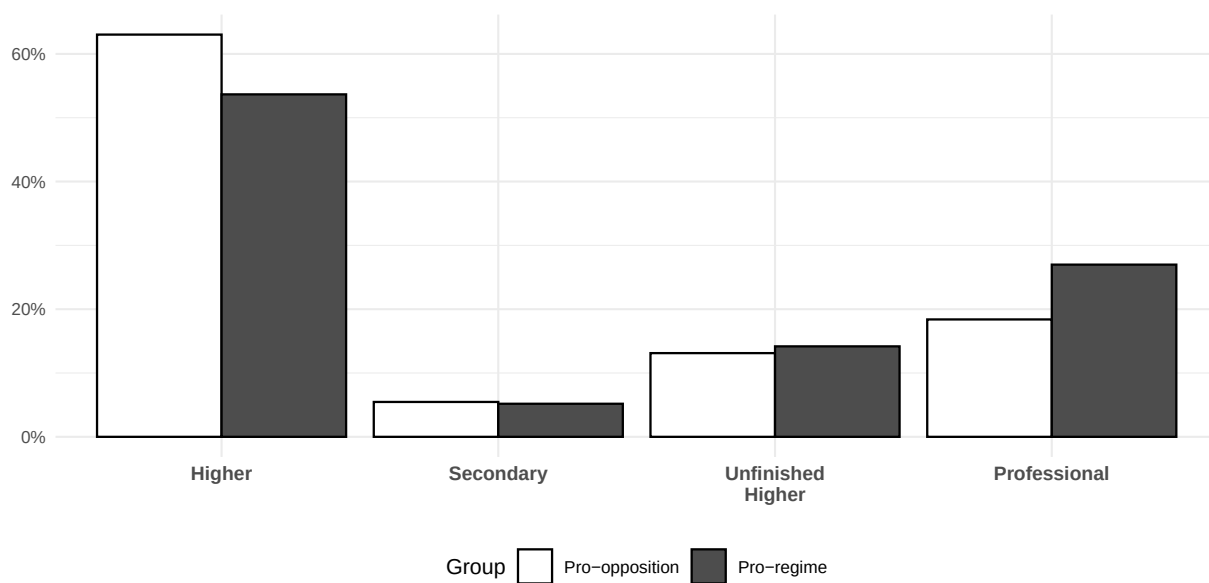


Figure A3

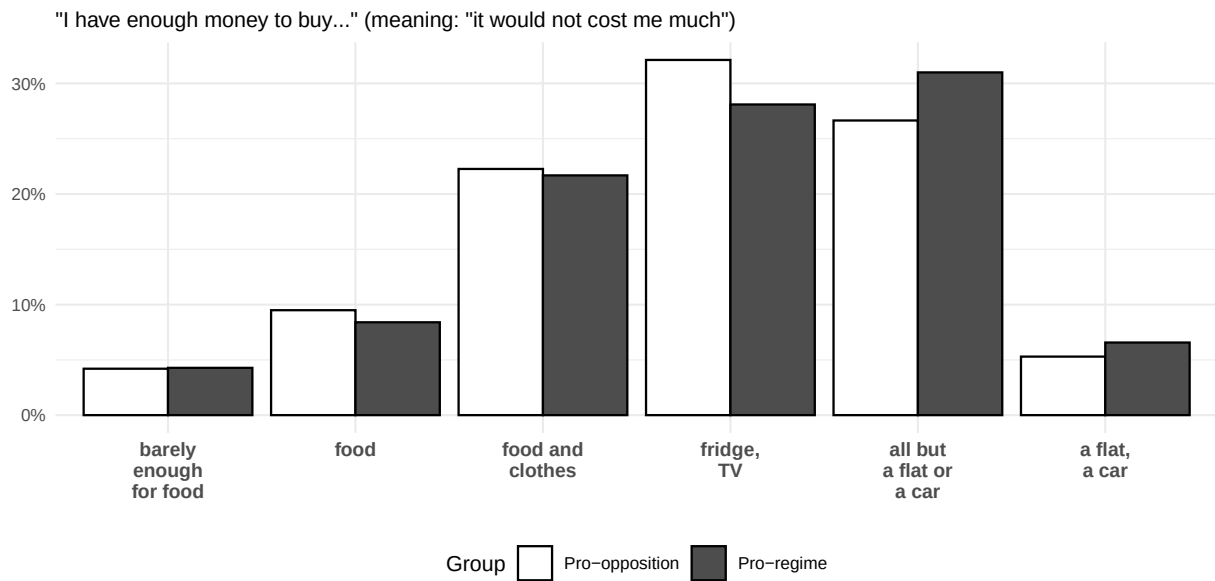


Figure A4

Surprisingly enough, the groups are not significantly different in respect to education and income of the respondents, as reported in Figure A3 and Figure A4. Even though on average the respondents classified as pro-opposition are more educated and seem to be more well-off financially, the differences are marginal. In fact, the share of “wealthy” respondents - those that can easily afford buying a flat or a car - is higher in the pro-regime group.

Election participation

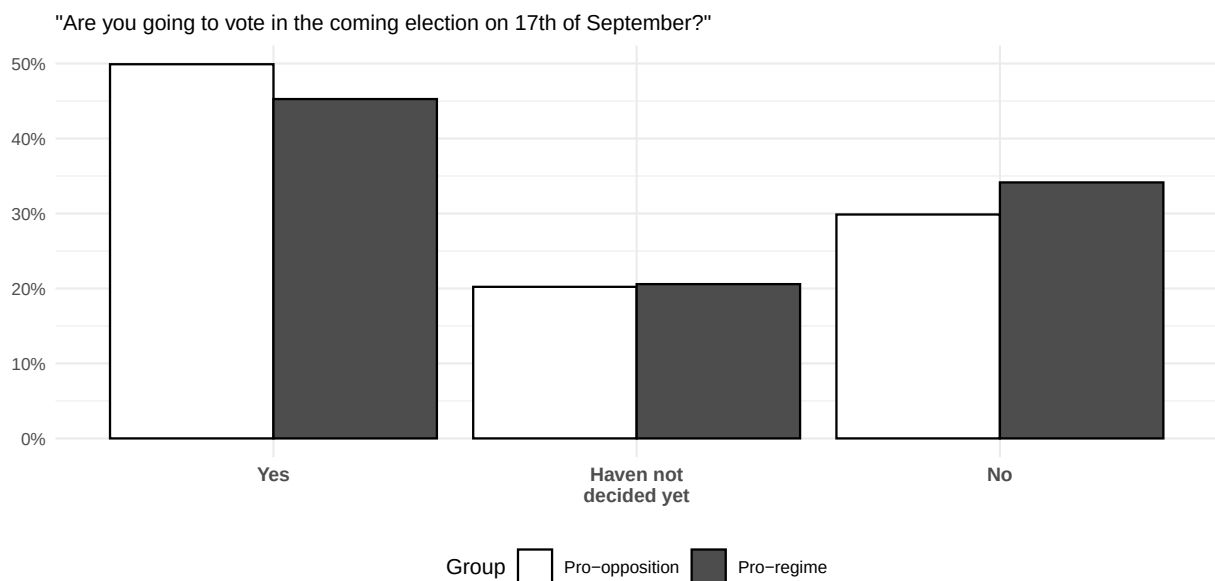


Figure A5

Turning to the electoral mobilization, we, yet again, see no drastic differences. As Figure A5 show, among the pro-opposition group, almost 50% percent expressed their willingness to vote in the coming election, while in the opposite group the respective number is 45%. The shares of those who had not decided whether to vote or not are roughly identical between the two groups, as reported in Figure A6. However, a larger proportion of the pro-opposition respondents reported not having decided which party or candidate they were going to vote for: its share equals to 41% percent in comparison to 31% in the pro-regime group.

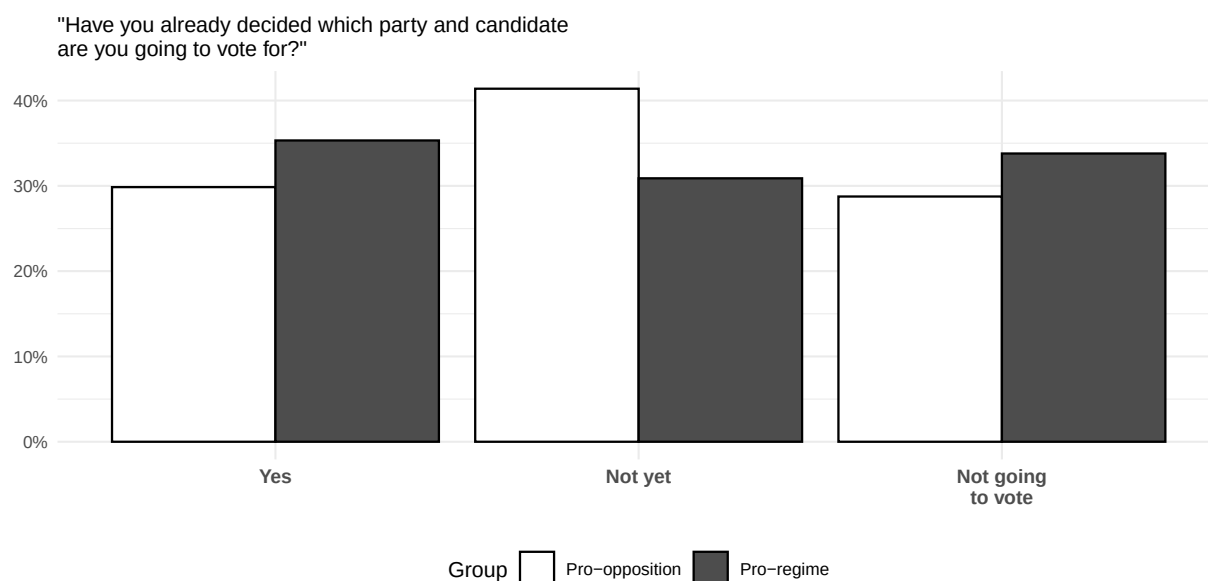


Figure A6

All in all, two groups are not significantly different from each other in respect to education, income or willingness to vote. However, it is important to stress that these number are not representative: they provide information only about the sample, not about the general population of Saint Petersburg.

Smart Voting and political parties

Having compared the pro-opposition and the pro-regime respondents, I now turn to the former group on its own. In contrast to pro-regime respondents, those identified as opposition minded were asked several additional questions, such as the attitude towards five main political parties, the knowledge about the Smart Voting project and the willingness to follow its recommendations on the election day.

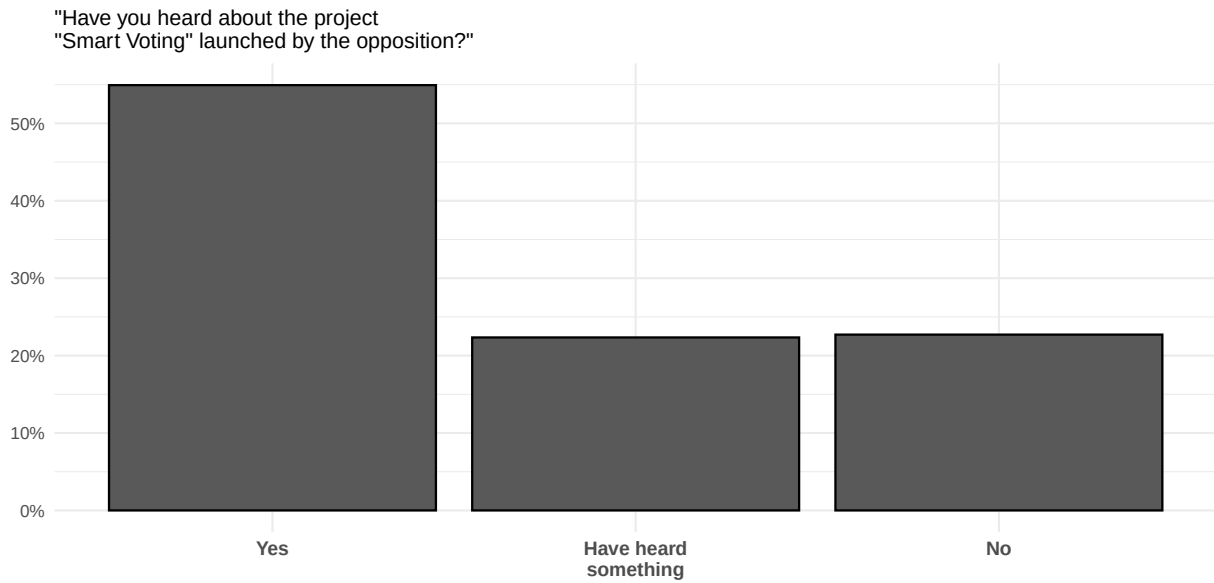


Figure A7

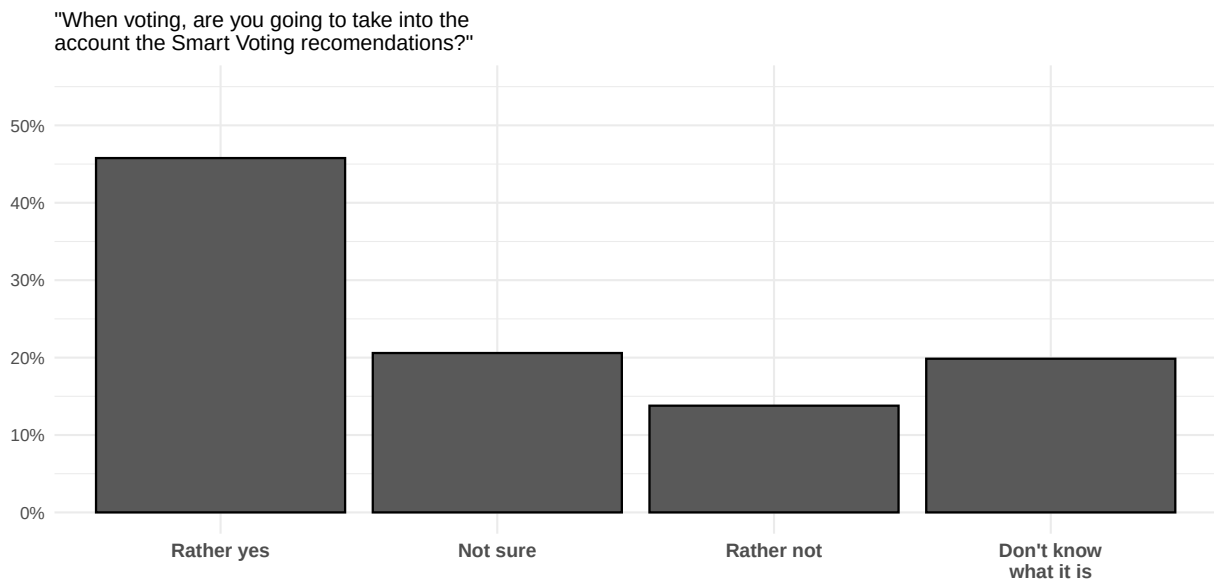


Figure A8

As shown in Figure A7, among the opposition minded respondents from my sample, 55% knew about the Smart Voting project and 22% reported that they “heard something” about it. Next, as Figure A8 shows, 46% claimed that they were likely to follow the project’s recommendations regarding the vote choice, 21% were undecided and 14% answered “rather not”. It is important to note that these questions, as well as the question about the parties, respondents answered on their own, without an enumerator seeing their choices. Therefore, these data should reflect the actual distribution of preferences rather accurately, and issues associated with the social desirability bias (or, to put it differently, preferences falsification) can be partially mitigated.

Figure A9 shows the distribution of pro-opposition respondents' ratings towards five main political parties - the United Russia, the Just Russia, Communist Party of Russian Federation, the Liberal Democratic Party of Russia and Yabloko - the results are rather unsurprising. The most well-rated party is Yabloko - the only one that can be considered truly oppositional and the only one not present in the State Duma. Similarly, the United Russia - the dominant party with a super majority in the parliament - is rated the lowest. However, in general the parties are given rather low scores: the mode of the score for all parties except Yabloko is 1. Yet even Yabloko was rated less than 3 points by 43% of respondents. All of this is consistent with the observation that Russians in general place little trust in political parties.

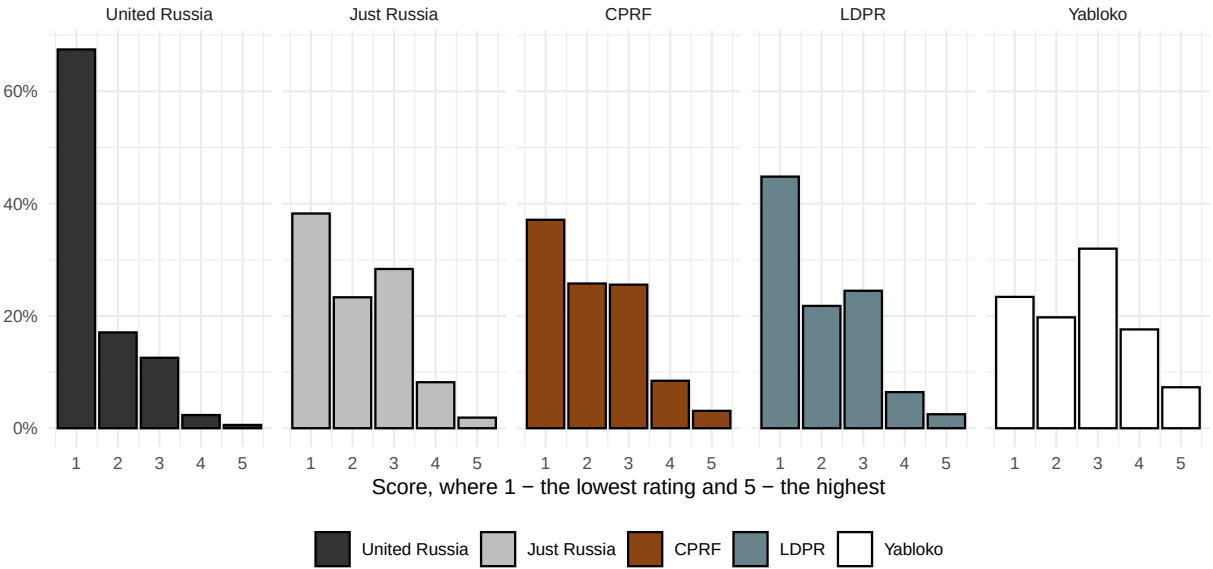


Figure A9

Smart Voting and image of vulnerability treatment

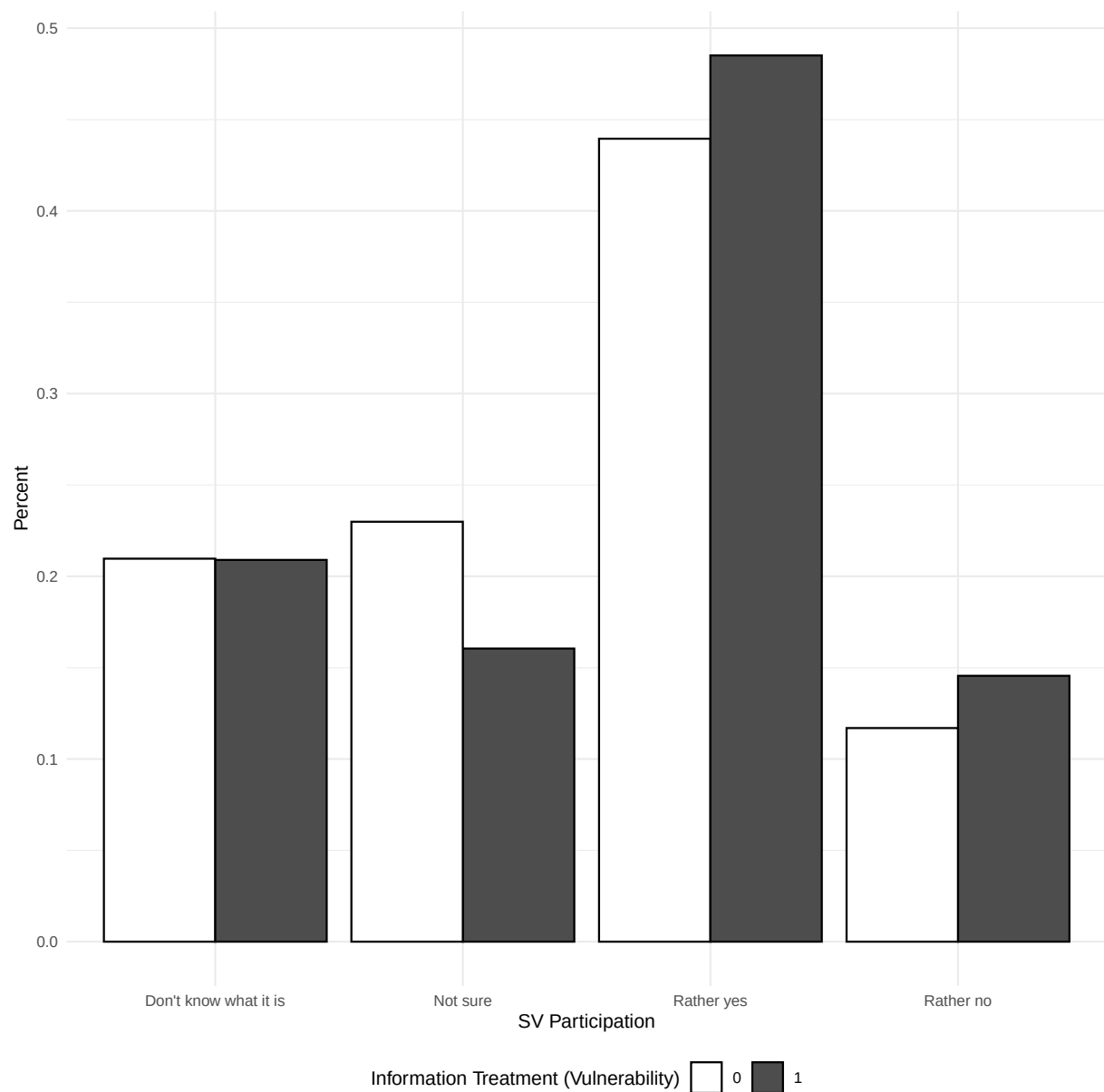
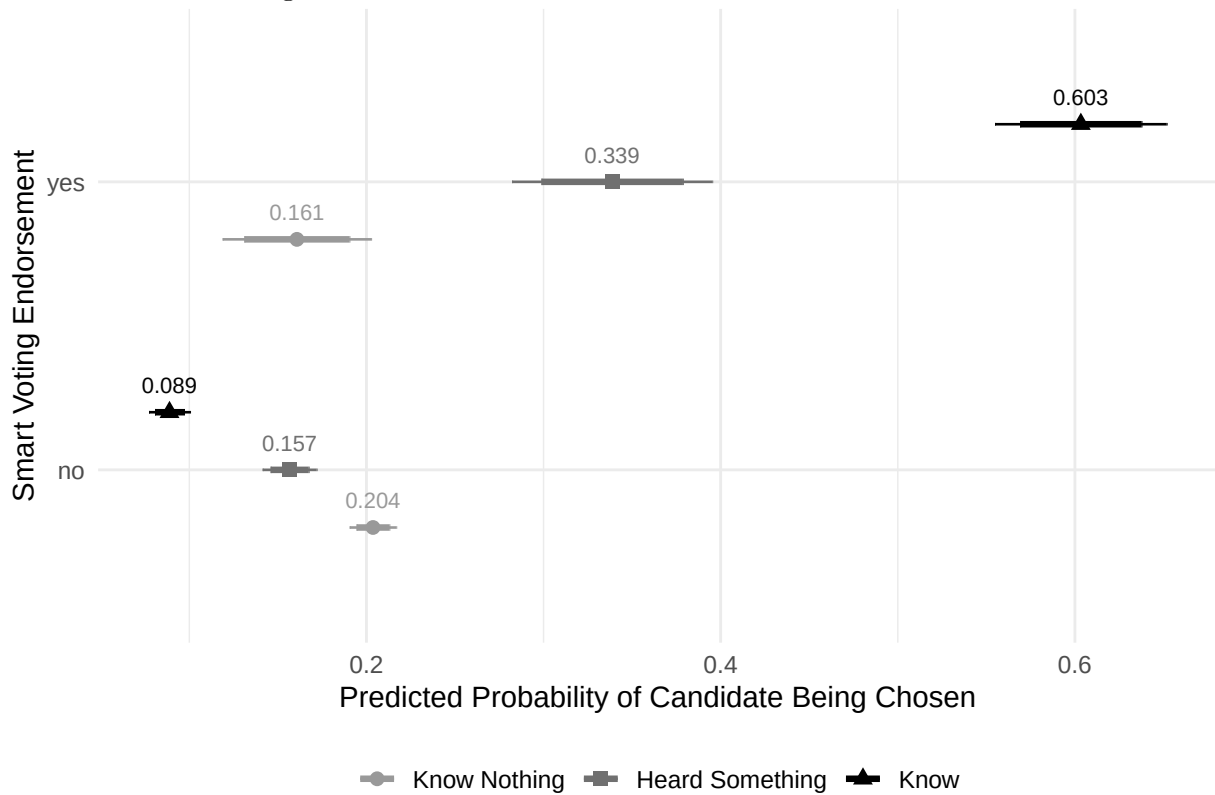


Figure A10

Effect of Smart Voting by Smart Voting Knowledge

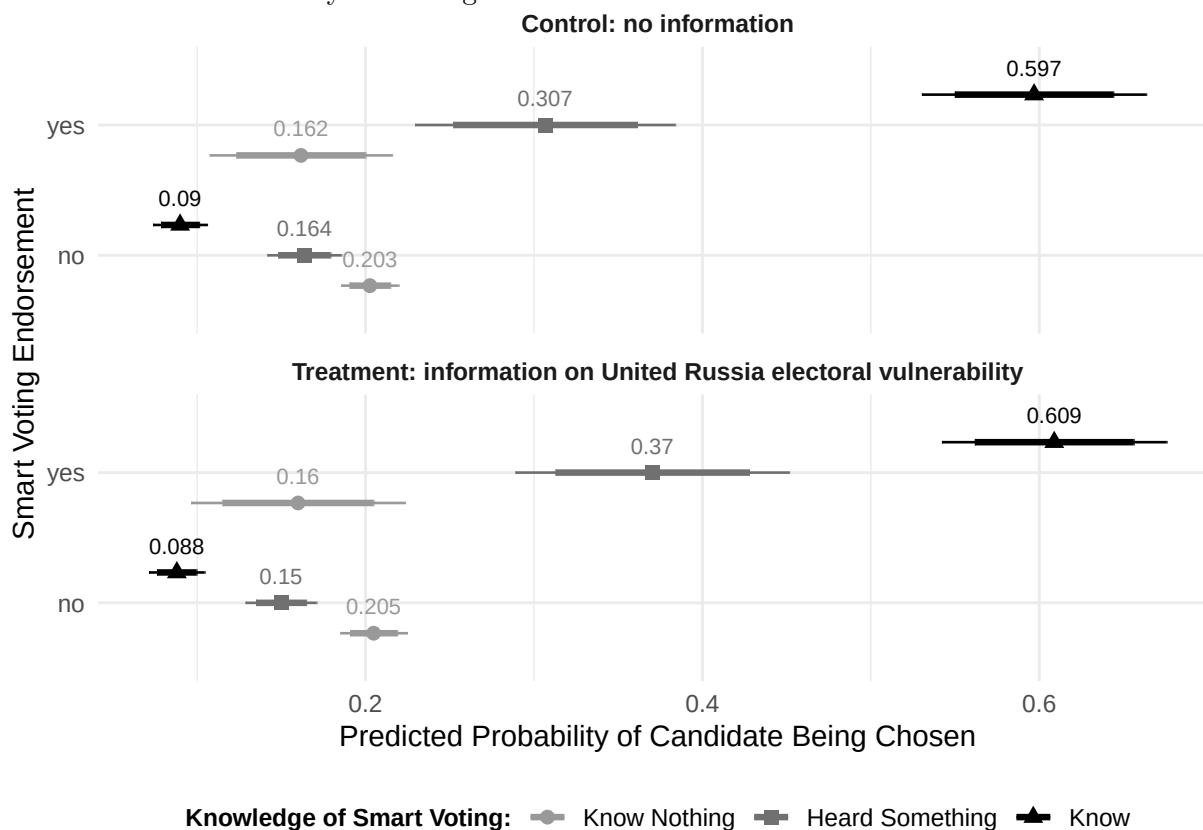
Figure A11: Predicted Probability of Candidate Selection by Smart Voting Endorsement and Knowledge Level



- (a) The figure presents predicted probabilities of candidate selection derived from logistic regression, averaged across experimental conditions, with standard errors clustered by respondent. Points represent predicted probabilities; horizontal bars indicate 95% confidence intervals (thin bars) and 83.4% confidence intervals (thick bars). Probabilities are shown for different levels of prior knowledge about Smart Voting.

Effect of Vulnerability Treatment by Smart Voting Knowledge

Figure A12: Predicted Probability of Candidate Selection by Smart Voting and Information Treatment by Knowledge



- (a) The figure presents predicted probabilities of candidate selection derived from logistic regression with standard errors clustered by respondent. Points represent predicted probabilities; horizontal bars indicate 95% confidence intervals (thin bars) and 83.4% confidence intervals (thick bars). The dashed line marks the baseline probability of selection (0.2). Confidence intervals not intersecting this line indicate statistically significant effects. Overlapping 83.4% confidence intervals between groups indicate the absence of statistically significant differences in the effect of Smart Voting endorsement across experimental conditions at the 5% level.

Survey questionnaire

A Short Description

What: A survey containing a conjoint experiment + a list experiment.

How: The survey will be delivered offline, using electronic tablets; the survey is scheduled to be conducted on September 10-17, 2021.

Aim: To understand how effective the Smart Voting project is in coordinating opposition voters against the incumbent party and what moderates its effect.

The Approach Script

Hi! My name is %NAME%, I am a student from %UNIVERSITY__NAME%. We are conducting research on voting behavior of citizens of Saint-Petersburg. Could you please answer a few questions? It will not take more than 7 minutes.

The Consent Form In order to proceed, please review the consent form.

This is a research project being conducted by Ivan Fomichev, a PhD student at the European University at Saint-Petersburg and the European University Institute. We are not working for any candidate in the following elections; this is a purely academic project.

Your participation in this survey is voluntary, and we do not provide any benefits to the participants. You may refuse to take part in the research or exit the survey at any time. You are free to decline to answer any particular question for any reason.

Since this is a survey about electoral preferences, we will ask about your opinion on political issues, which might cause discomfort. You can always refuse to answer or abort the survey at any moment.

We guarantee confidentiality and anonymity. We do not collect personal data such as phone numbers, names, or any other identifying information. No video or audio recordings are made.

We do not anticipate any risks for participants beyond those encountered in daily life.

If you have any questions or complaints about the survey, you can contact the principal investigator, Ivan Fomichev, by calling +7-981-748-45-12 or via email at ifomichev@eu.spb.ru.

CONSENT: Do you agree to participate in the survey?

☐ I consent

☐ I do not consent

Quota Block

1. **Gender** (filled by the enumerator)

2. **District residency:**

☐ I live in this district (continue the survey)

☐ I live somewhere else (stop the survey)

3. **Age:** (respondent states their age)

General Questions about Politics and Elections

4. **Have you heard about the upcoming September elections for the State Duma and the Legislative Assembly of St. Petersburg?**

☐ No, I have not

☐ I heard something

☐ Yes, I am aware of that

☐ [Refuse to answer]

5. **Are you going to vote in this election?**

☐ Yes, I am

☐ I have not decided yet

☐ No, I am not

☐ [Refuse to answer]

6. **If yes, have you decided on a party or candidate?**

☐ Yes, I know both the candidate and the party

☐ I decided only about the party

☐ No, I have not decided

☐ [Refuse to answer]

Politician Rating

7. **Please rate the following politicians on a five-point scale:**

- Alexander Beglov (Governor of Saint-Petersburg)
- Vladimir Putin (President of Russia)
- Grigori Yavlinsky (Founder of “Yabloko” party) + “I don’t know him”
- Alexei Navalny (Opposition politician) + “I don’t know him”

Candidate Choice Experiment [Respondent were asked to fill the rest of the survey themselves so that the enumerator doesn’t see the answers]

[Treatment] 8.1. The candidates' performance varied significantly in the last election: in some districts United Russia candidates won by a large margin, while in others the elections were very close. For example, in the 15th single-member district, where we are now, the United Russia candidate Boris Ivchenko attracted, 20% of the vote, while his closest rival - 18%. Below there is a list of fictional candidates with several characteristics: gender, occupation etc. If you were to vote in the following election, which one do you think you would choose?

- Ivan Ivanov (former school principal, nominated by Just Russia, supported by Smart Voting)
 - Daria Smirnova (environmental activist, nominated by Yabloko)
 - Dmitry Dmitriev (CEO of a private company, nominated by United Russia)
 - Anna Aleksandrova (householder, nominated by LDPR)
 - Piotr Petrov (chief medical officer, nominated by CPRF)
- ☐ I wouldn't vote for any

[Control Group] 8.2. I have a list of fictional candidates with some characteristics: gender, party, occupation etc. Had these candidates been real, which one do you think you would have chosen? *(same list as above)*

List Experiment

9. How many of the following factors affect your voting choice? *(do not specify which ones)*

- Food price inflation
- Candidates' TV debate performance
- Candidates' positions on COVID-19 restrictions
- *(treatment only)* Smart Voting recommendations

10. Please rate the following political parties on a five-point scale:

- United Russia
- A Just Russia — Patriots — For Truth
- Yabloko

- CPRF
- LDPR

Smart Voting Awareness

11. **Have you heard about the Smart Voting project?**

- ☐ Yes, I know its purpose
- ☐ I heard something but am unsure
- ☐ No, I have not

12. **If yes, will you consider its recommendations?**

- ☐ Yes, most likely
- ☐ Maybe
- ☐ No, highly unlikely

Demographic Block

13. **Highest level of education completed:**

- ☐ Secondary
- ☐ Upper-secondary
- ☐ Short-cycle tertiary
- ☐ University degree
- ☐ PhD

14. **Household financial well-being**

Please select the option that best describes your financial situation:

- ☐ We have barely enough money for food.
- ☐ We can buy food, but buying clothes can be more problematic.
- ☐ We can buy food and clothes without difficulty, but need to save for more serious purchases.

- ☐ We can buy a fridge or a TV without much effort, but buying more expensive items requires planning.
- ☐ We can buy almost anything except a flat or a car.
- ☐ We can buy a flat or a car without difficulty.

Thank you for your time! Goodbye!

Appendix B: Spatial Effects

Covariate table for preserved (stable) and excluded (unstable) clusters

Variable	Statistic	Units	
		Unstable Unit	Stable Unit
ER_share_2021	Mean	19.0435	19.53371
	SE	0.1492734	0.1475779
area_km2	Mean	2.114039	2.109732
	SE	0.02400726	0.02304287
mean_night_lights	Mean	42.80612	37.66142
	SE	0.3752388	0.3833603
median_price_m2	Mean	119673.3	86320.58
	SE	883.2674	531.2641
voters_density	Mean	6613.991	6380.13
	SE	218.7334	128.3555
within_city_price_m2	Mean	0.9650313	0.9528
	SE	0.00191748	0.001875912

Average change in the number of registered voters for preserved (stable) and excluded (unstable) clusters

Variable	Statistic	Units	
		Unstable Unit	Stable Unit
voters_difference	Mean	2681.44583	365.891569
	SE	43.03486	4.083685

Buffer clusters in the control group

Table B1: Estimated Difference Between Treated and Untreated Units in the Turnout Change (2021 vs. 2016 Elections), with Buffer Clusters in the Control Group

	Treatment					
	500m		1000m		Relative	
All Protests						
Estimate	-1.019**	-0.76	-1.373***	-1.194**	-1.345**	-1.143*
SE	(0.489)	(0.522)	(0.395)	(0.465)	(0.6)	(0.617)
Obs.	10200	10086	10200	10086	10200	10086
Pension Reform						
Estimate	-0.571	-0.241	-1.136***	-0.906**	-0.758	-0.466
SE	(0.507)	(0.536)	(0.423)	(0.426)	(0.587)	(0.614)
Obs.	10018	9904	10018	9904	10018	9904
Anti-Corruption						
Estimate	-1.405**	-1.318**	-1.604***	-1.512***	-2.409***	-2.58***
SE	(0.678)	(0.648)	(0.448)	(0.505)	(0.908)	(0.883)
Obs.	9638	9524	9638	9524	9638	9524
Covariates	No	Yes	No	Yes	No	Yes

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B2: Estimated Difference Between Treated and Untreated Units in the UR Vote Share Change (2021 vs. 2016 Elections), with Buffer Clusters in the Control Group

	Treatment					
	500m		1000m		Relative	
All Protests						
Estimate	-1.099**	-0.946**	-1.209***	-1.049***	-1.233**	-1.113**
SE	(0.438)	(0.447)	(0.347)	(0.37)	(0.49)	(0.55)
Obs.	10200	10086	10200	10086	10200	10086
Pension Reform						
Estimate	-0.694	-0.489	-0.978***	-0.783**	-0.669	-0.484
SE	(0.45)	(0.47)	(0.349)	(0.397)	(0.53)	(0.549)
Obs.	10018	9904	10018	9904	10018	9904
Anti-Corruption						
Estimate	-1.585***	-1.521***	-1.433***	-1.317***	-2.296***	-2.373***
SE	(0.534)	(0.53)	(0.401)	(0.427)	(0.735)	(0.766)
Obs.	9638	9524	9638	9524	9638	9524
Covariates	No	Yes	No	Yes	No	Yes

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B3: Estimated Difference Between Treated and Untreated Units in the Total Opposition Vote Share Change (2021 vs. 2016 Elections), with Buffer Clusters in the Control Group

		Treatment				
		500m		1000m		Relative
All Protests						
	0.08	0.186	-0.165	-0.145	-0.113	-0.03
Estimate						
SE	(0.205)	(0.213)	(0.171)	(0.177)	(0.22)	(0.242)
Obs.	10200	10086	10200	10086	10200	10086
Pension Reform						
	0.124	0.248	-0.157	-0.123	-0.088	0.019
Estimate						
SE	(0.202)	(0.221)	(0.172)	(0.174)	(0.236)	(0.255)
Obs.	10018	9904	10018	9904	10018	9904
Anti-Corruption						
	0.179	0.203	-0.171	-0.194	-0.113	-0.207
Estimate						
SE	(0.242)	(0.267)	(0.179)	(0.21)	(0.321)	(0.32)
Obs.	9638	9524	9638	9524	9638	9524
Covariates	No	Yes	No	Yes	No	Yes

Note:

Standard errors are clustered on the level of the treatment assignment and reported in parentheses.

Calculated using bootstrap method with 1000 iterations

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Enlarged buffer and restricted sample estimation

Table B4 reports results from robustness checks designed to address concerns about potential spillover effects between treated and control clusters. In addition to the original specification—where clusters within 500 meters of a protest site are treated, adjacent clusters are excluded (buffer), and remaining clusters serve as the control group—I estimate models using an enlarged buffer. This alternative excludes from the control group all clusters located within 1 km of protest sites and their immediate neighbors. I also restrict the sample in some specifications to include only cities where protests took place, ensuring comparisons are made solely within protest-affected urban contexts.

The table presents results for three main outcomes: turnout, vote share for United Russia (UR), and vote share for opposition parties. Across specifications, the negative effect of protest proximity on turnout and UR vote share remains robust, with slightly larger (in absolute terms) and more precise estimates under the enlarged buffer specification. No significant effect is found

on the opposition vote share in any model.

Overall, these robustness checks confirm that the main findings are not driven by spillover bias due to the construction of the control group.

Table B4: Effect of Protest Proximity on Electoral Outcomes using Varying Buffer Zones and Samples

[H]								
Outcome	Buffer	Sample	Estimate	Std. Error	95% CI	n	Covariates	Type
Turnout	original	original	-1.122**	0.57	(-2.23, -0.01)	8264	Yes	All protests
Turnout	enlarged	original	-1.534***	0.58	(-2.67, -0.4)	7428	Yes	All protests
Turnout	enlarged	restricted	-1.468**	0.65	(-2.75, -0.19)	5322	Yes	All protests
Share UR	original	original	-1.133**	0.48	(-2.07, -0.19)	8264	Yes	All protests
Share UR	enlarged	original	-1.398***	0.49	(-2.36, -0.44)	7428	Yes	All protests
Share UR	enlarged	restricted	-1.491***	0.54	(-2.54, -0.44)	5322	Yes	All protests
Share Opp.	original	original	0.012	0.23	(-0.43, 0.46)	8264	Yes	All protests
Share Opp.	enlarged	original	-0.136	0.24	(-0.6, 0.33)	7428	Yes	All protests
Share Opp.	enlarged	restricted	0.023	0.27	(-0.51, 0.56)	5322	Yes	All protests

Regions with above and below average UR share in 2016

Table B5: The List of Regions with UR Vote Share Below or Above Nation Average

Below Average	Above Average
Astrakhan Oblast	Bashkortostan
Sverdlovsk Oblast	Kalmykia
Chelyabinsk Oblast	Kemerovo Oblast
Irkutsk Oblast	Tatarstan
Saint Petersburg	Volgograd Oblast
Leningrad Oblast	Rostov Oblast
Krasnoyarsk Krai	Nizhny Novgorod Oblast
Moscow Oblast	Stavropol Krai
Moscow	North Ossetia–Alania
Tver Oblast	Mordovia
Perm Krai	Tula Oblast
Yamalo-Nenets Autonomous Okrug	Lipetsk Oblast
Pskov Oblast	Saratov Oblast
Vologda Oblast	Tuva Republic

Primorsky Krai	Kabardino-Balkaria
Oryol Oblast	Chuvashia
Karelia	Dagestan
Kaliningrad Oblast	Tambov Oblast
Vladimir Oblast	Bryansk Oblast
Yaroslavl Oblast	Karachay-Cherkessia
Altai Krai	
Orenburg Oblast	
Murmansk Oblast	
Amur Oblast	
Khanty-Mansi Autonomous Okrug — Yugra	
Belgorod Oblast	
Mari El	
Tomsk Oblast	
Khabarovsk Krai	
Altai Republic	
Kaluga Oblast	
Jewish Autonomous Oblast	
Sakhalin Oblast	
Arkhangelsk Oblast	
Kamchatka Krai	
Republic of Khakassia	
Smolensk Oblast	
Kursk Oblast	
Kirov Oblast	
Samara Oblast	
Penza Oblast	
Kurgan Oblast	
Republic of Sakha (Yakutia)	
Republic of Komi	

Ulyanovsk Oblast

Tyumen Oblast

Voronezh Oblast

Novgorod Oblast

Adygea

Udmurtia

Ivanovo Oblast

Kostroma Oblast

Novosibirsk Oblast

Zabaykalsky Krai

Ryazan Oblast

Appendix C: War Fatalities and Social Media

Demographics of school group subscribers

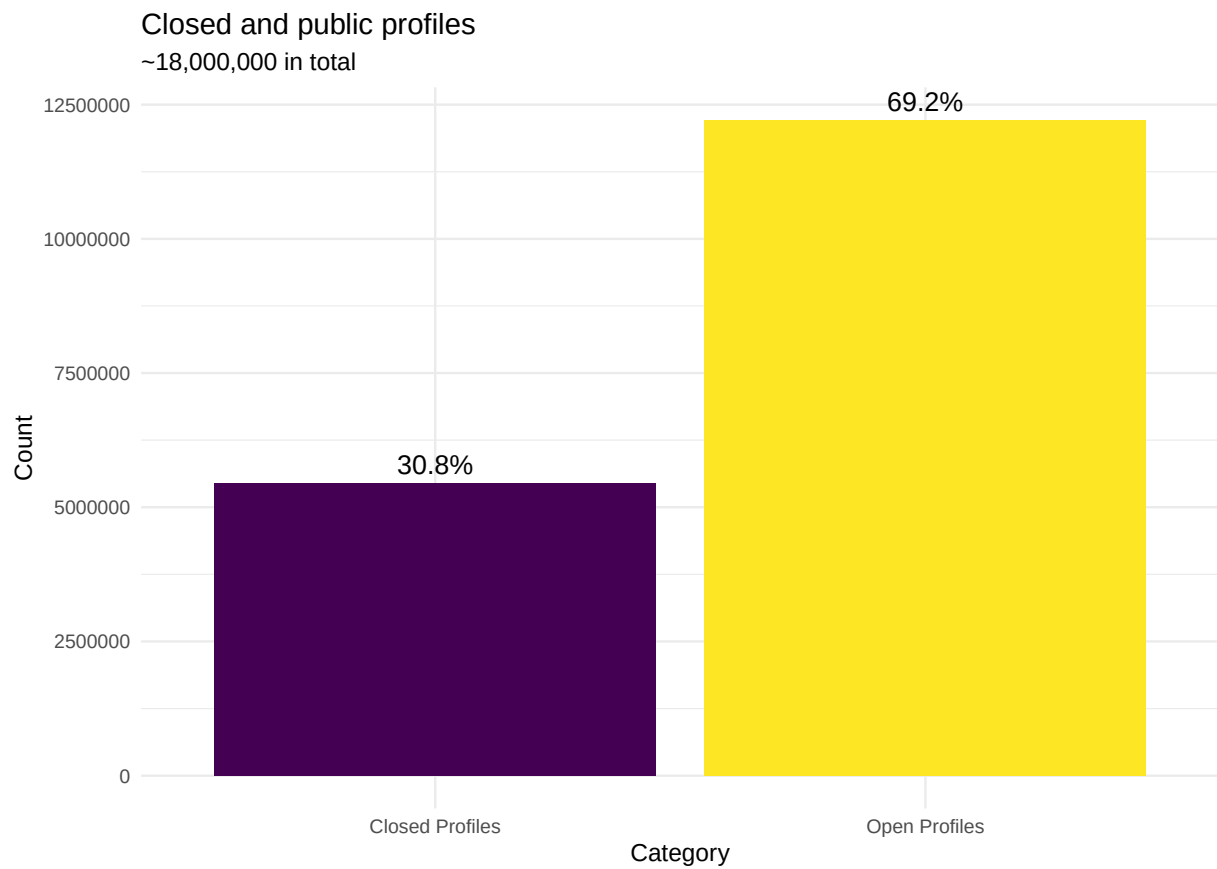


Figure C1: The distribution of closed and open profiles among school groups' subscribers

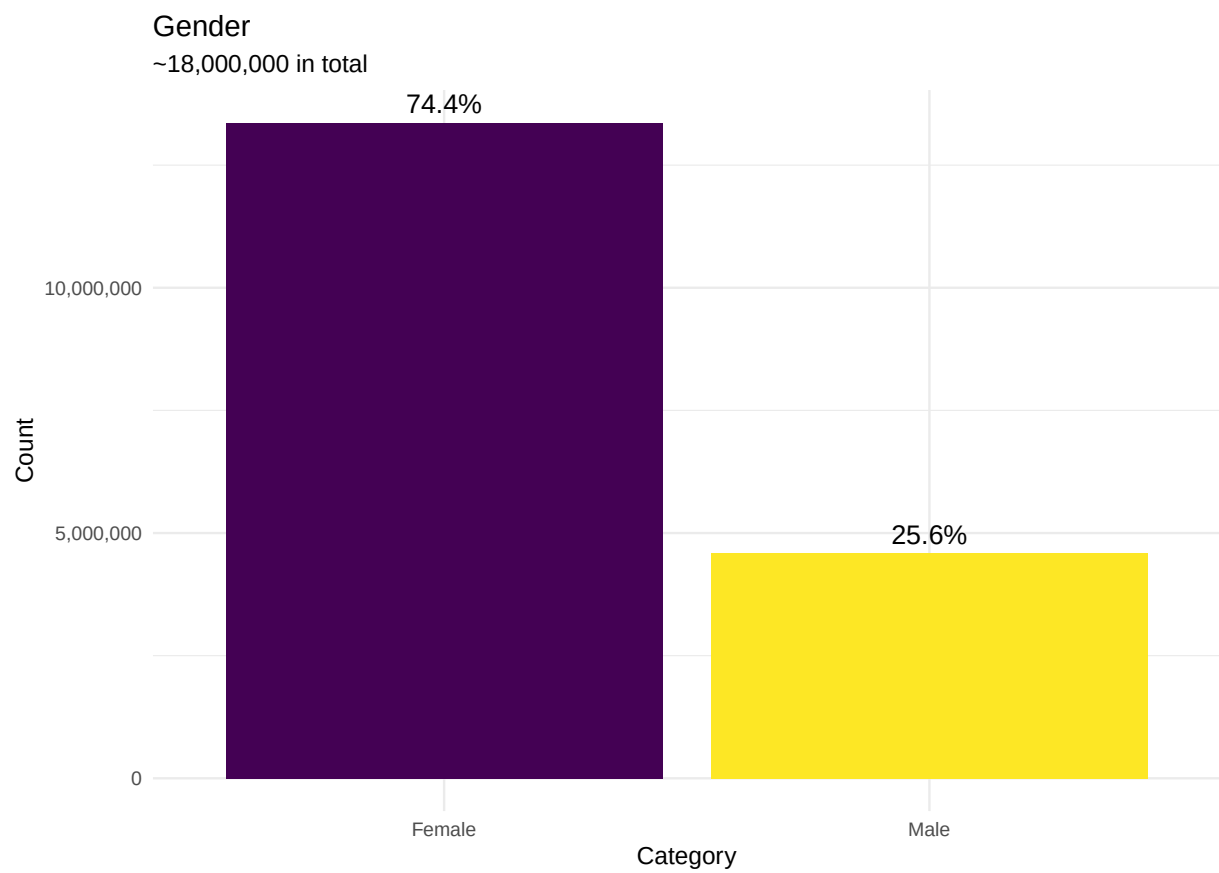


Figure C2: Gender distribution among school groups' subscribers

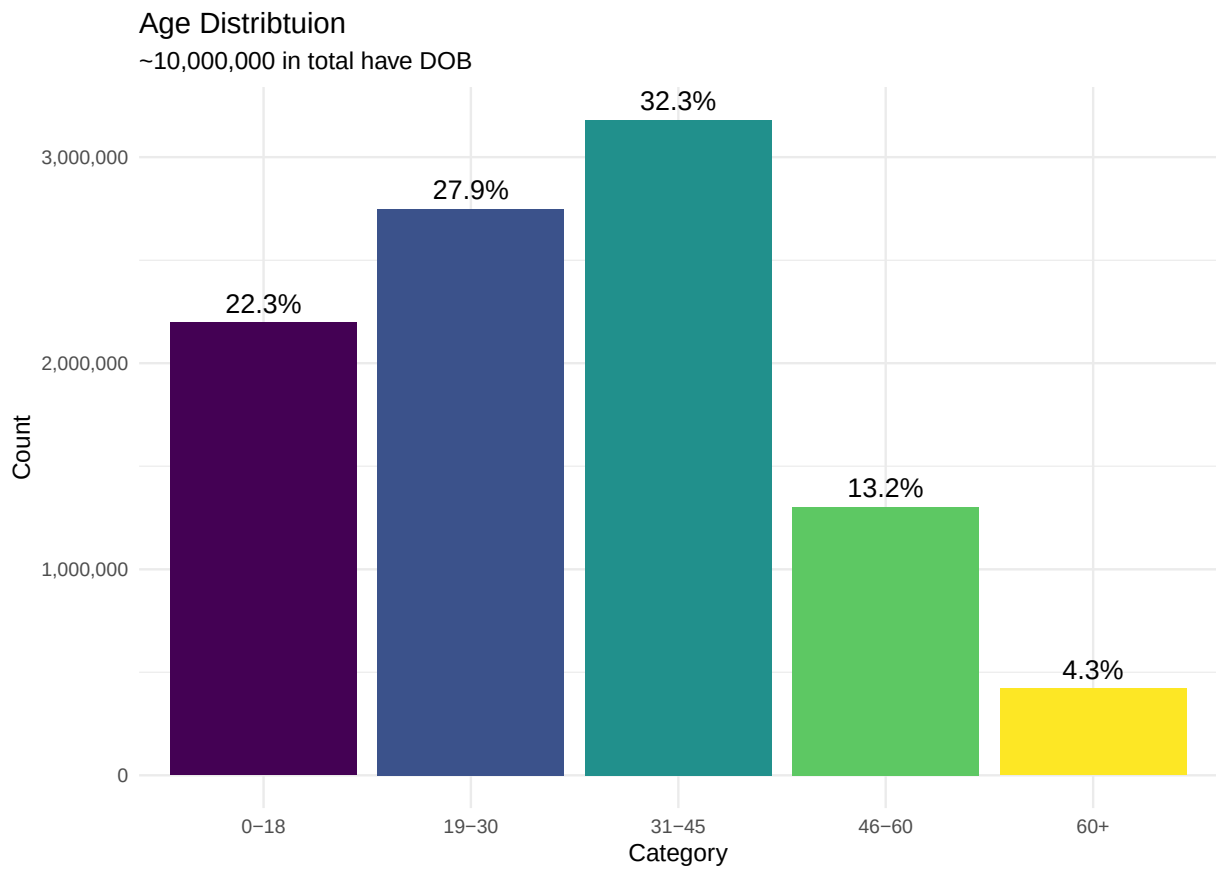


Figure C3: Age distribution among school groups' subscribers

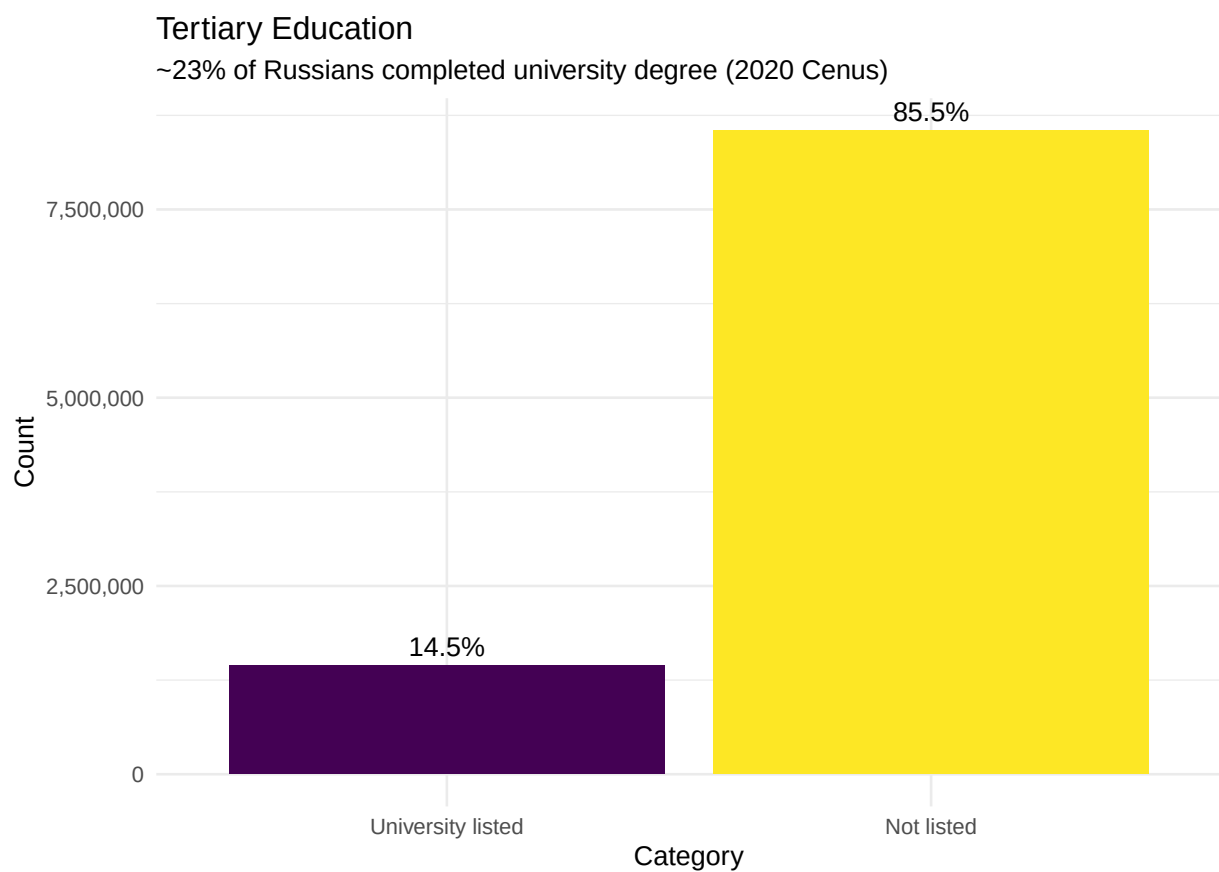


Figure C4: Percentage of subscribers with listed tertiary education

Example posts in schools groups



Figure C5: Neutral Post



OGBOU "Biryuchenskaya Secondary School" | Official group

12 Jun at 7:08 · Government organization

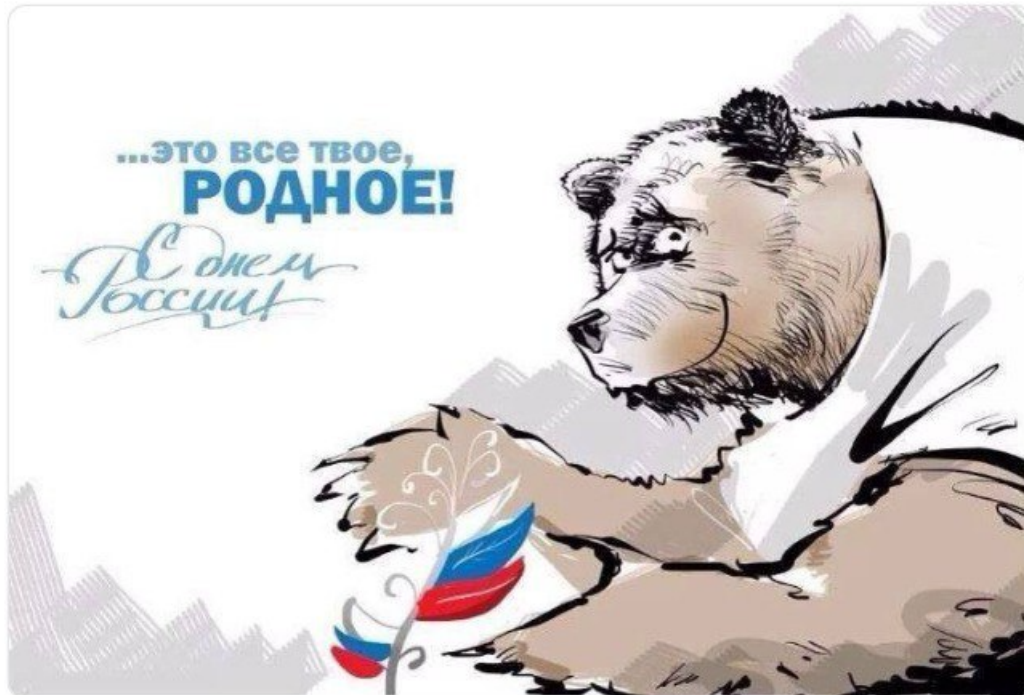


Today, June 12, we celebrate the day of our country, the day of our Russia!

What do we call Motherland?

Every person has a homeland. This is the country where he was born. We are citizens of Russia, our Motherland is the Russian Federation.

[Show more](#)



25



809

Figure C6: Patriotic Post

The President of the Russian Federation announced the launch of a new national project "Family". He announced this while delivering a message to the Federal Assembly.

The national program will provide financial support to regions with low birth rates.

#OGBOUBiryuchenskayaSOSOofficial group

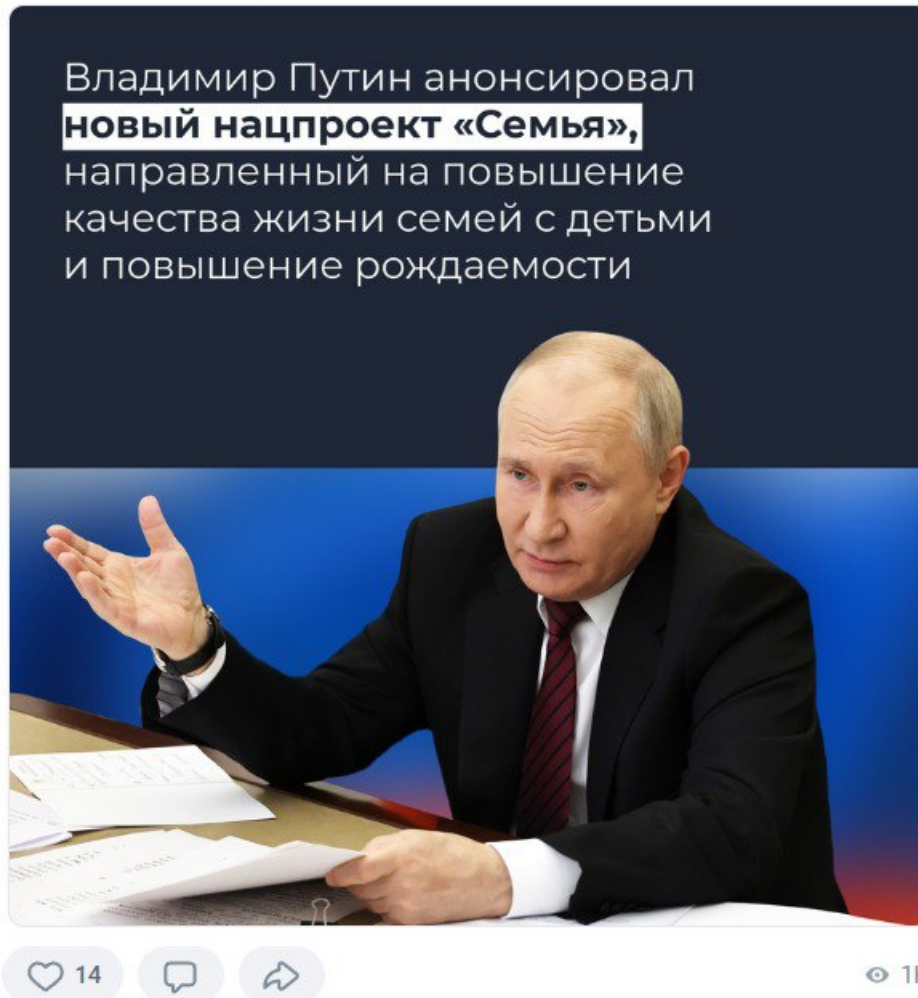


Figure C7: President Post



Figure C8: Special Military Operation Post

Keyword and Zeroshot comparison

Note: Number of posts in municipality-month identified by keywords on horizontal axis. Number of posts weighted by the zero-shot quasi-probabilities on the vertical axis. Zero-shot label for the *National symbols* is *Patriotism*.

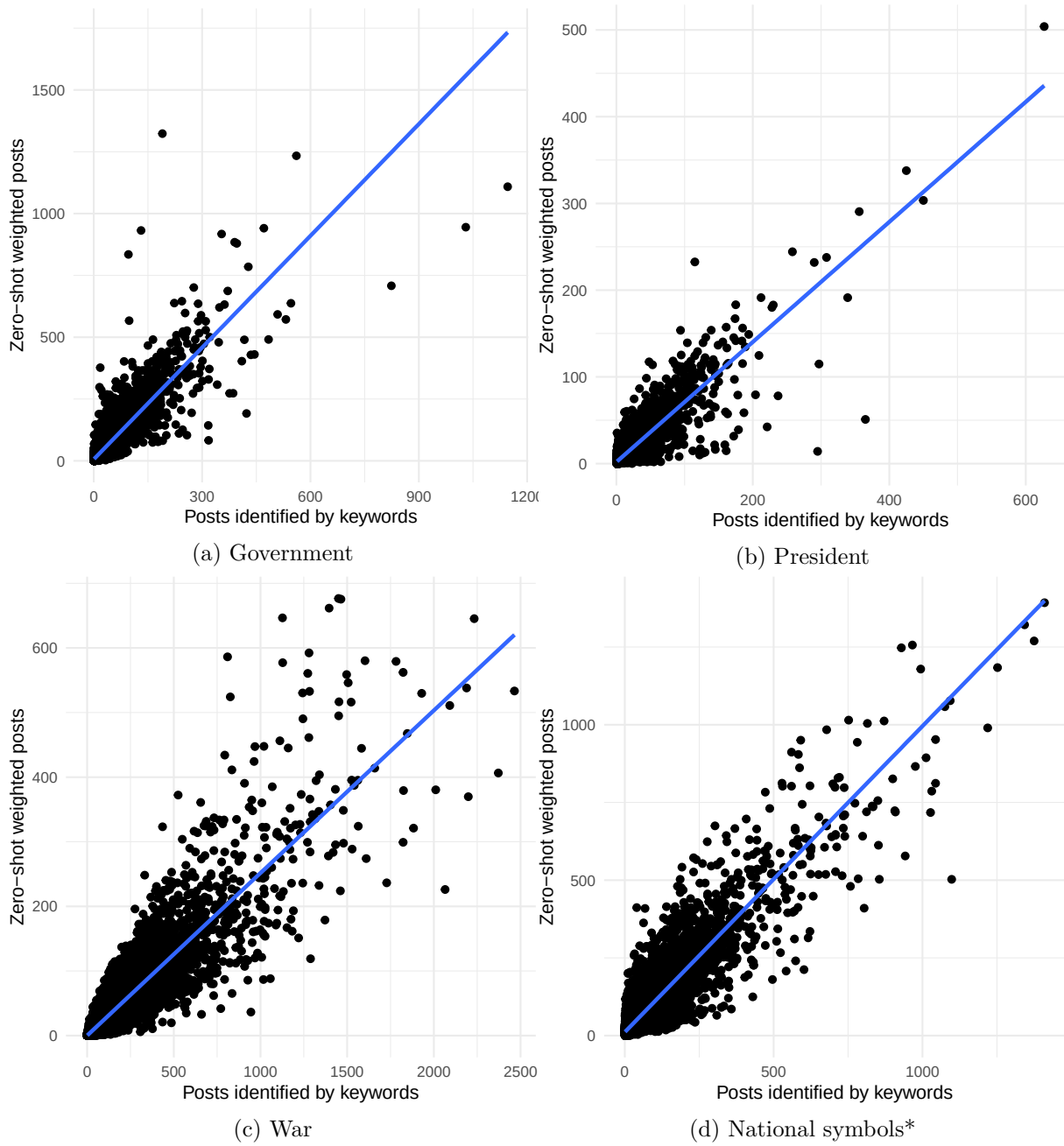


Figure C9: Posts identified by keywords and zero-shot

War fatality reports

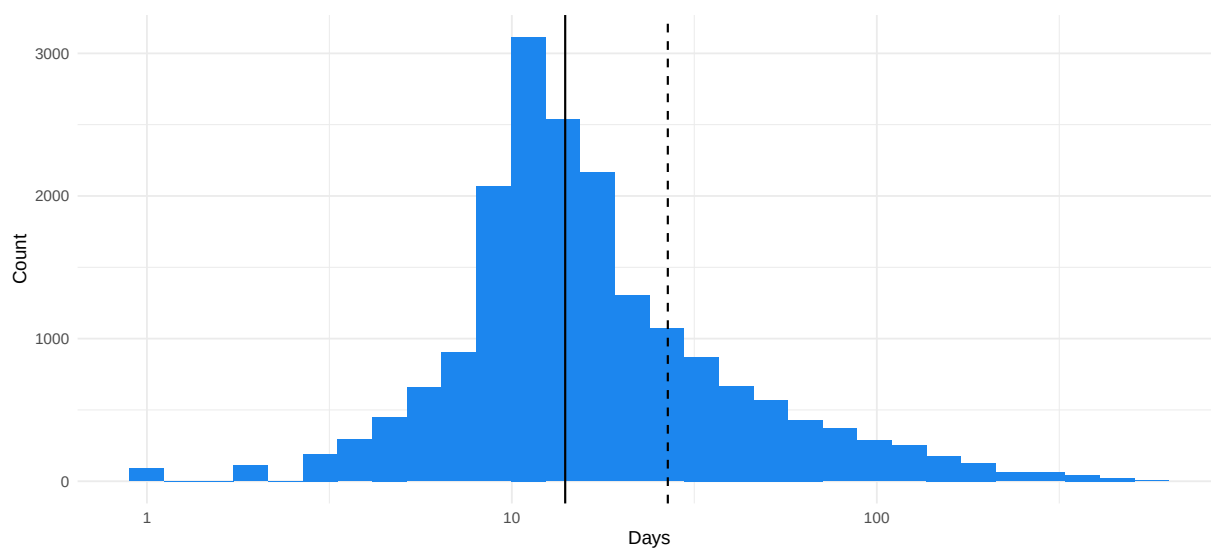


Figure C10: Days between a soldier's death and a death act

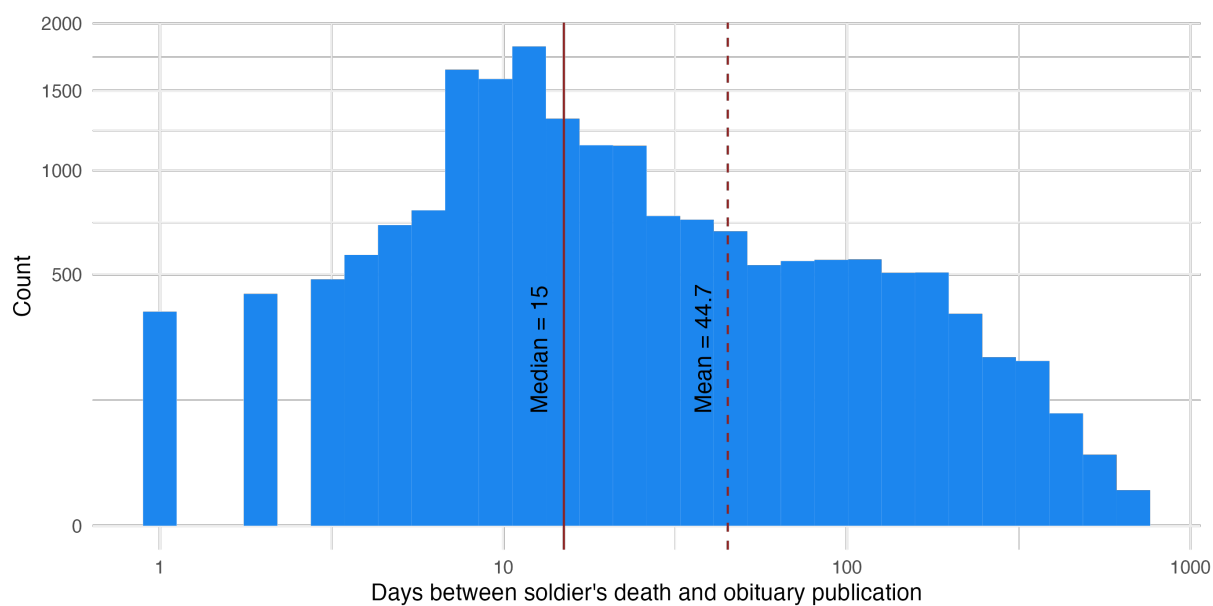


Figure C11: Dynamics of monthly KIA reports. **Note:** Blue bars denote the total number of KIA reports in a given month. Red bars denote the number of municipalities that received the first KIA report in a given month.

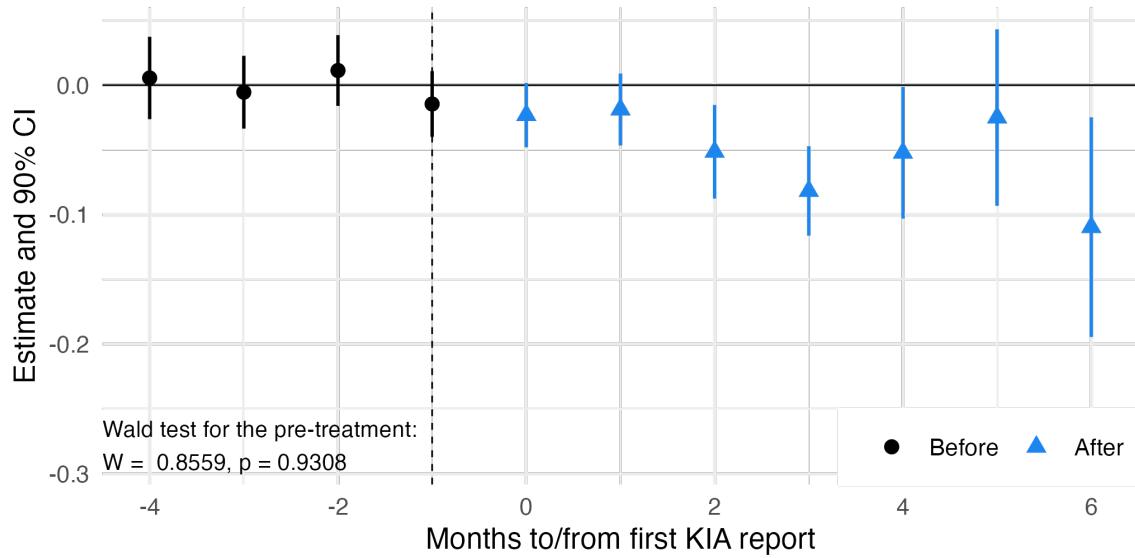
Table C1: The effect of cumulative KIA reports on engagement.

Topic	$\Delta \log$ Likes per post				
	War in Ukraine (1)	Authorities (2)	Patriotism (3)	Authorities (4)	Patriotism (5)
Cumulative KIA	0.006 (0.005)	-0.002 (0.002)	0.0009 (0.0007)	-0.002** (0.0009)	0.0008 (0.0007)
Classifier	Keywords	Keywords	Keywords	Zero-shot	Zero-shot
N Municipality	1,687	2,030	2,035	2,038	2,038
N Month	13	13	13	13	13
Observations	5,345	21,990	23,853	24,886	24,886
R ²	0.435	0.282	0.205	0.340	0.275
Municipality FE	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓

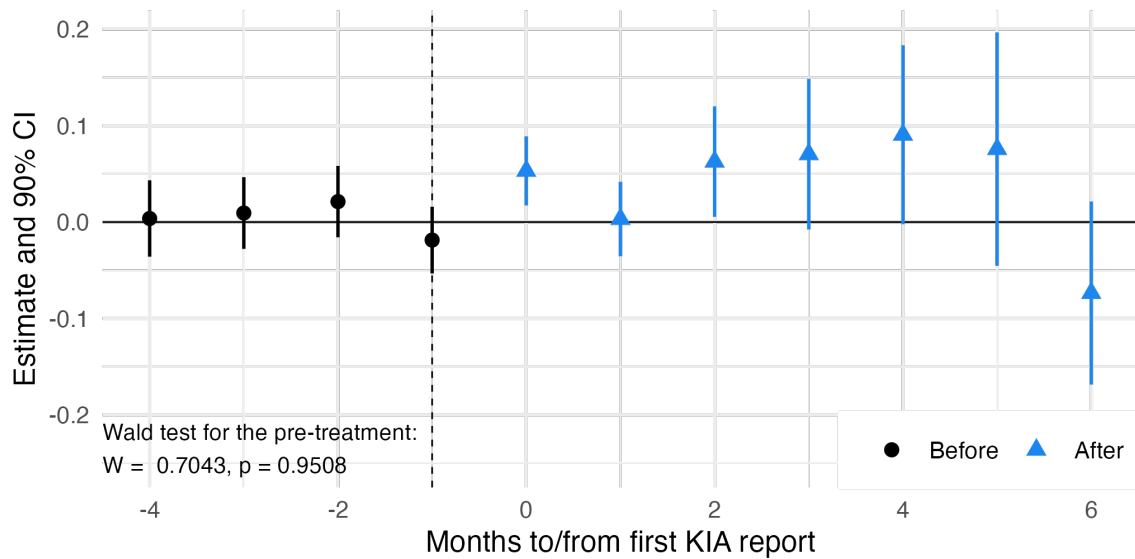
Additional results

Cumulative KIA reports

Event-study graphs for zero-shot topics



(a) Effect on engagement with *Patriotism & Military* topic



(b) Effect on engagement with *Authorities* topic

Figure C12: Event study graphs for the effect of the first KIA report (zero-shot topics). *Note:* Dependent variable: $\Delta \log$ Likes per post. Independent variable: *Post KIA*. Estimator: Callaway and Sant'Anna (2021). Control group: Not yet treated. Dots report the point estimates. Vertical bars report the 90% confidence intervals. Standard errors clustered at municipality level.

Extended sample estimates

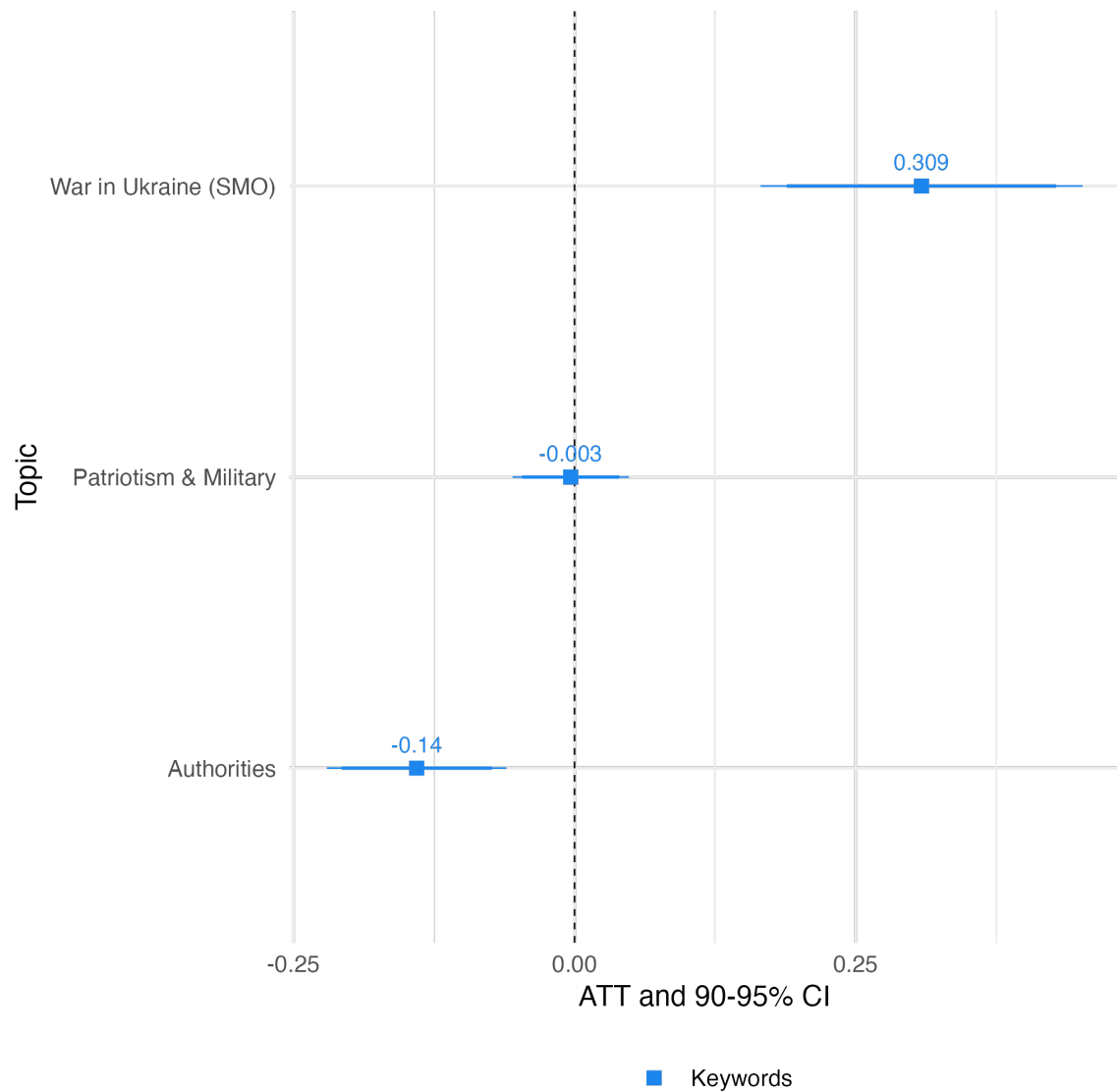
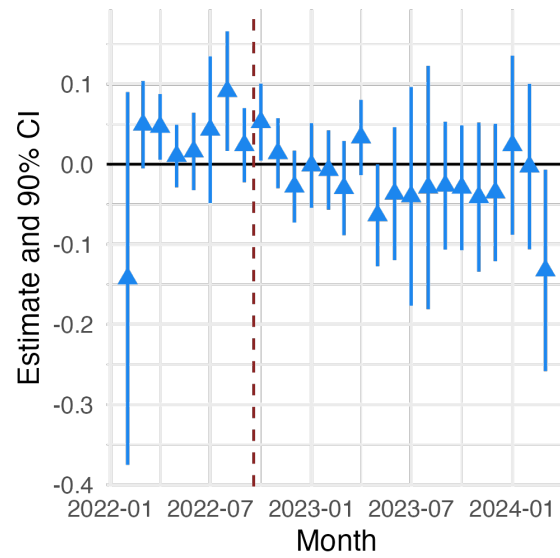
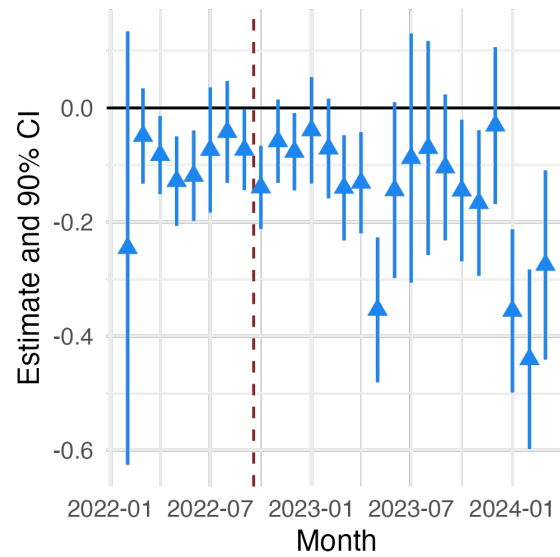


Figure C13: Change in engagement by topic relative to *Neutral* content after the first KIA report. *Note:* Dependent variable: $\Delta \log \text{Likes}$ per post. Treatment: First KIA report month in the municipality. Dots show ATT coefficients from separate Callaway and Sant’Anna (2021) estimations by topic. Topic is defined along the vertical axis. Horizontal bars represent 90% and 95% confidence intervals. Standard errors clustered at the municipality level. Squares and circles indicate classification strategy.

Estimates aggregated by calendar months



(a) Effect on engagement with Patriotism & Military topic



(b) Effect on engagement with *Authorities* topic

Figure C14: Graphs for the effect of the first KIA report aggregated by calendar months (keywords). *Note:* Dependent variable: $\Delta \log$ Likes per post. Independent variable: *Post KIA*. Estimator: Callaway and Sant’Anna (2021). Control group: Not yet treated. Dots report the point estimates. Vertical bars report the 90% confidence intervals. Standard errors clustered at municipality level. Dashed vertical line indicated the start of the “partial mobilization” in late September 2022.

Robustness

Results with obituaries excluded

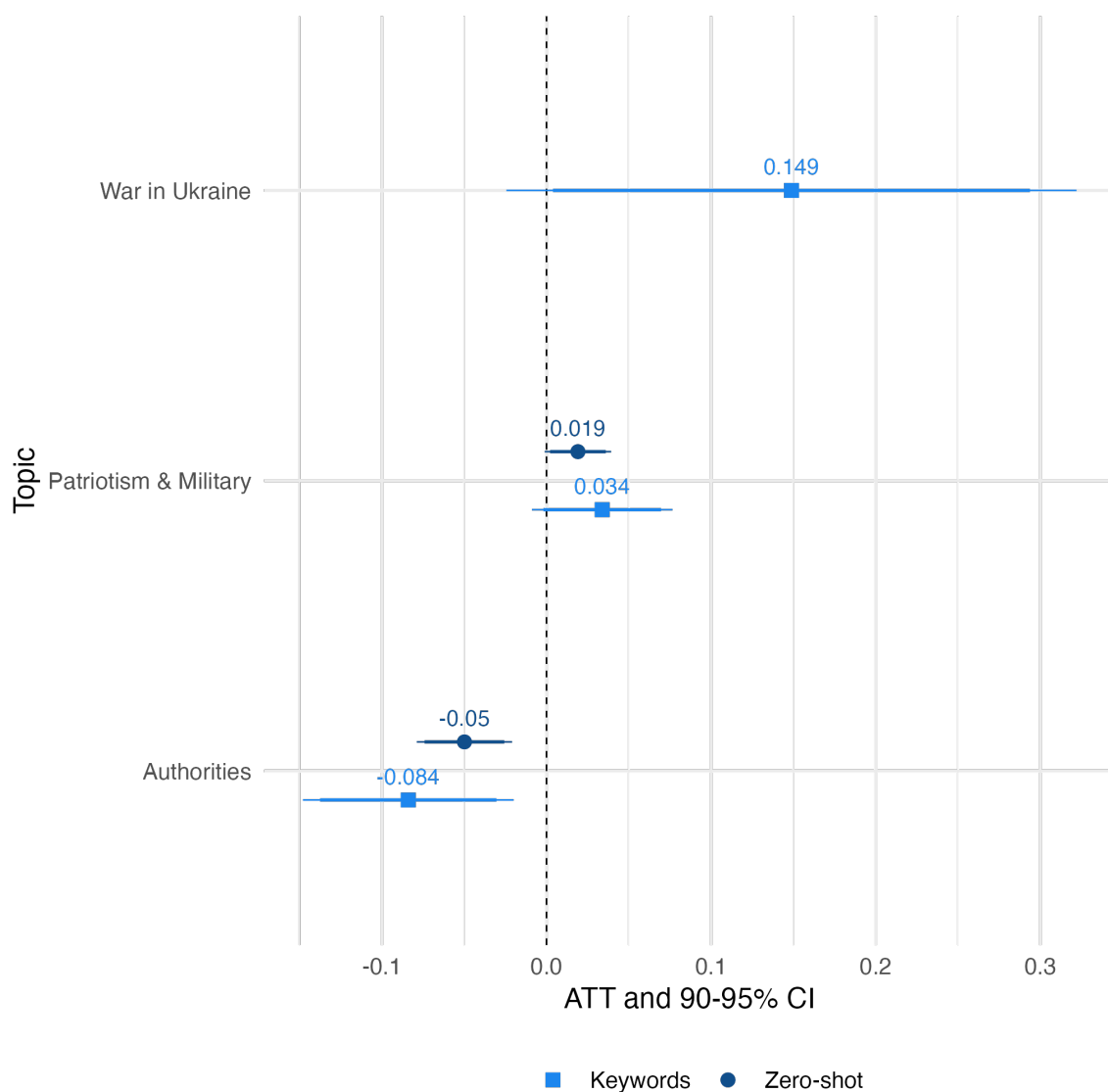


Figure C15: Change in engagement by topic relative to *Neutral* content after the first KIA report (no obituary posts). *Note:* Obituaries removed from the sample of the posts. Dependent variable: $\Delta \log \text{Likes}$ per post. Treatment: First KIA report month in the municipality. Dots show ATT coefficients from separate Callaway and Sant'Anna (2021) estimations by topic. Topic is defined along the vertical axis. Horizontal bars represent 90% and 95% confidence intervals. Standard errors clustered at the municipality level. Squares and circles indicate classification strategy.

Detailed topics

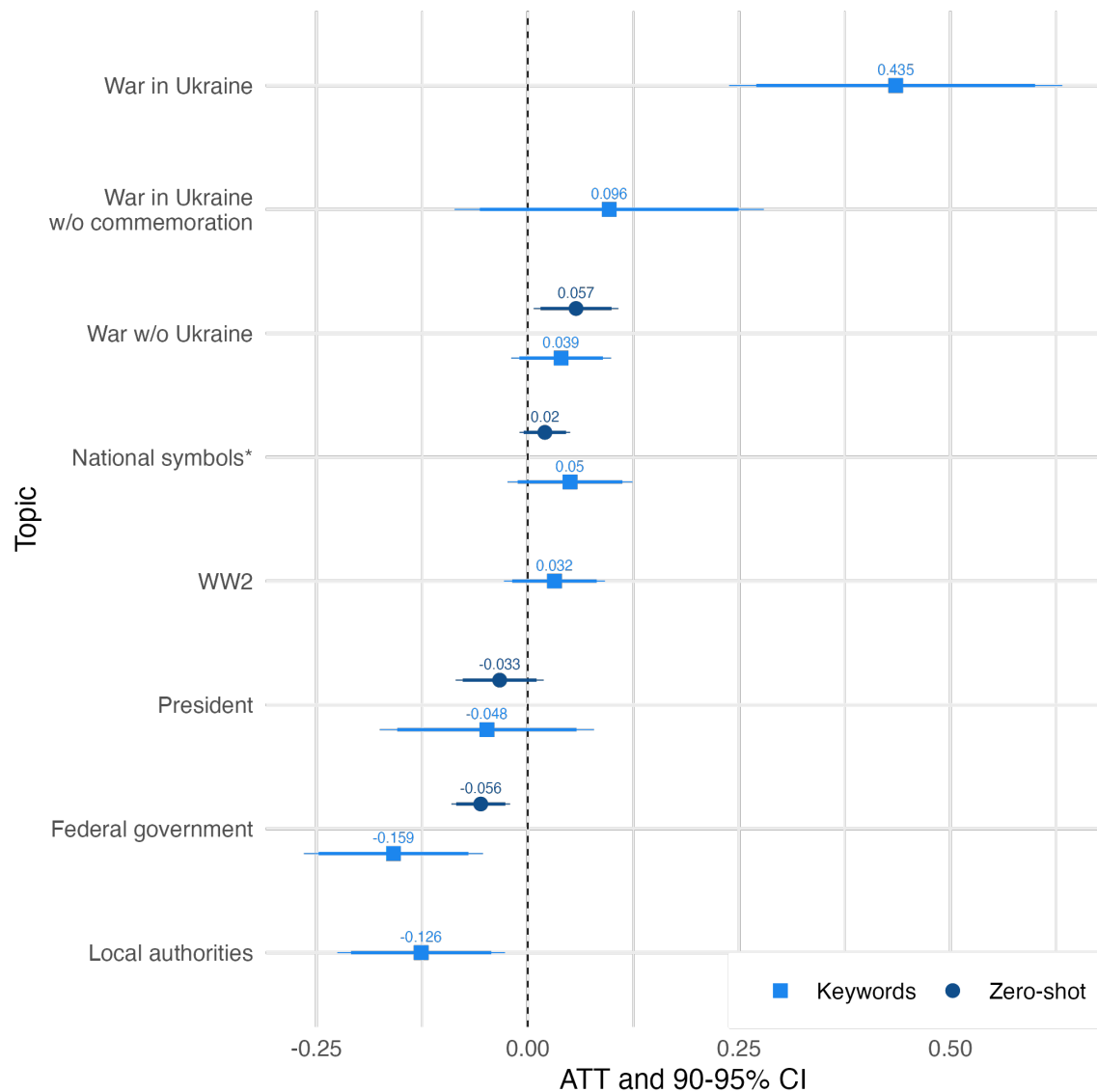
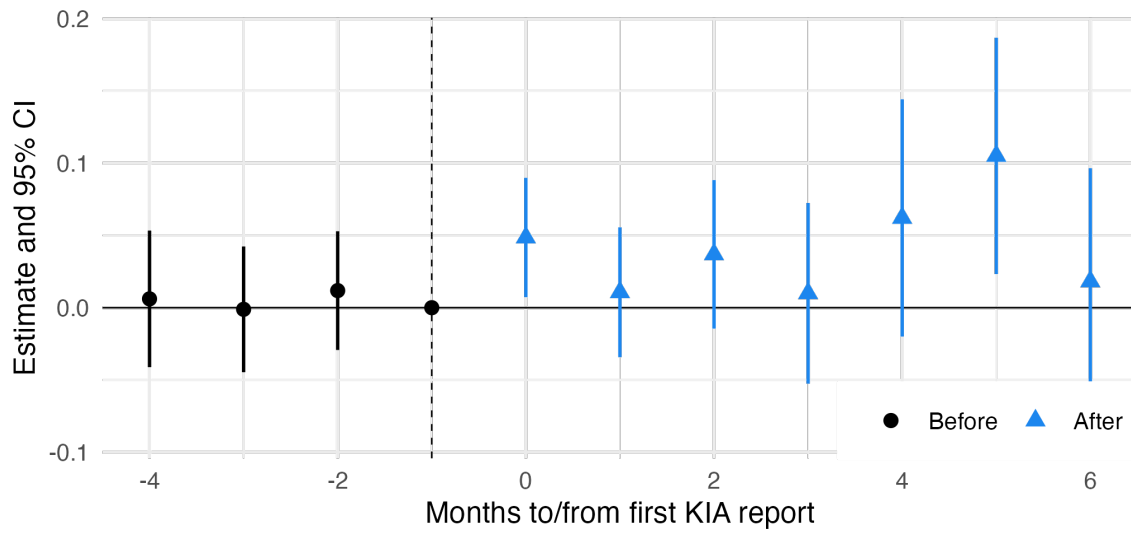
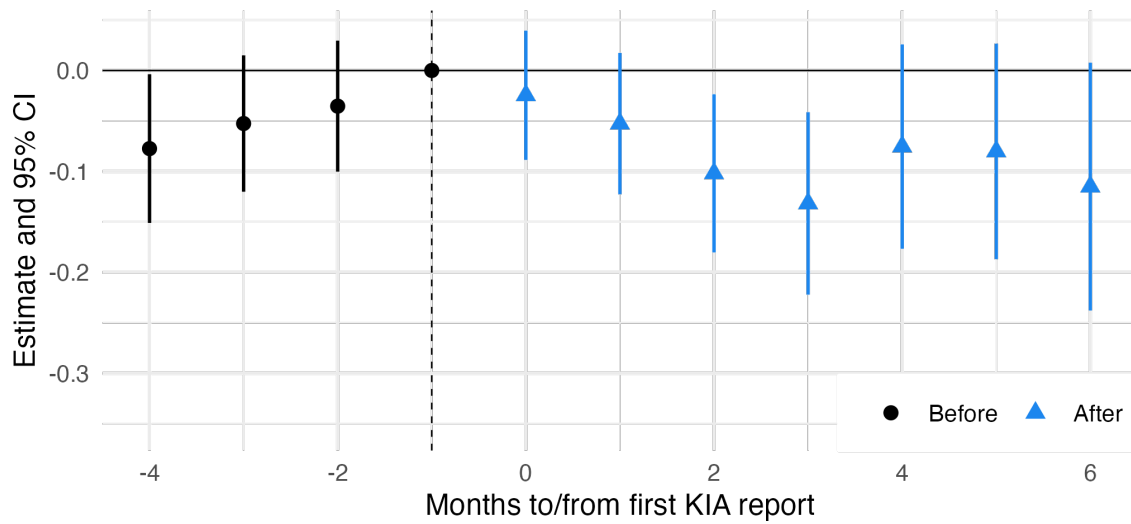


Figure C16: Change in engagement by topic relative to *Neutral* content after the first KIA report. *Note:* Dependent variable: $\Delta \log$ Likes per post. Treatment: First KIA report month in the municipality. Dots show ATT coefficients from separate Callaway and Sant'Anna (2021) estimations by topic. Topic is defined along the vertical axis. Horizontal bars represent 90% and 95% confidence intervals. Standard errors clustered at the municipality level. Squares and circles indicate classification strategy.

Alternative Estimator



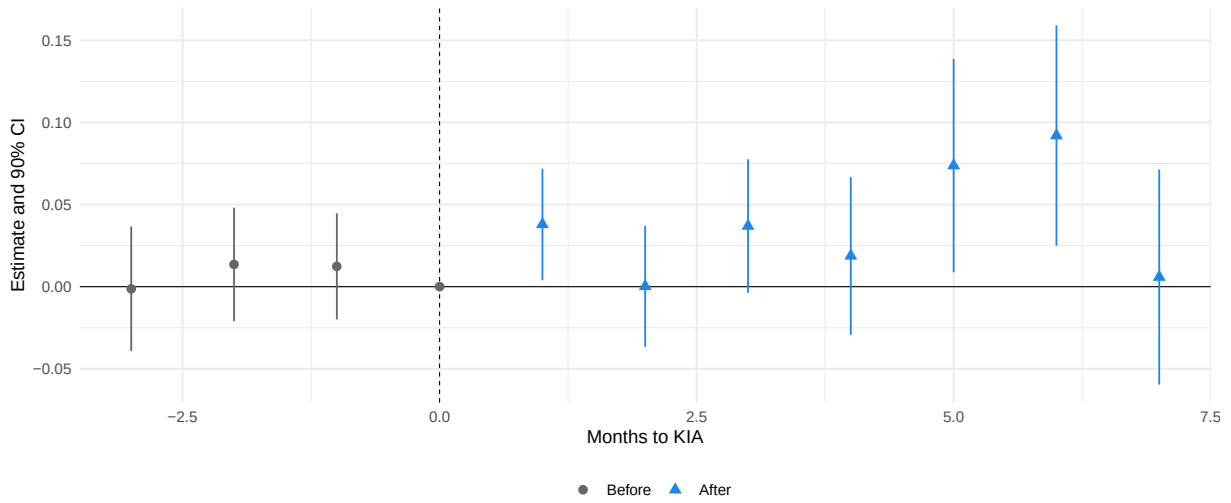
(a) Effect on engagement with *Patriotism & Military* topic



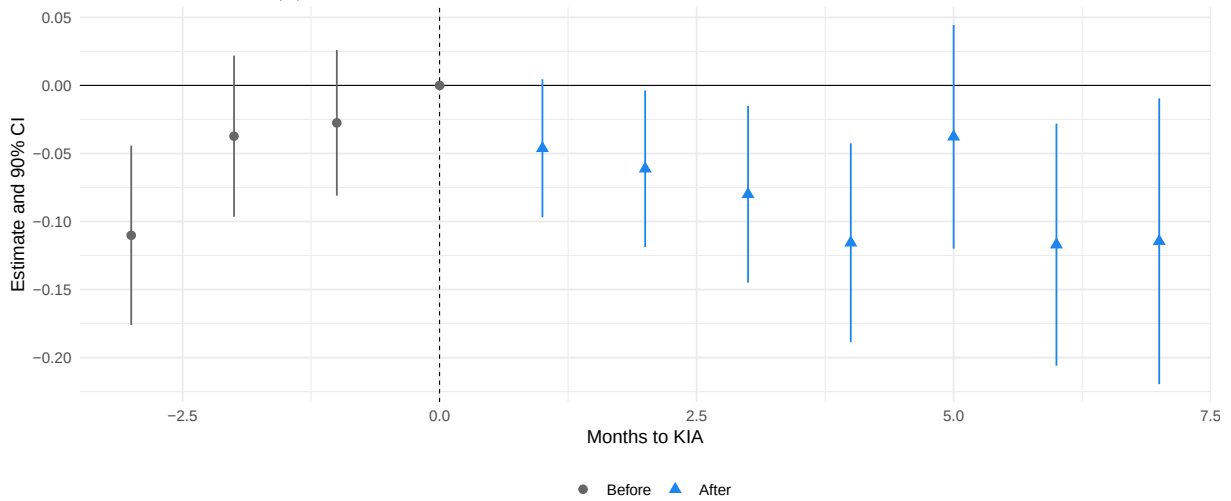
(b) Effect on engagement with *Authorities* topic

Figure C17: Event study graphs for the effect of the first KIA report (keywords). *Note:* Dependent variable: Δ log Likes per post. Independent variable: *Post KIA*. Estimator: Sun and Abraham (2021). Control group: Not treated by the end of the considered period (January 2022 – September 2022). Dots report the point estimates. Vertical bars report the 90% confidence intervals. Standard errors clustered at municipality level.

Repeated Treatment



(a) Effect on engagement with *Patriotism & Military* topic



(b) Effect on engagement with *Authorities* topic

Figure C18: Event study graphs for the effect of a KIA report (keywords). *Note:* Dependent variable: $\Delta \log$ Likes per post. Independent variable: *Post KIA*. Estimator: Chaisemartin and D'Haultfoeuille (n.d.), allowing for treatment switching on and off. Control group: Not treated at a given month with a similar treatment history. Dots report the point estimates. Vertical bars report the 90% confidence intervals. Standard errors clustered at municipality level.

Placebo studies

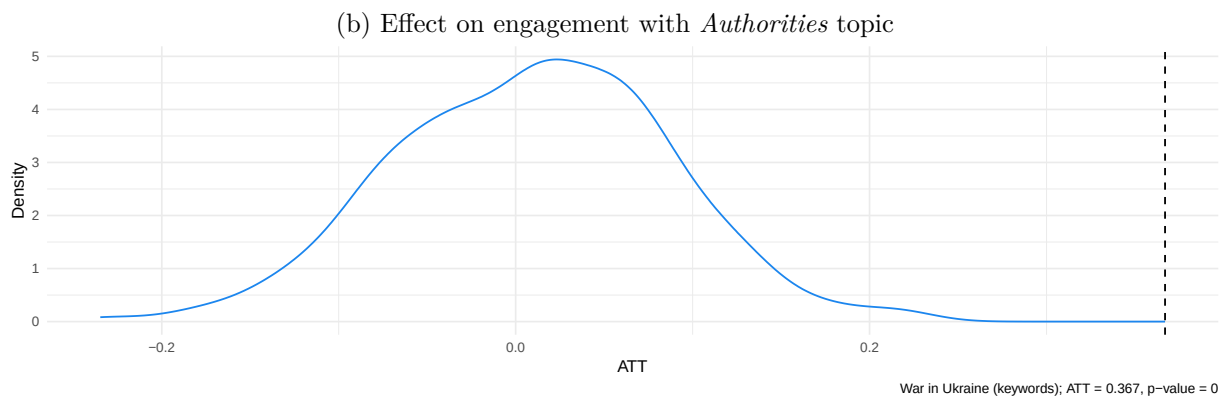
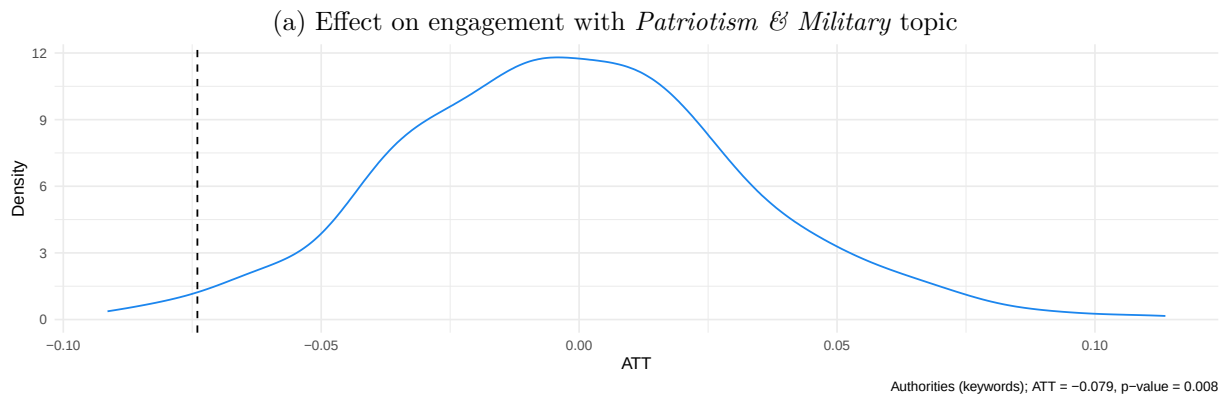
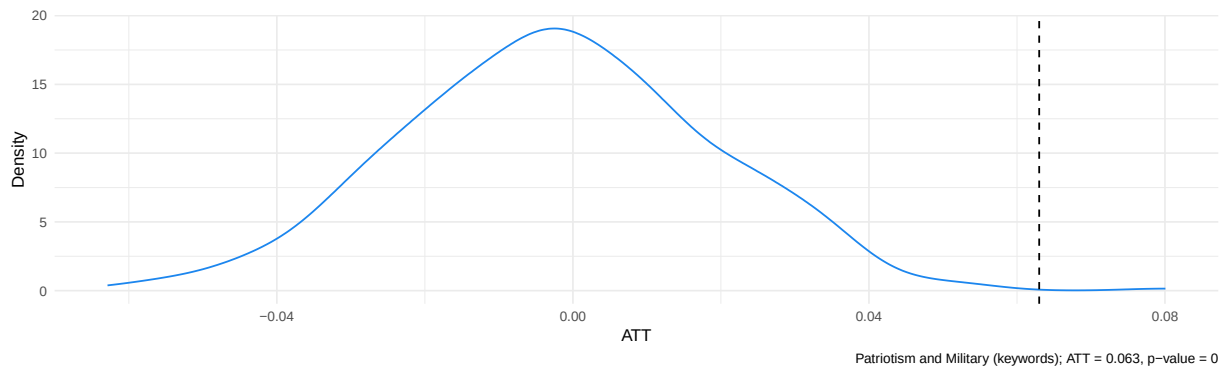
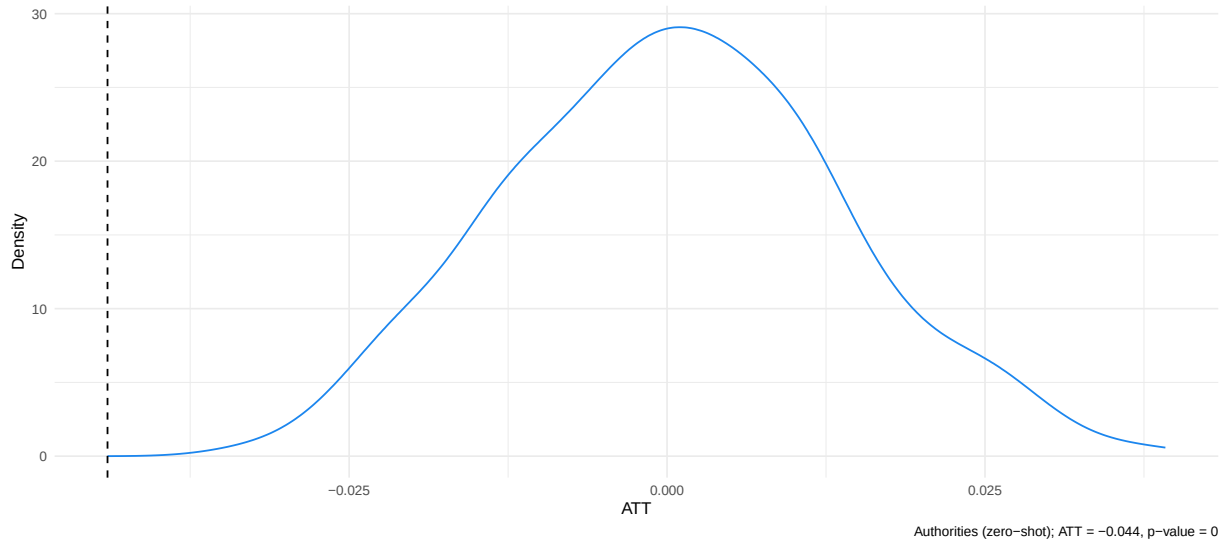
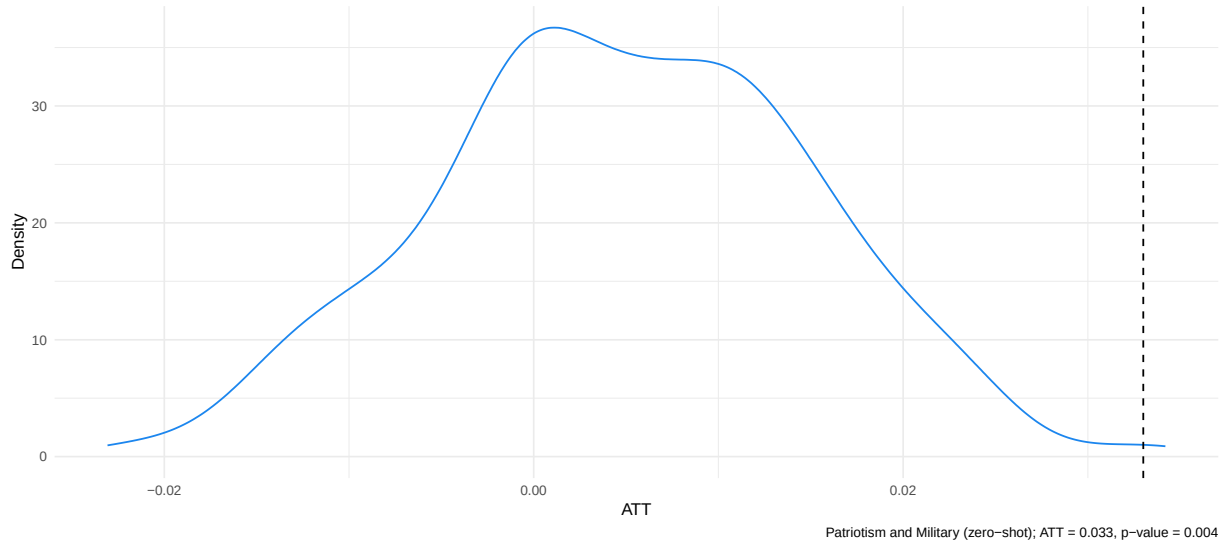


Figure C19: Monte-Carlo simulation for the effect of the first KIA report (keywords). *Note:* Dependent variable: Δ log Likes per post. Independent variable: *Post KIA*. Estimator: Callaway and Sant'Anna (2021). Control group: Not yet treated. Dashed vertical line reports the ATT estimate on the original sample. Standard errors clustered at municipality level.



(a) Effect on engagement with *Patriotism & Military* topic



(b) Effect on engagement with *Authorities* topic

Figure C20: Monte-Carlo simulation for the effect of the first KIA report (zero-shot). *Note:* Dependent variable: $\Delta \log \text{Likes per post}$. Independent variable: *Post KIA*. Estimator: Callaway and Sant'Anna (2021). Control group: Not yet treated. Dashed vertical line reports the ATT estimate on the original sample. Standard errors clustered at municipality level.

Table C2: Keywords and Topics. *Note:* Russian translation in *italics*. Keywords allow for flexible patterns at the end of strings to account for various grammatical forms. **Conversations* or *conversations on important topics* is a term used by the Russian Ministry of Education to refer to patriotic upbringing lessons.

Topic	Keywords
War in Ukraine	
Special military operation	SMO, special military operation, letter to a soldier, a hero's desk
Ukraine	Crimea, Sevastopol, Ukraine, Donbass, Luhansk, Kherson, Zaporizhzhia, Mariupol, Donetsk, Kyiv
Authorities	
President	president, Putin
Government	deputy, parliament, state дума, minister, ministry
Local authorities	head of region, governor
Patriotism and Military	
War	soldier, military, warrior, mobilization, special operation, protection, defender, hero, serviceman, frontline, valor, veteran, motherland, fighter, fighting
WW2	Great patriotic war, great victory, World War II, 1940s, victory
Patriotism	flag, national anthem, conversations*, lesson of bravery, heroism, duty

Descriptives

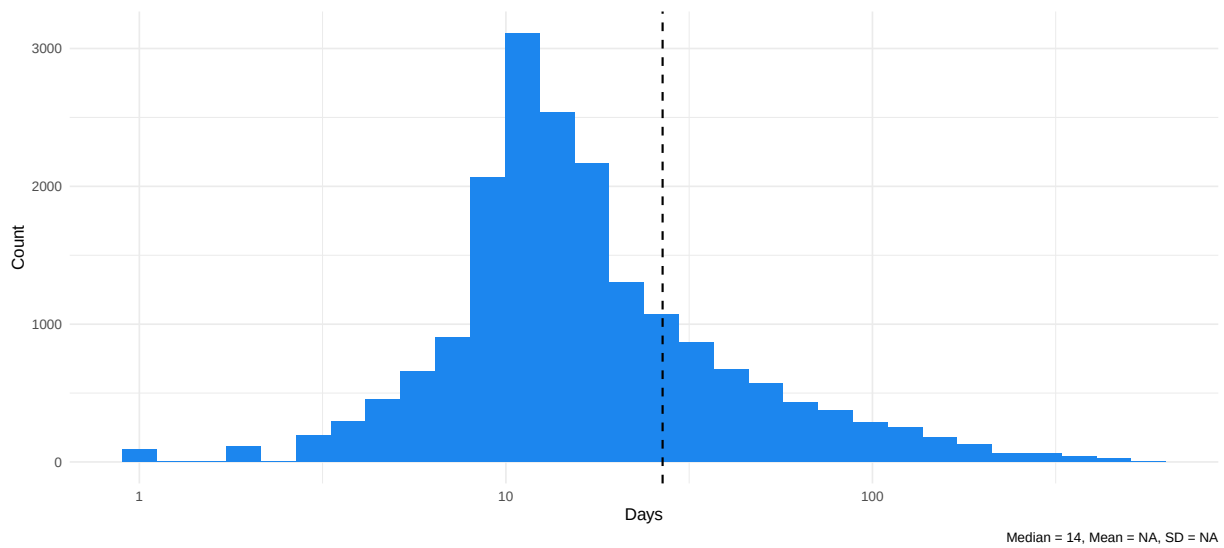


Figure C21: Days between a soldier's death and a death act

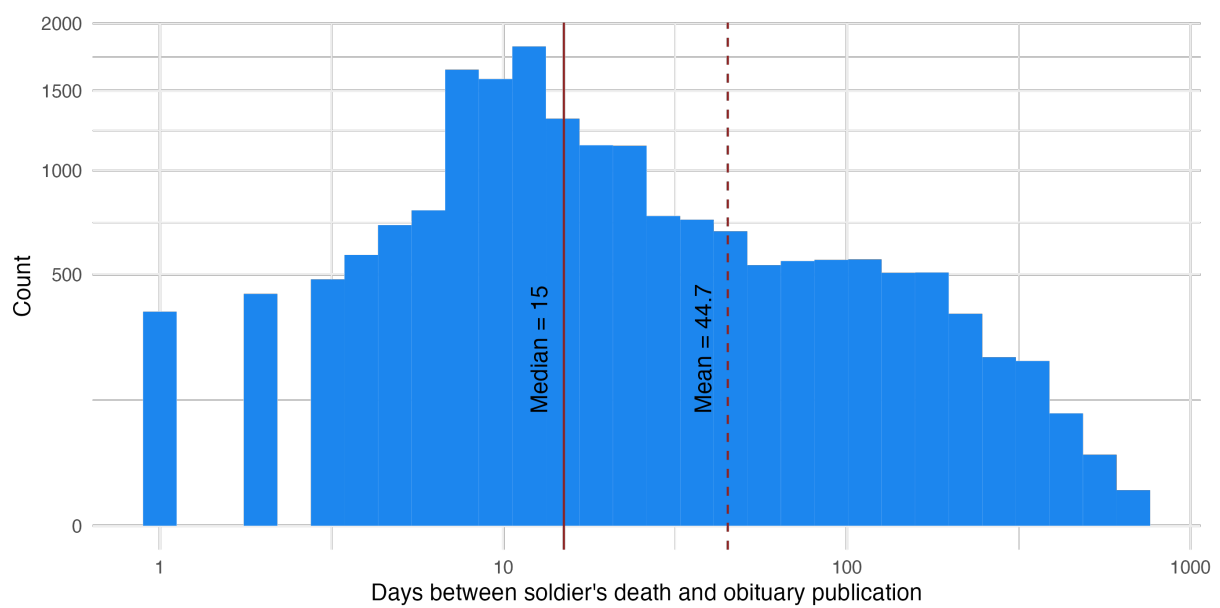


Figure C22: Dynamics of monthly KIA reports

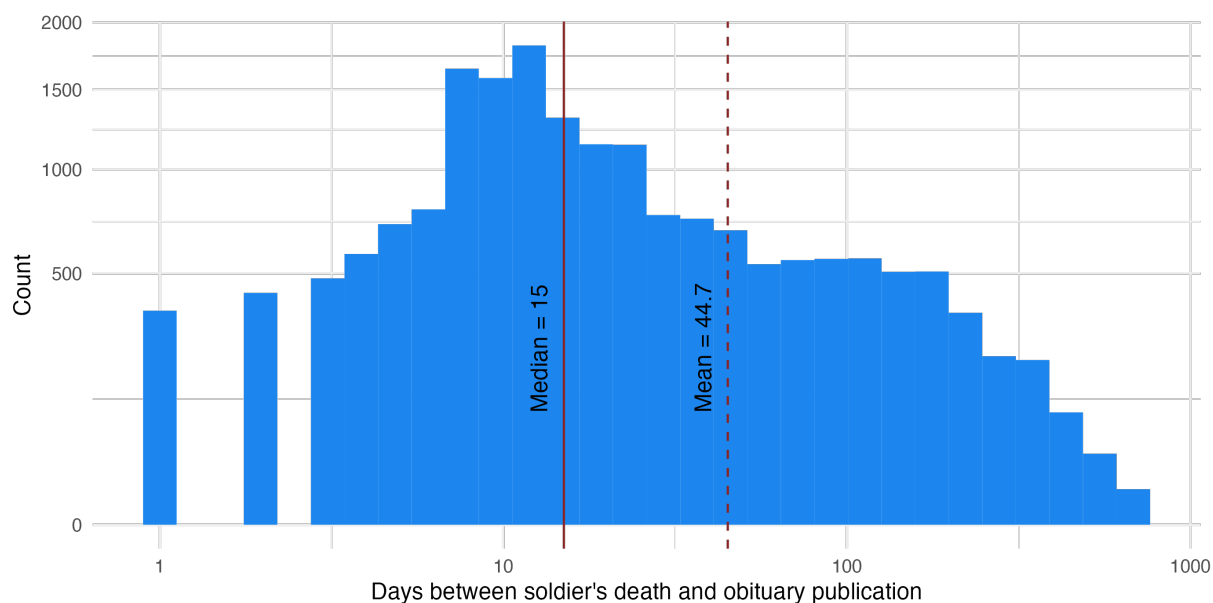


Figure C23: Date of first KIA reports

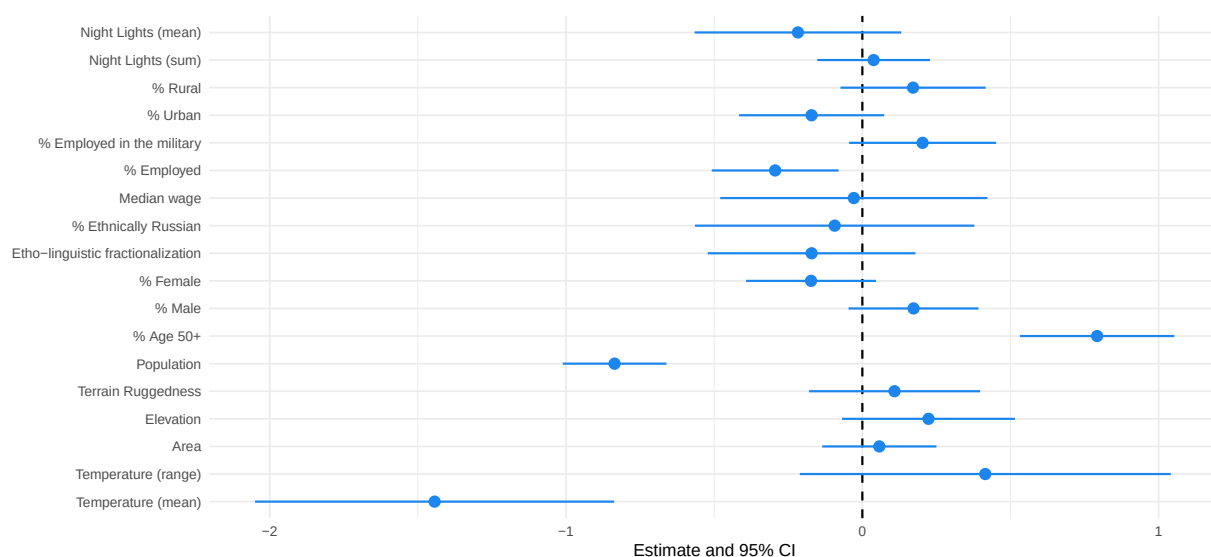
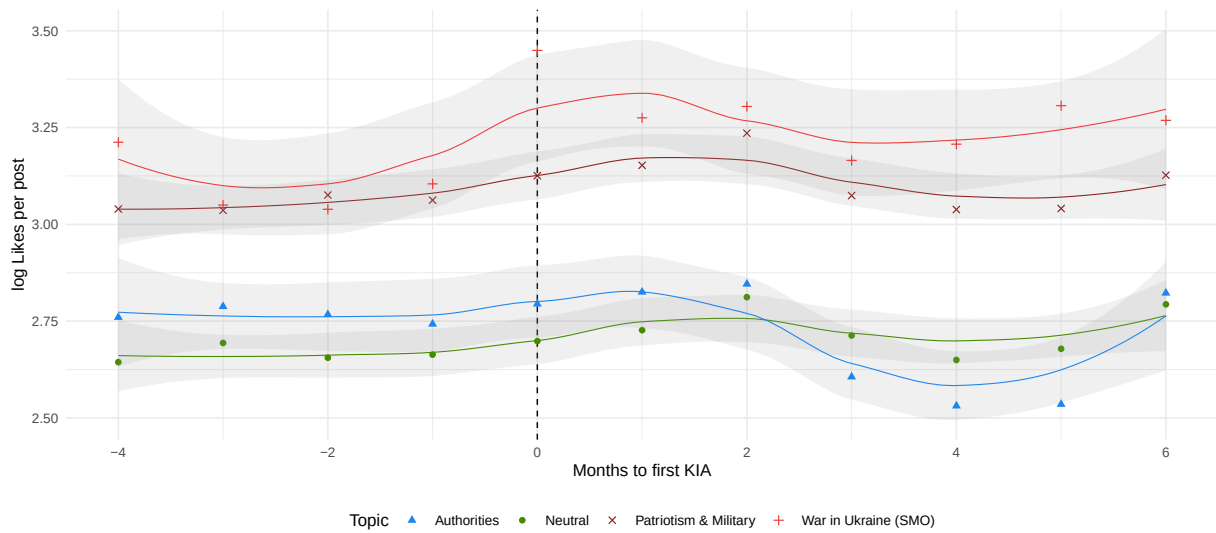
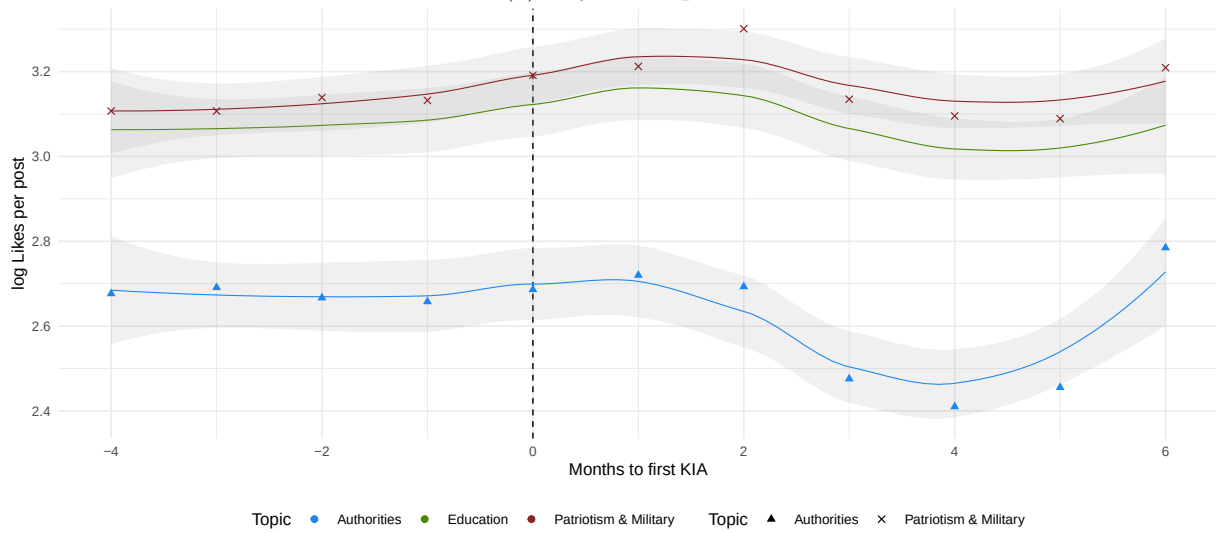


Figure C24: Correlates of the KIA report treatment with covariates. *Note:* Dependent variable: Month of the first KIA report. Independent variables: Covariates defined along the vertical axis and region fixed effect. Dots show the standardized coefficient from a separate regression. Negative coefficients correspond to earlier treatment time. Standard errors clustered at the region level. Data sources: Federal State Statistics Service of Russia (Rosstat); NASA VIIRS.



(a) Keyword topics



(b) Zero-shot topics

Figure C25: SM engagement around the date of the first KIA report

Supplemental Appendix

Topic and sentiment score correlations

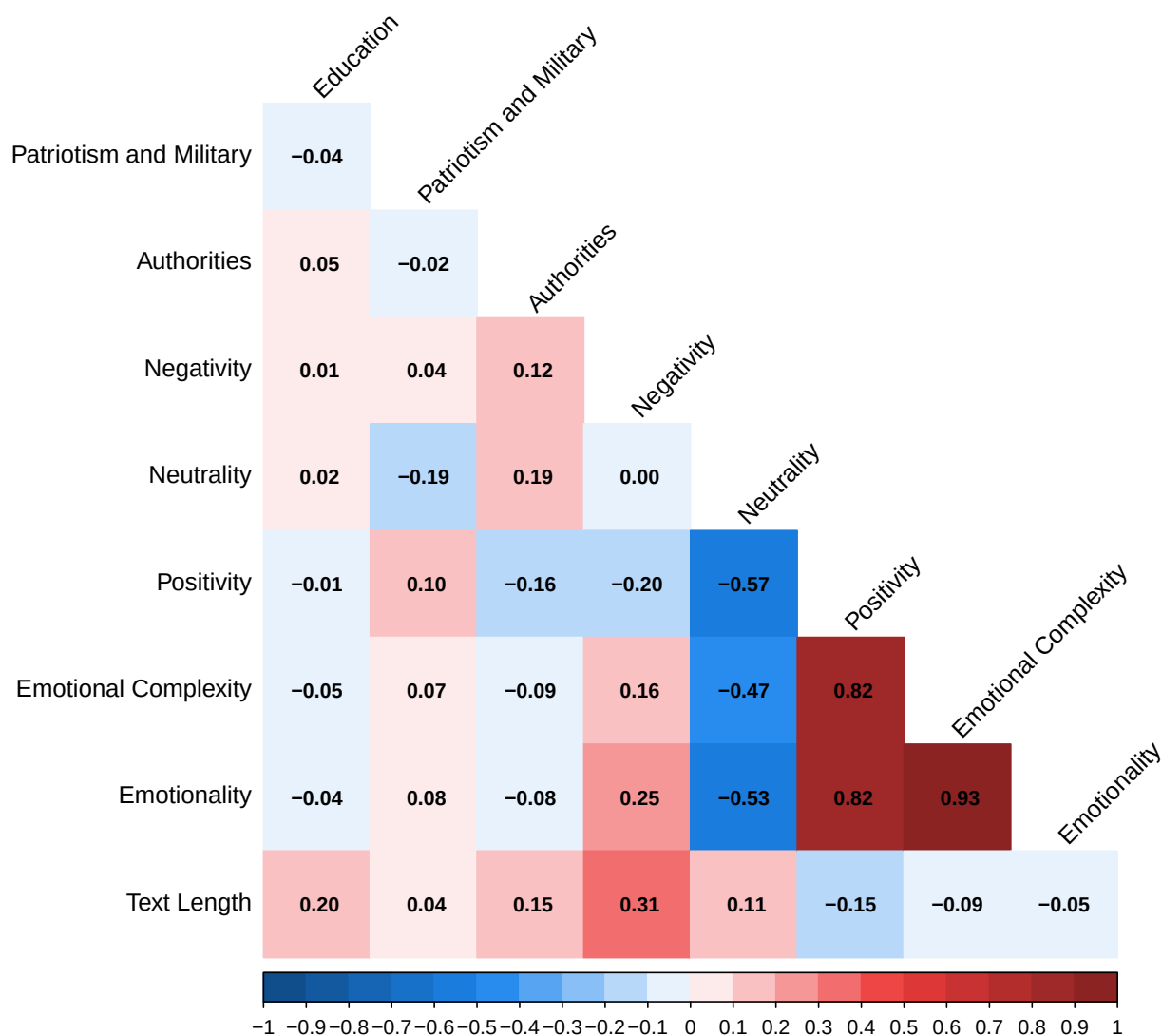


Figure C26: Correlation matrix of Zero-shot topic probabilities and sentiment scores

Table C3: Heterogeneity of the KIA reports effects with respect to alternative sentiment scores

Score	Likes per post			
	Neutrality (1)	Emotinality (2)	Complexity (3)	log Length (4)
Post KIA	2.98*** (0.288)	-1.52*** (0.198)	-0.754*** (0.177)	0.424 (0.657)
Topic relevance	-23.1*** (0.367)	18.6*** (0.497)	28.6*** (0.881)	-1.36*** (0.079)
Post KIA × Topic relevance	-5.55*** (0.396)	7.92*** (0.701)	11.7*** (1.29)	-0.120 (0.107)
Mean Likes	18.9	18.9	18.9	18.9
Mean Score	0.577	0.153	0.038	6.09
N School	12,971	12,971	12,971	12,971
N Month	13	13	13	13
Observations	2,014,787	2,014,787	2,014,787	2,014,787
R ²	0.319	0.276	0.275	0.272
School FE	✓	✓	✓	✓
Month FE	✓	✓	✓	✓

Emotional intensity and text length

Battlefield dynamics

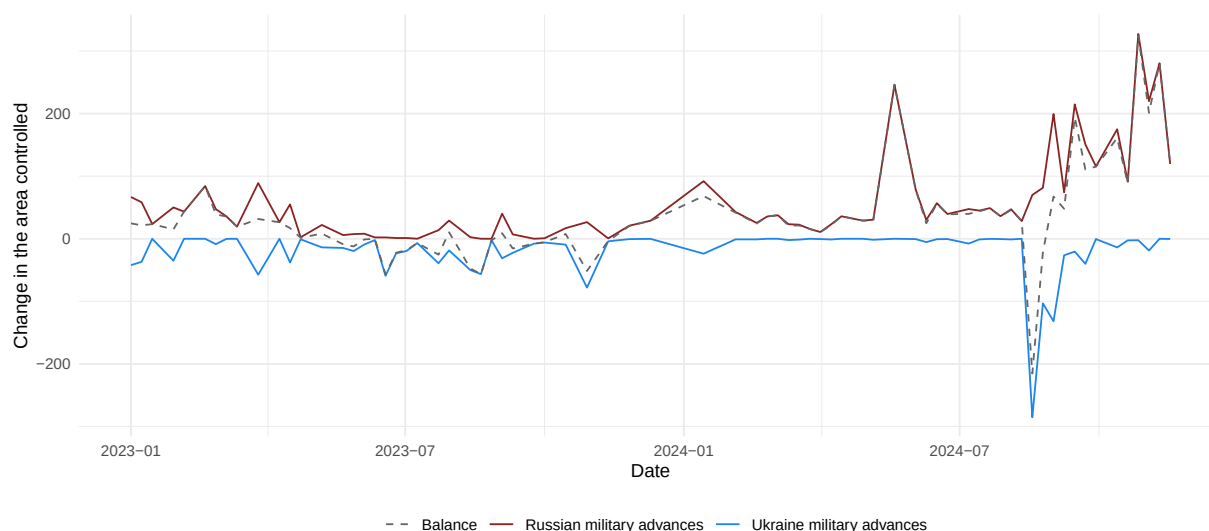


Figure C27: War in Ukraine battlefiled dynamics in 2023-2024. *Note:* Red and blue lines shows the changes in the area controlled by Russian and Ukrainian military in squared kilometers, respectively. Dashed line indicates the changes in the overall balance in terms of controlled territory. Note that the overall balance was very stable in 2023. Source: Meduza

Maps

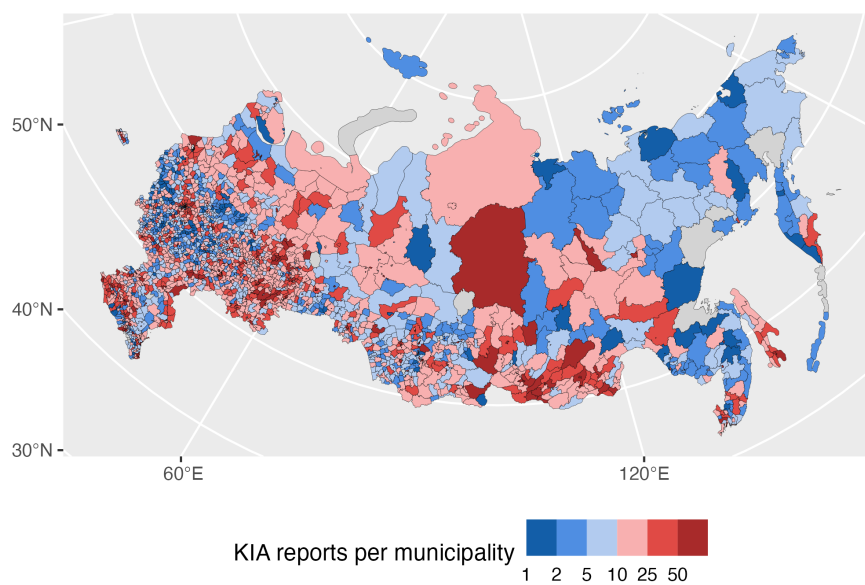


Figure C28: Map of confirmed military losses in Russian municipalities by March 2024

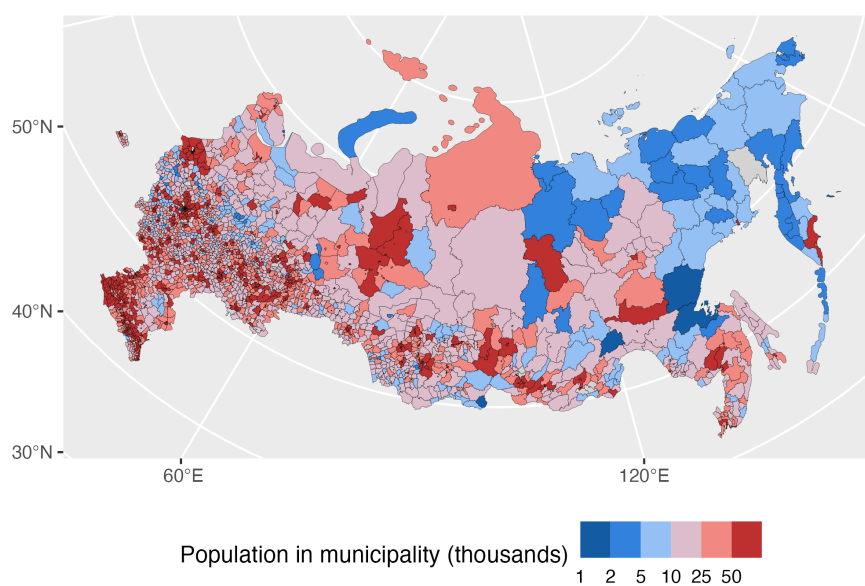


Figure C29: Map of population in Russian municipalities

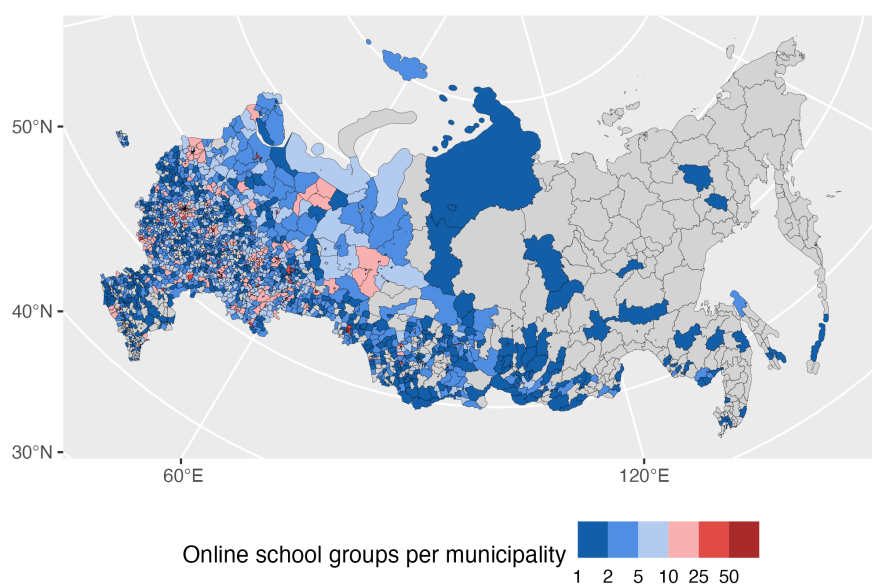


Figure C30: Map of online school groups in Russian municipalities in January 2022

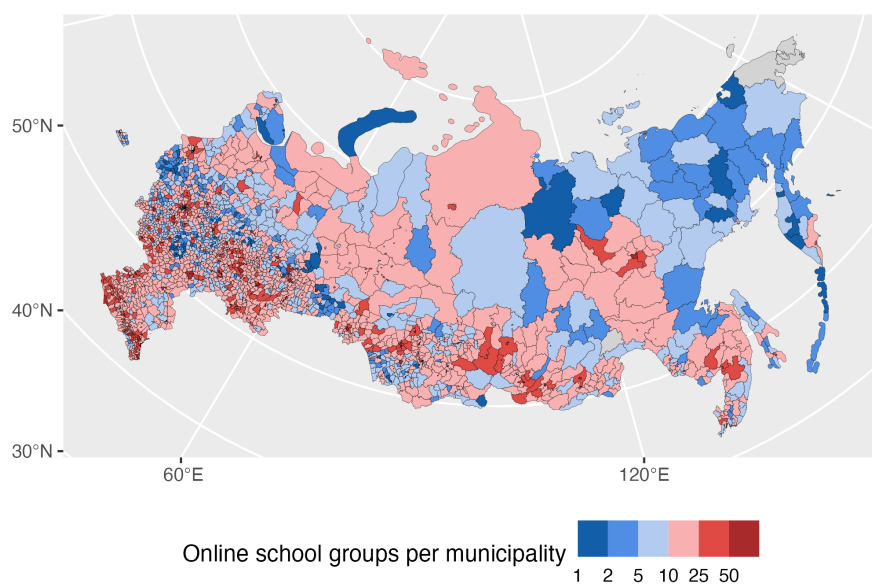


Figure C31: Map of online school groups in Russian municipalities in March 2024

Wordclouds



Figure C32: Wordcloud for *Education* topic from a random subsample of posts (zero-shot probability > 0.8)



Figure C33: Wordcloud for *War* topic from a random subsample of posts (zero-shot probability > 0.8)



Figure C34: Wordcloud for *Patriotism* topic from a random subsample of posts (zero-shot probability > 0.8)



Figure C35: Wordcloud for *President* topic from a random subsample of posts (zero-shot probability > 0.8)

report is then sent to the military commissariat at the servicemember's place of registration, which has to issue the death certificate to the family no later than 10 days after receiving the report. Therefore, a serviceman should be declared dead within 40 days since the supposed date of death. However, journalistic accounts suggest that the commanders of the military units have incentives to cover soldiers' deaths as the unit's KPI is calculated as a function of the number of people killed in action but not those missing.¹²

If no death certificate is available from a medical professional or the military, Russian civil law allows a court to declare a citizen dead. This requires five years of absence without information about the person's whereabouts. However, in cases where the individual went missing under life-threatening circumstances, this period can be reduced to six months. Before September 2023, servicemembers could only be declared dead no sooner than two years after the end of military operations. Following legal changes in September 2022, this waiting period was further reduced to a maximum of three months for those missing in the "Special Military Operation zone."¹³

¹²For instance, some military units have been reported for declaring soldiers who have not yet entered the battle missing "in advance" to appropriate their salary bank cards and increase the KPI of the unit (see "As a rule, they don't get there" 2024).

¹³Federal Law of 30.11.1994 N 52-FZ (as amended on 25.12.2023) "On the entry into force of Part One of the Civil Code of the Russian Federation" (as amended and supplemented, entered into force on 01.05.2024)